

THE
**INTELLIGENT
INVESTOR**
CRYPTO EDITION

*The Digital Gold Standard
& the Fair Value Revolution*



John Wright Gotts

The Intelligent Investor

Crypto Edition

John Wright Gotts

Founder & Creator

Digital Gold Foundation

Digital Gold Coin (DGD)

Digital Gold Standard Benchmark

Crypto Fair Value Formula

CFV CoinFunds

Author: Cryptocurrency Analysis

© 2026 Gotts Publishing

All Rights Reserved

To Benjamin Graham, Bram Cohen, Marc Andreessen, Nick Szabo, Elon Musk, Steve Jobs, Carl Menger, Ludwig von Mises, Friedrich Hayek, Murray Rothbard, and all of my heroes who think differently and change things.

—John Wright Gotts

“Here’s to the crazy ones. The misfits. The rebels. The troublemakers. The round pegs in the square holes. The ones who see things differently. They’re not fond of rules. And they have no respect for the status quo. You can quote them, disagree with them, glorify or vilify them. About the only thing you can’t do is ignore them. Because they change things. They push the human race forward. And while some may see them as the crazy ones, we see genius. Because the people who are crazy enough to think they can change the world, are the ones who do.”

—Apple’s “Think Different” commercial, 1997

“It is not the critic who counts: not the man who points out how the strong man stumbles or where the doer of deeds could have done better. The credit belongs to the man who is actually in the arena, whose face is marred by dust and sweat and blood, who strives valiantly, who errs and comes up short again and again, because there is no effort without error or shortcoming, but who knows the great enthusiasms, the great devotions, who spends himself in a worthy cause; who, at the best, knows, in the end, the triumph of high achievement, and who, at the worst, if he fails, at least he fails while daring greatly, so that his place shall never be with those cold and timid souls who knew neither victory nor defeat.”

—Theodore Roosevelt, 1910

$$\begin{aligned}
\text{CFV} = & 1.983 \text{ trillion} \times \\
& [0.70 \times (\text{Coin Adoption} / 80 \text{ million}) \\
& + 0.10 \times (\text{Coin AT} / 6.09 \text{ billion}) \\
& + 0.10 \times (\text{Coin ATV} / 13.49 \text{ trillion}) \\
& + 0.10 \times (\text{Coin Dev} / 905)]
\end{aligned}$$

INTRODUCTION	viii
PREFACE	1
1. Investment, Speculation, and Crypto	6
2. The Architectural Test	18
3. The Six Pillars of Perfect Money	30
4. The Foundation's Tracked Set	46
5. Bitcoin as Calibration Source	60
6. The Crypto Fair Value Formula	72
7. The Margin of Safety	86
8. Worked Coin Reports	101
9. Calibration Tests	114
10. Allocation Under Fiduciary Constraint	126
11. Allocation Under Personal-Capital Constraint	138
12. Rebalancing, Exit, and Holding	148
13. Reproducibility	163
14. Three Critics	174
15. Falsification Conditions	187
Appendix A: Benchmark Specification	201
Appendix B: Reproduction Protocol	214

Sources and Further Reading	224
Glossary	230
About the Author	243
Index	246

INTRODUCTION

“We are still in the early days of crypto. Despite Bitcoin launching more than a decade ago and some people already getting rich from it, the general public’s view of cryptocurrency’s promise is poor. Most people don’t see a reason to use it themselves and view much of its current use dimly. Even though the case for crypto is poor, its promise is real. Highly secure, reliable, permissionless money is an amazing technology which can revolutionize finance. The current underwhelming utility and bad reputation of crypto should be a mark of shame and an obligation to do better for us developers working on it. But it also presents an opportunity. Crypto will inevitably do more, and those who make the right investments in the future will reap great rewards.”

—Bram Cohen, Creator BitTorrent & XCH

“In 2016 and 2017, when the block size debate was raging, I argued that if BTC did not raise its block size, it would end up being used through custodians, reintroducing the third parties it was trying to eliminate. Unfortunately, these predictions ended up being correct and this came with disastrous consequences: there is no need to control BTC if you can control the custodians. This led to a speed-run of the traditional financial system, but not much freedom delivered. It’s time for a new round of innovation in the crypto ecosystem. Scalability, usability, privacy, and more do matter. We do our best to move the industry forward with eCash, and so do many others. The time has come to leave narrative behind and evaluate coins on their fundamentals again.”

—Amaury Séchet, Creator BCH & XEC

“The true value of a blockchain is not measured by its market cap. It is measured by the principles it stands on. A blockchain that matters long term is built on decentralization, genuine peer-to-peer transactions without middlemen, and open economic participation where anyone in the world can run a node, verify the ledger, and take part in the network. Those are the foundations that determine whether a blockchain serves humanity or simply enriches a small group of insiders. Simple market cap comparisons create what I call the market cap fallacy. They reduce the entire conversation to price and speculation and distract people from the fundamentals that actually matter. When markets are driven by that kind of thinking, the biggest frauds and the most centralized systems often rise to the top because they have the loudest marketing, the largest treasuries, or the most influence over exchanges and media narratives. Meanwhile, the truly decentralized networks that are evenly distributed across thousands of independent participants are often overshadowed and ignored. The result is a market that rewards centralization while pretending to celebrate decentralization, perpetuating the very systems of control that blockchain technology was meant to eliminate. Over the past decade, hundreds of extraordinary cryptographic innovations have emerged across this industry. Many of them have come from outside the Bitcoin ecosystem, including projects like DigiByte and other independent networks that have been quietly advancing the technology for more than a decade. Only a small handful of Layer-1 blockchains remain truly decentralized, independent, and resilient enough to withstand the pressure of nation states and centralized power structures. If we want to measure the real significance of a blockchain network, we must stop measuring price and start measuring principles. We must measure decentralization, resilience, real-world utility, and the freedom for anyone on earth to participate without permission. Those are the qualities that determine whether a blockchain is just another speculative token or a lasting technological breakthrough that can expand human freedom.”

—Jared Tate, Creator DGB

“In 2008, digital distributed ledgers were invented. This solved a very hard problem of how to have digital money outside of the control of governments, and just as importantly, resistant to hostile influence. This was used initially for Bitcoin, but soon, other brilliant innovators recognized the power of these ledgers for provenance, immutable documentation, timestamps, digital twins, authentication, proof of ownership of real-world assets like art and real estate, public communication, anchoring markets, swaps, ownership rights, tokenization, domain names, unique assets (NFTs), and so much more. With these new projects came more speed, more capacity, and often more privacy. The story and value of these top-tier projects need to be communicated widely, and this book does just that.”

—Tron Black, Founding Developer RVN

“Bitcoin was the standard bearer for permissionless money, for digital cash. It launched with no foundation, no VC money, no known founder, and yet managed to grow into a trillion-dollar asset based on the sheer amount of goodwill the world had for digital cash. For the past decade, however, Bitcoin has been stuck at digital gold: a collectible with hamstrung utility to actually act as a replacement for the fiat currency and banking system. And the whole crypto space has largely been stuck behind it. In 2017, when Bitcoin fully pivoted from money for the world, it represented 85% of the crypto market. Years later, it still commands much of investment in the space, largely under this false assumption that it still represented an uncensorable sound money the whole world could use. Crypto is no longer a new industry. We’re almost two decades in, and will soon need to onboard billions of everyday users. We just can’t afford to still be giving Bitcoin the lion’s share of the attention, capital, adoption, and more. We have to decouple and let the space truly flourish, before it’s too late and we end up stuck in centralized, permissioned systems.”

—Joel Valenzuela, Business Development DASH

“The crypto market still operates under the implicit assumption that Bitcoin’s performance dictates the health of every other chain. Yet Bitcoin was never engineered for high-throughput decentralized applications, DeFi, agents, or Internet-scale usage. Its lack of native horizontal scaling forces perpetual trade-offs between security, decentralization, and speed. Tying every project’s fate to BTC’s price chart means superior technology is routinely mispriced: a chain that can process 10,000× more transactions per second on the same class of hardware is still treated as a ‘beta version’ of Bitcoin. This correlation must end. Decoupling allows capital to flow to protocols based on engineering reality rather than narrative dominance. A disciplined valuation framework looks at capability first. When investors adopt this metrics-driven approach, valuation stops being a popularity contest and becomes an engineering assessment—exactly what the next phase of crypto adoption demands.”

—Robert Sasu, Core Developer EGLD

“Having worked closely with John for over two years, I have always been impressed by his exceptional knowledge of both crypto and crypto legislation — a rare combination that sets him apart as an entrepreneur. This book is truly the fruit of years of dedicated labour, and it shows on every page. Its contents are not only groundbreaking but a refreshing departure from the noise of the crypto market. The book introduces a crypto fund built on these very principles, giving investors a concrete and disciplined vehicle to put the theory into practice. Rather than chasing Bitcoin’s volatile highs and lows, it presents a compelling Fair Value model that brings a needed dose of analytical rigor to cryptocurrency investing.”

—Rien de Rooij, Chairman Digital Gold Foundation

“The transition to digital fiat currency is no longer speculative; it is structurally favored. The only question is how prepared we will be when it becomes the dominant global standard. As economies digitize, the absence of universally accepted valuation mechanisms creates friction, uncertainty, and mistrust. If digital currencies are to function effectively in global commerce, we must establish standardized, transparent methods to measure and compare value across platforms and borders. A Cryptocurrency Fair Market Value framework would provide the clarity needed for true interoperability, allowing nations, institutions, and individuals to transact with confidence rather than speculation. But valuation alone will not secure the future of digital finance. Trust is built not only on price discovery, but on the integrity of the trading environment itself. Today’s crypto ecosystem too often resembles a financial frontier, which has fragmented oversight, inconsistent exchange practices, and widespread exposure to manipulation and fraud. While reputable exchanges exist, the lack of uniform standards undermines the credibility of the entire system. If the industry fails to implement disciplined self-regulation, governments will inevitably step in, and history suggests their solutions may prioritize control over innovation. The path forward is clear: validated valuation standards combined with consistent exchange accountability. With both in place, digital currencies can achieve broad adoption without sacrificing their foundational promise of decentralization. The public does not seek chaos; it seeks fairness, transparency, and autonomy. By building trust proactively, the crypto community can ensure that digital finance evolves on its own terms, secure, interoperable, and truly decentralized.”

—Marv Hagedorn, Idaho Senator

“Government-issued currency has long faced a fundamental credibility problem. Fiat money is vulnerable to political manipulation and the consequences of unchecked printing. The marketplace responded by developing alternatives. Cryptocurrency emerged as one such alternative, but it has struggled with a problem of its own: valuation driven by emotion rather than substance. Hype, fear, and speculation have historically dictated prices, leaving holders without a reliable foundation for assessing what they actually own. The Crypto Fair Value (CFV) rating changes that. It introduces an independent, objective metric for evaluating cryptocurrencies, which is transparent, consistent, and resistant to manipulation. CFV shifts the basis of crypto valuation from sentiment to fact, giving holders a credible standard against which to measure their assets. Transitioning from emotional attachment to analytical assessment is never easy. Many investors have deep personal conviction in the coins they hold. But sound financial decision-making has always depended on honest, evidence-based evaluation. CFV provides exactly that, a stable, trustworthy framework that serves cryptocurrency holders far better than instinct or feeling alone ever could.”

—Mark Finchem, Arizona Senator

“Bitcoin was the breakthrough that proved decentralized money was possible, but it was never meant to be the final measure of the entire ecosystem. For cryptocurrency to reach its full potential, we must decouple from Bitcoin as the universal benchmark and begin evaluating each Layer-1 network on its own fundamentals, its technology, its utility, and the problems it solves. When every project rises or falls purely with Bitcoin’s price, innovation is obscured and real value becomes harder to recognize. True freedom in cryptocurrency comes when independent networks can compete, evolve, and empower individuals without being tethered to the shadow of the first chain.”

—Justin Wood, Oklahoma Representative

PREFACE

“An investment operation is one which, upon thorough analysis, promises safety of principal and an adequate return. Operations not meeting these requirements are speculative.”

—Benjamin Graham, 1949

This is the second volume of a four-volume corpus on cryptocurrency analysis and allocation. Volume 1, *Cryptocurrency Analysis*, established the framework: the architectural test that distinguishes a Layer-1 Coin from a token, the Six Pillars of Perfect Money that admit a coin to the Foundation’s tracked set, the December 2024 calibration of the Digital Gold Standard Benchmark from Bitcoin’s state at that moment, the Crypto Fair Value formula that converts a candidate coin’s measured fundamentals into an analytical reference value, and the institutional architecture the Digital Gold Foundation operates under. *Cryptocurrency Analysis* is the framework’s specification at full depth. It runs to over a thousand pages because the framework requires that depth to be defensible.

This volume is shorter. It is also different in purpose. *The Intelligent Investor (Crypto Edition)* applies the framework to the question Benjamin Graham asked in 1949 and that he answered for equity investing across the twentieth century: how should a capital allocator deploy capital under analytical discipline against an unfamiliar asset class?

Graham’s answer, in its essentials, is the answer this book translates to cryptocurrency. Compute an intrinsic value. Require a substantial margin between the intrinsic value and the market price. Take positions where the margin is adequate. Hold through divergence between the market’s voting-machine

output and the framework's weighing-machine reading. Recognize that the discipline does not guarantee outcomes; it produces defensible positions that can be examined when outcomes go against them.

Two readers are addressed throughout. First is the institutional allocator who manages capital under fiduciary duty and reports to a board: the fund manager, the family-office principal, the endowment officer, the pension administrator, the sovereign-wealth-fund executive, the state treasurer operating under statutory authority. Each operates under defensibility constraints the framework is designed to support. Second is the sophisticated individual who allocates personal capital under personal discipline. That discipline was built across the equity-investing decades, refined through Buffett's letters and Graham's two books and the operational record of holding positions across cycles. The individual has serious capital allocatable to cryptocurrency, has been skeptical of the asset class until the analytical infrastructure existed to support an allocation, and is willing to be persuaded by rigor.

Both readers have applied Graham's test to traditional asset classes and accepted what the test produces. Whether the test applies to cryptocurrency, and if so under what methodology and toward what allocation, is the question this book asks. Yes, the answer runs, under the methodology Volume 1 specified and this volume applies, toward an allocation that depends on the reader's specific constraints.

One thing the book is not, and a few things it does not claim. It is not a recommendation that any specific reader allocate any specific portion of capital to cryptocurrency. It is not a prediction that the framework's tracked coins will produce specific returns over specific horizons. It is not advocacy for cryptocurrency as an asset class, nor a critique of readers who

decline allocation. What the book offers is the analytical infrastructure that supports allocation under Graham's discipline, the operational schema for translating the framework's analytical outputs into deployed positions, and the documented procedures the reader's allocation operates against. Judgment that uses what the framework offers is the reader's to supply.

A reader who finds the framework persuasive can proceed to allocation under its discipline, at the pace and scale the reader's circumstances admit. A reader who finds the framework unpersuasive can decline allocation and continue with whatever discipline the reader has built across the asset classes the reader does allocate to. Both decisions are defensible. The framework is offered to readers who can apply it; readers who cannot are not the audience the framework is written for, and writing for one audience does not constitute a critique of those in the other.

The book proceeds in seven parts. Part I (Chapter 1) establishes the threshold: the conditions under which Graham's test admits cryptocurrency as an investable asset class. Part II (Chapters 2 through 4) develops what admits a cryptocurrency to the framework's analytical attention: the architectural test, the Six Pillars, and the Foundation's tracked set. Part III (Chapters 5 and 6) develops how the framework values an admitted coin: the Bitcoin calibration and the Crypto Fair Value formula. Part IV (Chapter 7) presents the framework's central concept: the Margin of Safety, in the operational form that converts the framework's analytical outputs into deployable positions. Part V (Chapters 8 and 9) applies the framework: worked Coin Reports for five tracked coins and calibration tests against Bitcoin at two moments, Ethereum, and XRP. Part VI (Chapters 10 through 12) develops position

management: allocation under fiduciary constraint, allocation under personal-capital constraint, and the rebalancing-and-exit mechanics that operate across measurement cycles. Part VII (Chapters 13 through 15) presents the framework's defense: reproducibility, three serious critics, and the falsification conditions the framework operates under.

Two appendices supply the technical specifications a reader applying the framework operationally requires: the Benchmark specification at full operational depth, and the reproduction protocol that supports independent verification of the framework's outputs.

A glossary supplies canonical definitions of the terms the framework operates under. An About the Author section identifies who is responsible for the framework, the Foundation, and the book. A Sources and Further Reading section directs readers toward the Austrian-tradition monetary literature, Graham's two principal works, and the institutional architecture references the framework rests on.

Most of the book is written in third-person voice. First-person voice is reserved for moments where the framework asks the reader to make a commitment that cannot be made by the analytical voice alone. Chapter 1 names the threshold and asks the reader to accept it. Chapter 7 presents the Margin of Safety and asks the reader to accept the discipline. Chapters 10 and 11 ask the institutional and personal allocators respectively to deploy capital under the framework's schema. Chapter 15 names the falsification conditions and what I personally stake on the framework's accountability. First-person passages are where the framework's author is speaking directly. Third-person passages are where the framework itself is speaking through the analytical procedures it specifies.

This framework is my work, produced through the Digital Gold Foundation across the years that culminated in the December 2024 calibration and the operational record the Foundation has accumulated in the cycles since. Its mistakes are mine; its defensibility is what the documented procedures support; its continued operation depends on the practitioners who deploy capital under it and on the adversarial reviewers who continue to verify the outputs against the published methodology. The book is offered to both. The work is the framework. The framework is what the book applies.

1. INVESTMENT, SPECULATION, AND CRYPTO

The threshold test, applied to the asset class

I. Graham's Test

Benjamin Graham wrote the sentence this book opens on in 1949: “An investment operation is one which, upon thorough analysis, promises safety of principal and an adequate return. Operations not meeting these requirements are speculative.” Twenty-three words. Three operative phrases. *Thorough analysis* commits the operator to a discipline before the position is taken. *Safety of principal* commits the operator to a margin against the price moving the wrong direction. *Adequate return* commits the operator to a defensible reason the position is expected to pay.

Graham was distinguishing investment from speculation. His distinction was not moral. Speculation, in Graham's framing, was not contemptible; it was simply a different activity, with different methods, different rewards, and different risks. What Graham insisted on was that the operator know which activity they were conducting. An investor who believed they were investing while actually speculating was the operator Graham was writing against. That operator had no defense when the position went wrong, because the position had never been defensible to begin with.

Graham's test was demanding in 1949. It is more demanding now, because the available investment universe has grown, the tools for analysis have grown, and the disciplines for distinguishing investment from speculation have grown alongside them. A 2026 capital allocator deploying capital to

public equities under Graham's test has a century of refined methodology to draw on. The test still applies; the methodology that satisfies it has matured.

For cryptocurrency, the test applies in the same form. Until recently, the methodology that satisfies it did not exist. A 2018 buyer of Bitcoin was speculating, not because Bitcoin lacked merit, but because no apparatus existed against which thorough analysis could be conducted, safety of principal quantified, or adequate return defended. The same holding by a 2020 buyer of Ethereum, the same holding by a 2024 buyer of any Layer-1 Coin: speculation. Each activity was defensible as speculation. None of it was defensible as investment, because the analytical preconditions did not exist for this asset class.

That has changed. Calibrated in December 2024, the Digital Gold Standard Benchmark supplies the reference standard Graham's discipline requires. Applied to a candidate coin against the Benchmark, the Crypto Fair Value formula produces a Composite Score, a Crypto Fair Value, and a Fair Coin Price. Computed from the Fair Coin Price and the candidate's current market price, the Margin of Safety quantifies the cushion against the position moving the wrong direction. Cross-provider validation requirements and publication-grade tolerance discipline produce analysis a third party can verify. Twelve Layer-1 Coins selected against the Six Pillars of Perfect Money make up the Foundation's tracked set, naming the candidates against which the analysis is conducted.

This book develops what those outputs mean for two readers. An institutional allocator manages capital under fiduciary duty and reports to a board. A sophisticated individual allocates personal capital under the discipline they have built across the equity-investing decades. Both have applied Graham's test to traditional asset classes and found it survives the application.

Whether the test applies to cryptocurrency, and if so toward what allocation, is the question.

Yes is the answer, toward an allocation that depends on the practitioner's specific constraints. This chapter establishes the threshold. The chapters that follow develop the discipline that operates on it.

II. Speculation Was the Honest Word

For most of cryptocurrency's history, speculation was the honest word for what was occurring. Early Bitcoin holders bought a cryptographic experiment whose monetary properties were unsettled, whose adoption was minimal, and whose price was the output of a thin market with limited liquidity. They were not conducting thorough analysis against verifiable inputs. They were participating in an experiment. Their participation was rewarded handsomely by the experiment's success, but the success was not the consequence of disciplined evaluation. It was the consequence of being early to something that worked.

A 2017 cycle extended the speculation by orders of magnitude. Initial coin offerings raised billions of dollars for projects whose technical foundations the buyers had not examined and could not have examined, because the disclosures were inadequate, the protocols were untested, and the analytical framework against which the offerings could have been evaluated did not exist. A 2017 ICO buyer was speculating. The activity was honest if the buyer named it honestly; it was not investment in Graham's sense.

A 2020-to-2024 institutional adoption cycle continued the pattern in different clothing. Corporate-treasury Bitcoin holdings beginning with MicroStrategy, the spot-ETF approvals, the sovereign-allocation discussions: these activities

supplied infrastructure that made larger positions operationally possible, but they did not supply the apparatus disciplined analysis requires. An institutional Bitcoin buyer in 2023 had better custody than the 2017 ICO buyer, but no more analytical defense for the position. Defense came down to what it had always come down to for the Bitcoin buyer: Bitcoin had appreciated historically, the buyer believed it would appreciate further, and the position was sized against the buyer's appetite for volatility. This is speculation. It is defensible speculation. It is not investment.

This book's framework rests on the recognition that cryptocurrency, until December 2024, did not admit Graham-style investment, because the methodology had not been built. Naming that honestly is the precondition for what follows. A reader who insists that pre-2024 cryptocurrency holdings were investments has redefined Graham to fit the activity rather than fitting the activity to Graham. The redefinition is available, but it costs the reader the discipline the original definition enforced.

III. The Calibration That Changed the Question

December 2024 is the moment at which a reference standard for cryptocurrency first became available. Bitcoin reached approximately \$100,000 per coin, a market capitalization of approximately \$1.983 trillion, and a network state whose four fundamental metrics could be measured through verifiable channels rather than estimated through informal ones.

Those four metrics are Adoption (approximately 80 million unique holders), Annual Transactions (approximately 6.09 billion), Annual Transaction Value (approximately \$13.49 trillion), and Active Developers (approximately 905 unique

contributors). They were not chosen because they were convenient. Each captures a distinct dimension of network monetary function. Adoption captures the breadth of voluntary holder participation. Annual Transactions captures the activity rate of the network as a medium of exchange. Annual Transaction Value captures the economic substance moving across the network. Active Developers captures the technical foundation on which continued operation depends. Together they describe what a Layer-1 cryptocurrency does as money.

Bitcoin's December 2024 state was the only one in cryptocurrency history at which all four could be measured through verifiable channels. Bitcoin in 2017 could not have been measured this way. Ethereum at any point in its history could not have been measured this way. Infrastructure allowing the four metrics to be measured (regulated exchange-traded products, audited custodians, entity-adjusted on-chain analytics, cross-validated developer accounting) is the infrastructure that matured around Bitcoin specifically across the years preceding the calibration. Measurement was possible from that state. It would not have been possible from any earlier state of any cryptocurrency.

This is the structural reason the Digital Gold Standard Benchmark is calibrated from Bitcoin. It is not a statement about Bitcoin's merit relative to other Layer-1 Coins. It is a statement about the measurement infrastructure that had concentrated around Bitcoin by the calibration date. Infrastructure existed because regulated capital flowing into Bitcoin had required it; measurement could be made because the infrastructure had matured to produce it; the Benchmark could be calibrated because the measurement was available. Any subsequent Layer-1 Coin can be measured against the

December 2024 Benchmark using the same four-metric methodology. The Benchmark is the reference. Bitcoin is not.

From December 31, 2024 forward, the Benchmark does not change. Not when Bitcoin's price moves. Not when Bitcoin's adoption count moves. Not when exchange-traded-fund flows reverse. Not when sovereign allocations expand or contract. The Benchmark is permanent, in the sense the international prototype kilogram in Sèvres was permanent from 1889 until 2019. That cylinder defined the unit. Every mass measurement everywhere on Earth was traceable to it. The cylinder itself was weighed against the standard the cylinder defined; the cylinder drifted; the drift was reported as a property of the cylinder, not as a revision of the definition. The cylinder did not get to exempt itself from the test.

The Benchmark operates on the same principle. Bitcoin produced it through its December 2024 state. From the calibration moment forward, the Benchmark is independent of Bitcoin, and Bitcoin is one Layer-1 Coin among others, measured on identical terms with every other coin the framework evaluates. Measuring Bitcoin against its own calibration produces, at later dates, the Composite Score that quantifies how Bitcoin has performed against the standard it produced. At the October 2025 peak, Bitcoin's Composite Score produced a Crypto Fair Value substantially below Bitcoin's market capitalization at that moment. The framework reported the reading. Bitcoin was overvalued against the standard it had produced. The framework did not exempt Bitcoin from the discipline it was built to administer.

This is what Graham's test requires. A reference against which a candidate coin's fundamentals are measured. A Margin of Safety, computed from the Fair Coin Price and the current market price, sizing the cushion the position carries.

A Composite Score trajectory across trailing quarters, integrated with the point-in-time Margin of Safety, supporting the holding decision once the position is taken. Three pieces, in operational form, applied to cryptocurrency against a permanent reference.

What I am claiming is precise. The discipline exists, it produces Graham-compatible outputs, and it is verifiable by adversarial third parties under the reproduction protocol the book's later chapters develop. What I am not claiming is that the discipline guarantees outcomes.

Graham's test never guaranteed outcomes; it guaranteed a discipline. An operation conducted under it still produces losses sometimes, because analysis can miss things, because cushions can be insufficient against tail events, and because expected returns can fail to materialize. The discipline does not promise the absence of loss. It promises that the loss, when it occurs, can be examined against the procedure that produced the position, and the procedure can be refined where the examination warrants. That is what Graham offered in 1949. It is what this book offers in 2026.

IV. What Bitcoin Is and Is Not

This chapter has named Bitcoin frequently. A position on Bitcoin requires explicit statement, because the position is consequential for everything that follows and because misunderstandings of it will produce misunderstandings of the framework.

Bitcoin is the asset from which the Digital Gold Standard Benchmark was calibrated in December 2024. Calibration was a single historical act. From the calibration moment forward, Bitcoin is one Layer-1 cryptocurrency among others, measured

against the Benchmark on identical terms with every other Layer-1 Coin.

Bitcoin is not on the Foundation's tracked set. That set consists of twelve Layer-1 Coins for which the Foundation produces and publishes quarterly Coin Reports: Cardano, Digital Cash, DigiByte, MultiversX, Internet Computer, Near Protocol, Ravencoin, Chia, eCash, Monero, Nano, and ZClassic. The Foundation also supports Digital Gold (DGD), its own coin, which is structurally always at its Crypto Fair Value at each level by design and is therefore presented as a separate item rather than as a thirteenth tracked coin. Bitcoin is not in the tracked set, and the CFV CoinFunds the Foundation supports do not hold Bitcoin.

The reason is not ideological. It is analytical. Bitcoin's contribution was the December 2024 calibration. That contribution is permanent and historically singular; no further work on Bitcoin advances the framework's purpose. Layer-1 Coins on the tracked set are coins whose properties advance the technical and monetary functions Bitcoin's design left unsolved: the base layer of seven transactions per second, the absence of cryptographic privacy at the protocol level, the proof-of-work energy intensity, the limited smart-contract expressiveness, and the holder-concentration pattern at the 93%-of-supply level in the top two percent of addresses. These are properties of Bitcoin specifically, and Layer-1 Coins whose architectures address one or more of them are the coins on which the framework is operated.

Bitcoin is the Model T of cryptocurrency. The Model T proved that the automobile was possible. The Model T did not remain the standard against which subsequent automobiles were evaluated. Standards for automobiles are what an automobile must accomplish (transportation, safety, fuel efficiency,

durability), and subsequent automobiles are evaluated against those standards, not against the Model T. Bitcoin proved that decentralized digital money was possible. Standards for decentralized digital money are what such money must accomplish (the Six Pillars of Perfect Money this book's Chapter 3 develops), and subsequent Layer-1 Coins are evaluated against those standards, not against Bitcoin. Measuring Bitcoin against the Benchmark is the framework demonstrating that its standard discriminates on Bitcoin the same way it discriminates on every other Layer-1 Coin. Evaluated against the modern automobile standard, the Model T would not be a modern automobile. Evaluated against the modern Layer-1 Coin standard, Bitcoin is not a modern Layer-1 Coin.

A reader committed to Bitcoin as the singular cryptocurrency worth analytical attention is not the reader this book is written for. That commitment is defensible on its own terms, and the framework does not require conversion. What the framework does require is that whoever proceeds past this chapter accept the separation between Bitcoin's role as calibration source and Bitcoin's status as one Layer-1 Coin among others. The two roles are operationally distinct. The framework's measurements depend on the distinction.

V. The Two Readers

This book is written to two specific readers. Each has applied Graham's test to traditional asset classes and accepted what the test produces. Each has reached cryptocurrency through different routes and faces different operational constraints. The framework produces the same analytical outputs for both. Operational consequences differ, and the book develops the differences explicitly in Chapter 10 and Chapter 11.

First is the institutional allocator. An institutional allocator manages capital under fiduciary duty. The capital may belong to a fund, a family office under an investment-committee structure, an endowment, a pension, a sovereign-wealth fund, or a state treasury operating under statutory authority. Allocators report to a board, an investment committee, or a constituency. Positions are defensible to the reporting constituency or they are not held. This defensibility constraint is operationally binding in a way personal capital allocation is not.

For the institutional allocator, the framework supplies an analytical posture toward cryptocurrency that is structurally similar to the posture institutional allocators take toward equities, fixed income, and other established asset classes. Coin Reports are the documentation the investment committee receives. Margin of Safety thresholds are the discipline against which the committee evaluates the position size. A reproducibility protocol is the assurance that the allocation rests on inputs a third party can verify. Falsification commitments are the conditions under which the committee can be told the position requires reduction or exit. Defensibility is built into the methodology rather than asserted around it.

Second is the sophisticated individual. A sophisticated individual allocates personal capital under personal discipline. That discipline was developed across the equity-investing decades, refined through Buffett's letters and Graham's two books and the operational record of holding positions across cycles. The sophisticated individual has serious capital allocatable to cryptocurrency, has been skeptical of the asset class until the methodology existed to support an allocation, and is willing to be persuaded by rigor.

The capital is the individual's own; the reporting constituency is the individual themselves; the fiduciary duty is to family or estate rather than to a board.

For the sophisticated individual, the framework operates under a different operational constraint. Coin Reports are documentation the individual reviews for their own purposes. Thresholds size positions. A reproducibility protocol is what the individual relies on when their own measurement is incomplete. Falsification commitments are the conditions under which the individual would reduce or exit positions. The discipline operates the same way it does for the institutional allocator; operational consequences differ because the constraints differ.

Where the framework produces the same output for both readers, the book says so. Where the output produces different operational consequences for fiduciary capital and personal capital, the book develops the differences in the chapters that address allocation. The two readers are not in tension. They stand in the same analytical relationship to the framework. Differences are downstream of the analytical outputs, in the allocation decisions the operational constraints shape.

A third reader appears throughout without being addressed: the state treasurer or public fiduciary operating under statutory authority. Arizona Senate Bill 1649 names that reader. A state treasurer applying the framework operates under constraints that combine the institutional allocator's fiduciary duty with statutory requirements and political accountability the private institutional allocator does not face. The state treasurer is well-served by *Cryptocurrency Analysis* Chapter 14 and Appendix C, where the Foundation's institutional architecture and the model legislative agenda are developed at the depth the role requires.

This volume references the state-treasurer application where the references inform the institutional allocator's posture, but does not develop the state-treasurer's working procedure as a third reader. The institutional allocator and the sophisticated individual are the readers this book is written to.

VI. The Threshold the Chapter Has Set

A reader who has worked through this chapter has the threshold. Graham's test applies to cryptocurrency under the methodology this book develops. That methodology rests on a permanent reference standard calibrated from Bitcoin's December 2024 state but independent of Bitcoin from the calibration moment forward.

What the threshold supplies is the analytical posture the rest of the book builds on. Cryptocurrency is an asset class. Until December 2024, it did not admit Graham-style investment, because the methodology had not been built. That methodology exists now. Applied to a tracked Layer-1 Coin, it produces outputs against which a defensible position can be taken. Whether the position is investment or speculation depends on whether the analysis is thorough, whether the Margin of Safety is adequate, and whether the expected return is defensible against documented inputs. Discipline is what the framework supplies. Operational judgment that uses it is the reader's to supply.

A reader who applies that judgment is doing what Graham did in 1949, with an apparatus Graham could not have written and an asset class he did not live to see, but under a test he composed for any operator who would conduct an investment seriously. The test is the threshold. The threshold is what this chapter has set out.

2. THE ARCHITECTURAL TEST

The first filter: what a Layer-1 Coin is, and what it is not

I. Three Conditions

A Layer-1 Coin is a cryptocurrency that operates its own blockchain, supports its own native wallet, and is the protocol's native coin. These three conditions are necessary together. A candidate that fails any one is a token, not a coin, and the framework does not value it.

These conditions are stated as a procedure the reader can apply to a candidate cryptocurrency in under five minutes. The check does not require specialized tools. It requires the candidate's public documentation, a block explorer, and a wallet that supports the chain natively. Where the documentation, the explorer, and the wallet converge on the candidate satisfying all three conditions, the candidate is a Layer-1 Coin. Where any of the three disagrees with the other two, the candidate is not.

The architectural test is the first of the framework's filters. A candidate cryptocurrency that passes the architectural test proceeds to the Six Pillars of Perfect Money, developed in Chapter 3. A candidate that fails the architectural test is outside the framework's scope before any monetary evaluation is warranted. This filter operates on its own terms and produces a binary output: coin or token. That output is not graduated. A cryptocurrency does not partially satisfy the architectural test.

What follows develops each condition in operational depth, demonstrates the test on representative candidates, and

addresses the cases where the test produces a result the candidate's marketing material would dispute.

II. Condition One: Its Own Blockchain

First, the candidate must operate its own blockchain. That blockchain must be the candidate's own, not a host chain on which the candidate has issued itself as a token.

A blockchain is a distributed ledger maintained by a consensus mechanism. The ledger records the protocol's transactions. The consensus mechanism is what produces agreement among the network's participants about which transactions are valid and in what order they occurred.

Different protocols use different consensus mechanisms: proof-of-work, proof-of-stake, proof-of-space-and-time, directed-acyclic-graph constructions, hybrid architectures combining two or more approaches. Which mechanism a protocol uses is not what matters for the architectural test. What matters is that the mechanism is the candidate's own, and that the ledger it maintains is the candidate's own ledger.

A cryptocurrency that issues itself as an ERC-20 token on Ethereum does not have its own blockchain. It has a smart contract deployed on Ethereum's blockchain. That smart contract's state is Ethereum's state; the consensus that validates the contract's operations is Ethereum's consensus; the security that protects the contract from attack is Ethereum's security. If Ethereum fails, the ERC-20 token fails with it. The token has no independent chain.

The same is true of every other token standard that exists. BEP-20 tokens on Binance Smart Chain. TRC-20 tokens on Tron. SPL tokens on Solana. In each case, the host-chain prefix

names the chain that owns the consensus. The token is a smart-contract artifact on that chain. It is not its own blockchain.

A practitioner can verify the first condition through three steps.

First, examine the candidate's public documentation for the chain identifier. A coin will name its chain (Bitcoin's chain is Bitcoin; Cardano's chain is Cardano; Monero's chain is Monero). A token will name the host chain on which it has been deployed (USDC operates on multiple host chains; Chainlink operates principally on Ethereum).

Second, locate the candidate on a block explorer. A coin's transactions appear on the coin's own explorer (block.io for Bitcoin, cardanoscan.io for Cardano, monero.com for Monero).

A token's transactions appear on the host chain's explorer, with the token contract address as a parameter (a Chainlink transaction appears on Etherscan with the LINK contract address; not on a Chainlink-specific explorer).

Third, examine the consensus mechanism that produces the block the candidate's transactions are settled in. A coin's transactions are settled by the consensus mechanism that secures the coin's own chain. A token's transactions are settled by the host chain's consensus mechanism.

These three steps converge on the answer. Where the chain identifier, the explorer, and the consensus mechanism all point to a chain that bears the candidate's name and serves the candidate as its primary settlement layer, the first condition is satisfied. Where they point to a different chain that the candidate is using as a host, the condition is not satisfied.

III. Condition Two: Its Own Native Wallet

Second, the candidate must support its own native wallet derivation architecture. That wallet must be the candidate's own, with key derivation specific to the candidate's protocol.

A wallet is the cryptographic apparatus through which a holder controls coins on the chain. The wallet generates the keys that authorize transactions, derives the addresses to which the chain's coins can be sent, and signs the transactions that move the coins to other addresses. Its cryptographic operations are protocol-specific. Bitcoin's wallet uses Bitcoin's elliptic-curve parameters and Bitcoin's address derivation. Ethereum's wallet uses Ethereum's elliptic-curve parameters and Ethereum's address derivation. Monero's wallet uses Monero's stealth-address construction. Cardano's wallet uses Cardano's hierarchical-deterministic derivation under Cardano-specific parameters.

A coin has its own wallet. That wallet is the protocol-native instrument for holding the coin. Its existence is independent of any host-chain infrastructure. A practitioner who holds the coin on a wallet specific to the coin's protocol is holding the coin directly, on the coin's own chain, with the keys the coin's own protocol requires.

A token does not have its own wallet in this sense. A token has a contract address on a host chain, and a holder of the token uses the host chain's wallet to interact with the token's contract. The wallet's cryptographic operations are the host chain's operations; the holder is operating on the host chain, not on a chain specific to the token. A token's existence inside the holder's wallet is a balance recorded in the smart contract's state, accessible to the wallet through the host chain's interface

but not constituted by any cryptographic property the wallet derives for the token specifically.

Why this distinction matters: a practitioner who holds a coin natively holds it with the security guarantees the coin's protocol provides. A practitioner who holds a token holds it with the security guarantees the host chain provides for tokens, mediated by the smart contract that issues the token. These are different security postures, and the practitioner who confuses them is exposed to risks the framework's analysis cannot mitigate. A coin held natively is exposed to the coin's protocol-level risks. A token held through a smart contract is exposed to those risks plus the smart contract's risks plus the host chain's risks. The combined exposure is what the architectural test's second condition isolates.

Verification of the second condition: the candidate must publish, or have published on its behalf, a wallet that supports the protocol natively. That wallet may be developed by the protocol's core team, by an independent contributor whose work the protocol's documentation references, or by a hardware-wallet vendor that has integrated the protocol's derivation. What matters is that the wallet exists and that holding the candidate in the wallet means holding the candidate on the candidate's own chain with the candidate's own keys. A candidate whose only available "wallet" is a host-chain wallet pointing at the candidate's smart contract is not satisfying the second condition.

IV. Condition Three: The Protocol's Native Coin

Third, the candidate must be the protocol's native coin. It must be the coin the protocol's consensus mechanism rewards, the coin in which the protocol's transaction fees are denominated,

and the coin the protocol treats as its primary unit of account at the protocol level.

A blockchain that supports its own consensus mechanism rewards the validators that maintain the chain with a coin. That reward is the protocol's economic incentive for the validators to perform the consensus function. The coin the protocol rewards is the protocol's native coin. Bitcoin's protocol rewards miners with bitcoin (BTC); Cardano's protocol rewards stake-pool operators with cardano (ADA); Monero's protocol rewards miners with monero (XMR). The reward coin is the protocol's native coin because the protocol cannot operate without rewarding validators and the reward is paid in this specific coin.

Transaction fees are the second indicator. A blockchain charges its users a fee to include their transactions in the next block. That fee is denominated in a specific coin, paid to the validators who include the transaction. Bitcoin's transaction fees are paid in BTC; Ethereum's transaction fees (gas) are paid in ether; Monero's transaction fees are paid in XMR. The fee currency is the protocol's native coin because the protocol cannot operate without users paying for inclusion, and the fee is paid in this specific coin.

A protocol-level unit of account is the third indicator. A blockchain that maintains a ledger denominates the ledger's balances in some unit. That unit is the protocol's native coin. The Bitcoin ledger denominates balances in BTC; the Cardano ledger denominates balances in ADA; the Monero ledger denominates balances in XMR. The protocol records its transactions in this coin and the chain's state at any moment is the distribution of this coin's balances across the addresses the chain recognizes.

These three indicators converge on the protocol's native coin. Where the reward coin, the fee coin, and the unit of account are the same coin, the candidate is the protocol's native coin and the third condition is satisfied. Where the candidate is rewarded by the chain but transaction fees on the chain are paid in a different coin, or where the candidate is the unit of account but not the reward, or where any other inconsistency appears, the candidate is not the protocol's native coin and the third condition fails.

The third condition is what distinguishes a coin from a wrapped representation of a coin, a stablecoin issued on a chain whose native coin is a different asset, or a governance token whose holders vote on protocol decisions but who are not rewarded for maintaining the chain. Wrapped Bitcoin (wBTC) is not the native coin of any chain it operates on; it is a tokenized representation of bitcoin held in custody. USDC is not the native coin of any chain it operates on; it is a stablecoin issued under its own contract on host chains whose native coins are different assets. UNI is the governance token of Uniswap, an application that runs on Ethereum; UNI is not Ethereum's native coin (ether is), and UNI is not the native coin of any other chain.

V. The Test Operationalized

A practitioner who has worked through the three conditions has the architectural test in operable form. The sequence: identify the candidate. Examine its public documentation for the chain identifier, the wallet, and the native coin designation. Locate the candidate on a block explorer. Verify that the explorer the candidate appears on is the candidate's own, not a host chain's explorer with the candidate as a contract parameter.

Examine the consensus rewards, the transaction fees, and the unit of account; verify that the three converge on the candidate. A candidate that passes all three conditions is a Layer-1 Coin. A candidate that fails any one is a token.

This check can be applied to thousands of candidates in operational time. Most candidates fail. Cryptocurrency space at the time of this writing contains tens of thousands of distinct cryptocurrencies by various counts. The vast majority are tokens that fail the first condition because they have been issued on a host chain. A substantial minority among the remaining candidates fail the second condition because their wallet support is host-chain mediation rather than native derivation. A smaller minority fail the third condition because they are not what the protocol's consensus rewards or what the protocol's transactions are denominated in.

Candidates that pass all three conditions are the Layer-1 Coins in the technical sense. Perhaps a few hundred such protocols are in operation at any given moment, with the count varying over time as new protocols launch and existing protocols cease operation. This architectural filter reduces the universe of cryptocurrencies the framework's subsequent filters will evaluate from many tens of thousands to a few hundred. Reduction at this scale is the operational purpose of the test.

That reduction is not normative. A candidate that fails the architectural test is not thereby a bad investment; it may be a fine speculation, or a useful instrument for some specific purpose, or an outright fraud. The framework simply does not value such candidates, because the methodology the framework operates depends on the candidate having the architectural properties of a Layer-1 Coin. A practitioner who allocates capital to tokens, stablecoins, governance instruments, or other non-coin cryptocurrencies is operating outside what the

framework values. Discipline appropriate to those allocations is not the discipline this book documents.

VI. Cases Where the Test Disagrees with Marketing

The architectural test produces results that occasionally disagree with the marketing language used to promote a candidate. Where the test and the marketing disagree, the test is the authoritative output. Three representative cases follow.

First case: Ethereum and the question of what Ethereum's native coin is. Ethereum operates its own blockchain. That blockchain supports a wallet that derives keys for ether (ETH) under Ethereum's parameters. Ether is the coin the protocol's consensus rewards (under proof-of-stake, validators receive ether), the coin in which transaction fees are paid (gas, denominated in gwei, which is a unit of ether), and the protocol's unit of account at the protocol level. Ether is the protocol's native coin. Ethereum the asset (the cryptocurrency a practitioner can hold) is ether. The architectural test does not produce different outputs for Ethereum-the-platform and ether-the-coin; it identifies ether as the Layer-1 Coin and the platform's smart-contract capabilities as features of the chain.

Second case: stablecoins issued on the chain of a Layer-1 Coin. USDC, USDT, and analogous stablecoins are issued as tokens on Ethereum, Solana, Tron, and other host chains. A stablecoin issuer's marketing may describe the stablecoin as "a cryptocurrency" without naming the host chain that issues it. The architectural test produces an unambiguous output: a stablecoin issued as a token on a host chain is a token, not a coin, and the framework does not value it. A practitioner who holds USDC on Ethereum is holding an Ethereum-based smart-contract balance, not an independent Layer-1 Coin.

The framework values neither USDC nor any other stablecoin under any architectural arrangement, because stablecoins as a category fail the first condition (they are issued as tokens on host chains, with no exceptions the framework has identified) and additionally fail subsequent monetary-property tests the Six Pillars develop.

Third case: wrapped representations of Layer-1 Coins on other chains. Wrapped Bitcoin (wBTC) is an ERC-20 token on Ethereum that represents a Bitcoin balance held by a custodian. wBTC marketing material may describe the token as “Bitcoin on Ethereum” or as a way for a holder of bitcoin to participate in Ethereum’s decentralized-finance applications.

The architectural test produces the same output: wBTC is an ERC-20 token, not a coin, and the framework does not value it. A practitioner who holds wBTC is holding a tokenized claim against a custodian who holds the underlying bitcoin. Custodial risk, smart-contract risk, and host-chain risk are all present.

A practitioner who wishes to hold bitcoin under the framework’s discipline holds bitcoin on Bitcoin’s chain, in a Bitcoin-native wallet, exposed to Bitcoin’s protocol-level risks and no others. Wrapped representations are operationally distinct from native holdings and are not the candidates the framework values.

A pattern runs across all three cases. Marketing language describes cryptocurrencies in the categories the marketing wishes to position them. The architectural test classifies cryptocurrencies in the categories their actual architecture places them. Where marketing and architecture disagree, the framework follows architecture.

VII. The Architectural Test and the Reader's Existing Portfolio

A reader approaching cryptocurrency from a traditional capital-allocation discipline has likely encountered cryptocurrencies through commercial channels that do not distinguish between coins and tokens with the precision the architectural test requires. Exchange listings, financial-press coverage, and retail investment platforms typically present cryptocurrencies as undifferentiated assets in a single category. Existing exposure, if any, may be to tokens the architectural test classifies as outside what the framework values.

This book does not require the reader to liquidate existing positions in cryptocurrencies the framework does not value. The framework's scope is limited to the Layer-1 Coins on the Foundation's tracked set, plus DGD. Other cryptocurrencies may have value, may produce returns, and may serve purposes the reader's allocation discipline accommodates. What the framework requires is that the reader who applies the framework distinguish between cryptocurrencies the framework values and cryptocurrencies it does not, and apply the framework only to the former.

An institutional allocator with existing stablecoin holdings used for treasury operations is operating in a different discipline when conducting those operations, and the framework does not produce outputs that would inform the stablecoin holdings. A sophisticated individual with an existing position in a memecoin acquired during a previous cycle is operating in a different discipline on that position, and the Margin of Safety calculations do not apply. Discipline appropriate to those holdings is the discipline the reader brought to them when they were taken.

The framework does not replace that discipline; it adds a different discipline for the specific cryptocurrencies that satisfy the architectural test and proceed to the framework's subsequent filters.

VIII. What the Test Has Established

This architectural test is the first filter. It distinguishes coin from token through three conditions: own blockchain, own native wallet, own native coin. The conditions are necessary together. The output is binary. The check is operational and can be applied by the practitioner in under five minutes against a candidate's public documentation, a block explorer, and a wallet that supports the chain natively.

Reduction the filter accomplishes is the precondition for what follows. The framework operates on Layer-1 Coins. The architectural test names which cryptocurrencies are Layer-1 Coins. Cryptocurrencies that fail the test fall outside what the framework evaluates and require different analytical disciplines the book does not develop. Cryptocurrencies that pass the test proceed to the Six Pillars of Perfect Money, where the framework's monetary evaluation begins. The architectural test is necessary but not sufficient. The Six Pillars complete the admission the test begins.

A reader who has worked through this chapter has the test. Applied to a candidate, it produces an answer. Coin or token. The framework values one and not the other, and the difference matters before any further evaluation is warranted.

3. THE SIX PILLARS OF PERFECT MONEY

The second filter: the monetary properties a Layer-1 Coin must demonstrate

I. From Architecture to Money

Chapter 2 established what makes a cryptocurrency a Layer-1 Coin. A coin that passes that test has its own blockchain, its own native wallet derivation, and is the protocol's native coin. That test is necessary, and it is binary. Either the candidate is a Layer-1 Coin or it is a token.

The architectural test does not address whether the coin functions as money. A coin can pass the architectural test and still fail to deliver the monetary functions that justify holding it as an investment. The Six Pillars of Perfect Money are the second filter. They evaluate whether a coin's monetary properties support the analytical work the rest of the framework will conduct on it.

These Pillars derive from the Austrian tradition. Carl Menger established in the late nineteenth century that money emerges from the market's collective preference for whatever commodity best satisfies the conditions commerce imposes. Ludwig von Mises extended the analysis to demonstrate that sound money requires not only the right physical properties but the right institutional conditions, specifically the absence of monopolistic control by any authority whose interests diverge from the holders of the money. Friedrich Hayek extended the argument further in his 1976 work on the denationalization of money, establishing that competing currencies produce the market discipline that single-issuer monopolies cannot.

Murray Rothbard synthesized the tradition in his work on the history and theory of money.

These Six Pillars distill what that tradition identified as the conditions money must satisfy. Five Pillars are admission gates: a coin must satisfy each of the five to enter the Foundation's tracked set. One Pillar is developmental: a coin's performance on it is reported but does not gate admission, because the property it measures is one no Layer-1 Coin in the current tracked set has fully achieved.

What follows develops each Pillar in operational depth, names what satisfaction looks like and what failure looks like, and explains why the framework treats five as gates and one as developmental.

II. The Six Pillars Named

Here are the Pillars, in the order this chapter develops them:

Scarcity. The condition that the supply of the money is fixed or grows on a known, predictable schedule that cannot be altered by the discretion of any central authority.

Stable Pricing. The condition that the money's purchasing power varies within bounds that allow it to function as a unit of account across the timescale of ordinary commerce.

Free Adoption. The condition that the use of the money is not imposed by any government, central bank, or other central authority, but arises from the voluntary decisions of holders.

Decentralized Governance. The condition that the rules governing the money's operation are not subject to the discretion of any identifiable party, but emerge from the consensus mechanism the protocol operates under.

Freedom to Transact. The condition that holders can move the money to other parties without permission from any intermediary that could deny, censor, or reverse the transaction.

Adequate Circulation. The condition that the supply in operation across the network is sufficient to support the network's transactional purposes without forcing fractional-reserve substitutes to emerge.

Five of the six are admission gates: Scarcity, Free Adoption, Decentralized Governance, Freedom to Transact, and Adequate Circulation. A coin that fails any of these five is not admitted to the Foundation's tracked set. The sixth, Stable Pricing, is developmental. The reasoning is straightforward: no Layer-1 Coin in the current tracked set has achieved the price stability of established sovereign currencies, and gating admission on stability would empty the tracked set. The framework reports each coin's price-volatility profile without admitting or rejecting coins on the basis of volatility alone.

What follows develops each Pillar in turn.

III. Scarcity

Scarcity requires that the money's supply be fixed or grow on a known, predictable schedule that cannot be altered by the discretion of any central authority. This Pillar evaluates whether the supply schedule is determined by the protocol's rules rather than by the decisions of an identifiable party.

A coin's supply schedule is determined by its protocol. Bitcoin's protocol caps the supply at 21 million coins, with new issuance halving approximately every four years until the cap is reached. Digital Cash's protocol caps the supply at 18.9 million coins under a similar halving schedule.

DigiByte's protocol caps the supply at 21 billion coins under its own emission curve. Each coin's supply schedule is cryptographically enforced at the protocol level. That schedule cannot be altered without a coordinated protocol change that the network's participants would have to agree to and that the consensus mechanism would have to admit.

Satisfaction of this Pillar requires that the supply schedule be enforced by the protocol and not subject to the discretion of any central authority. A protocol may admit emission tails, masternode collateral requirements, governance-pool allocations, or other structural features that affect the supply over time, provided those features are themselves protocol-level and not discretionary. A coin whose issuance authority can be exercised by an identifiable party (a development team, a foundation board, a corporate issuer) fails Scarcity, because the supply is subject to discretion the holder cannot verify will be exercised in the holder's interest.

Scarcity is binary at the protocol level. Either the supply schedule is enforced by the protocol or it is not. A coin satisfies the Pillar or fails it; there is no graduated middle. That binary character is what makes Scarcity a gate. The framework cannot value coins whose supply is subject to discretionary issuance, because the calculation depends on the supply being a fixed quantity the coin's other metrics can be measured against.

Fiat currencies do not satisfy Scarcity. The United States dollar, the euro, the Japanese yen, and every other fiat currency operate under discretionary issuance by their central banks. That discretion is the structural property that distinguishes fiat from sound money under the Austrian tradition. This Pillar's role in the framework is to enforce that distinction at the asset-class level.

IV. Stable Pricing

Stable Pricing requires that the money's purchasing power vary within bounds that allow it to function as a unit of account across the timescale of ordinary commerce. This Pillar evaluates whether the coin's price exhibits the stability that monetary function requires.

Among the six, Stable Pricing alone is reported developmentally rather than as an admission gate. The reasoning is empirical. No Layer-1 Coin in the current tracked set has achieved the price stability of established sovereign currencies. Volatility profiles of every coin the framework evaluates are substantially higher than those of the United States dollar, the euro, or any other major sovereign currency. A version of the framework that gated admission on stability would either admit no coins, or would admit only coins whose stability had been achieved through mechanisms (custodial reserves, central-issuer pegs, algorithmic stabilization) that violate other Pillars.

Honesty is the framework's posture here. The Pillar exists because monetary theory requires it. It is developmental because empirical reality has not yet produced Layer-1 Coins that satisfy it. Each coin's volatility profile appears in its Coin Report. A practitioner can examine the profile and weigh the coin's stability against the coin's satisfaction of the five gating Pillars. A coin's failure to achieve stability is not disqualifying; it is information the practitioner integrates with the framework's other outputs.

In some future protocol version, Stable Pricing may move from developmental to gating, when the Layer-1 Coin tracked set contains coins that demonstrate the operational stability the

Pillar requires. That moment has not yet arrived. The Pillar's developmental status is the framework's recognition of the current state.

V. Free Adoption

Free Adoption requires that the use of the money not be imposed by any government, central bank, or other central authority, but arise from the voluntary decisions of holders. This Pillar evaluates whether the coin's holder base has been assembled through voluntary acquisition rather than through legal tender requirements, employer or counterparty obligations, or other forms of coercion.

Satisfaction requires that no coercive mechanism produce the coin's adoption. A holder acquires the coin because the holder has decided to hold it. That decision may be motivated by speculation, by ideology, by anticipated returns, by privacy concerns, by remittance utility, or by any other reason the holder has. The decision is the holder's. No party with authority over the holder has compelled the acquisition.

Every Layer-1 Coin in the current tracked set satisfies Free Adoption. Satisfaction here is not narrow; it is broad. Coins the framework values have been acquired voluntarily by their holders across the years and cycles of cryptocurrency's existence.

That voluntary character of the holder base is the structural property that distinguishes Layer-1 Coins from sovereign currencies, whose adoption is supported by legal tender requirements, by tax obligations denominated in the currency, and by the regulatory infrastructure that mandates the currency's use in domestic commerce.

This Pillar enforces the distinction at the analytical level. A cryptocurrency whose adoption depends on government mandate (a central bank digital currency that residents are required to use for tax payments, a state-issued token that public-sector employees are paid in) does not satisfy Free Adoption. Such a cryptocurrency may have analytical interest for other purposes, but the framework does not value it, because the Pillar's failure indicates the absence of the voluntary holder base the framework's monetary evaluation depends on.

The same reasoning applies to stablecoins backed by sovereign-currency reserves. A stablecoin's adoption is largely a function of the sovereign currency's adoption; the stablecoin operates as a wrapped representation of the sovereign currency, with the same adoption properties the sovereign currency has. Stablecoins fail the architectural test as a category, and they would additionally fail Free Adoption if they passed the architectural test, because the voluntary character of their adoption is mediated by the sovereign currency they wrap.

VI. Decentralized Governance

Decentralized Governance requires that the rules governing the money's operation not be subject to the discretion of any identifiable party, but emerge from the consensus mechanism the protocol operates under. This Pillar evaluates whether the coin's protocol decisions are made by a distributed network of participants rather than by a centralized authority.

Satisfaction requires that no identifiable party be able to unilaterally alter the protocol's rules. A protocol may admit governance procedures (on-chain voting, validator-coordinated upgrades, foundation-sponsored proposals subject to network

ratification), provided those procedures distribute decision-making across the network's participants rather than concentrating it in a controlling party.

Empirical evaluation of Decentralized Governance is multidimensional. The framework examines the distribution of validators or miners that maintain the chain, the distribution of the coin's supply across holders, the operational history of protocol upgrades and how they were ratified, and the institutional arrangements around the protocol's continued development. A coin that scores well on each of these dimensions satisfies the Pillar; a coin that exhibits substantial concentration on any one fails.

Validator distribution is the first dimension. A coin whose consensus is maintained by a small number of identifiable validators concentrates control in those validators. A coin whose consensus is maintained by thousands or tens of thousands of independent validators distributes control across a broader base. Bitcoin's mining is concentrated in a small number of large pools; the pool operators have substantial influence over which transactions are included in blocks and which protocol changes are accepted. Several Layer-1 Coins in the tracked set distribute validation more broadly than Bitcoin does, and the framework's evaluation reflects the distinction.

Holder distribution is the second dimension. A coin whose supply is concentrated in a small number of addresses concentrates economic power in those holders. Bitcoin's supply is heavily concentrated: approximately two percent of addresses hold approximately ninety-three percent of the supply. That concentration is the consequence of Bitcoin's early-distribution pattern, the accumulation of large holders during the price discovery period, and the inability of small holders to acquire meaningful positions at later prices.

Concentration of this kind is a Pillar consideration regardless of its causes.

Operational history of protocol upgrades is the third dimension. A coin whose upgrades have been ratified by genuine network consensus differs from a coin whose upgrades have been imposed by a development team and accepted by the network through deference. Several Layer-1 Coins in the tracked set have operational histories that distribute upgrade authority across the network's participants; others have histories closer to centralized development.

Institutional arrangements are the fourth dimension. A coin maintained by an open development community with multiple contributing parties differs from a coin maintained by a single foundation or a single corporation.

That institutional arrangement does not by itself determine Decentralized Governance, but it informs the Pillar's evaluation.

A coin satisfies Decentralized Governance when the four dimensions converge on distribution across the network's participants. A coin fails when any one of the four exhibits concentration that would allow an identifiable party to alter the protocol's rules without genuine network consensus.

VII. Freedom to Transact

Freedom to Transact requires that holders be able to move the money to other parties without permission from any intermediary that could deny, censor, or reverse the transaction. This Pillar evaluates whether the coin's protocol admits transactions to be conducted without permission from any party in a position to refuse.

Satisfaction requires that the coin's base-layer protocol permit peer-to-peer transactions between any two addresses on the network, without requiring permission from a validator, an intermediary, or a custodian. A protocol's consensus mechanism includes a transaction in the next block if the transaction is valid under the protocol's rules, regardless of who the sender or recipient is.

Base-layer property is what this Pillar evaluates. A coin satisfies Freedom to Transact when its base layer is permissionless. A coin fails when its base layer requires permission from a party with discretion over which transactions are admitted.

Several considerations complicate the evaluation in practice. First is the operational capacity of the base layer. A coin whose base layer admits seven transactions per second (Bitcoin) operates at a substantially different capacity from a coin whose base layer admits thousands of transactions per second (several Layer-1 Coins in the tracked set). A lower-capacity coin may be permissionless at the protocol level while admitting only a fraction of the transactions its holder base would conduct, with the remainder pushed to secondary infrastructure where permissionless character may not be preserved. The Pillar evaluates the base-layer property strictly; the framework reports the operational-capacity consideration as a Coin Report observation.

Second is the cryptographic privacy of transactions. A coin whose transactions are visible on a public ledger admits forensic analysis that can identify the parties to a transaction and the patterns of their transactional behavior. That analysis can be performed by any party with the technical resources, including parties with the capacity to act on what the analysis reveals. A coin whose transactions are cryptographically private (Monero's ring signatures, the privacy-preserving constructions

of several other tracked coins) resists this analysis at the protocol level. That cryptographic property is a stronger satisfaction of Freedom to Transact than the protocol-permissionlessness alone. The framework reports privacy as a Coin Report observation and considers it in the Pillar's evaluation.

Third is the centralized-exchange infrastructure that mediates much of cryptocurrency trading. The Pillar evaluates the coin's protocol, not the infrastructure that has developed around the coin commercially. A coin whose protocol is permissionless can still be traded through exchanges that impose know-your-customer requirements, withdrawal restrictions, and transaction monitoring. Exchange infrastructure does not affect the Pillar's evaluation; the protocol's properties do.

A coin satisfies Freedom to Transact when its base layer admits peer-to-peer transactions without permission. Satisfaction is strengthened by cryptographic privacy and by operational capacity sufficient to absorb the holder base's transaction volume. A coin fails when its base layer requires permission, regardless of how the surrounding infrastructure mediates access.

VIII. Adequate Circulation

Adequate Circulation requires that the supply in operation across the network be sufficient to support the network's transactional purposes without forcing fractional-reserve substitutes to emerge. This Pillar evaluates whether the coin's circulating supply, combined with its transaction throughput, can support the economic activity the network is designed for.

Satisfaction requires that the supply in operation be sufficient for holders to use the coin for its intended purposes without

the emergence of secondary instruments that fractionally represent the coin. The condition is satisfied when the coin's circulating supply, divided by its average transaction size, produces a throughput capacity adequate to the network's commercial usage.

Several coins in the tracked set satisfy Adequate Circulation broadly. Their protocols admit substantial transaction volumes at modest unit values, and the supply circulates among holders who use the coin for the purposes the protocol enables. Other coins have higher unit values and lower circulation rates, which strains Adequate Circulation. The Pillar evaluates whether the strain has produced the fractional-reserve substitutes that indicate inadequate circulation.

Empirical indicator of Adequate Circulation failure is the emergence of wrapped representations, custodial proxies, and similar instruments that fractionally represent the underlying coin. When holders cannot conduct their intended transactions on the base layer at acceptable cost and speed, the holders accept substitute instruments that operate off-chain or on host chains. Substitute instruments are not the coin; they are claims against the coin held by some intermediary. Proliferation of substitute instruments is what Adequate Circulation failure looks like in practice.

Bitcoin's Adequate Circulation is strained. The base layer admits seven transactions per second; the holder base is approximately eighty million; the operational pattern is predominantly long-term holding rather than active circulation; and substantial wrapped representations exist (wBTC on Ethereum, BTCB on Binance Smart Chain, and many others) that fractionally represent Bitcoin holdings held in custody.

Those wrapped representations are the indicator that Bitcoin's base layer is not absorbing the transaction volume its holder base would conduct natively.

Several Layer-1 Coins in the tracked set satisfy Adequate Circulation through architectural choices that increase base-layer throughput or reduce transaction costs to the point that off-chain substitutes do not need to emerge. The framework's evaluation of the Pillar considers both the protocol's design and the empirical pattern of substitute-instrument emergence around the coin.

IX. The Pillar-to-Metric Mapping

The Six Pillars are the framework's qualitative discipline. Four Composite Score metrics are the framework's quantitative discipline. These two disciplines operate together. The Pillars determine whether a coin is admitted to the tracked set. The metrics determine the coin's Composite Score, Crypto Fair Value, and Fair Coin Price once the coin is in the tracked set.

Mapping between the Pillars and the metrics is partial. The metrics do not measure all six Pillars, and the Pillars are not all reducible to metric measurements.

Adoption (70% of the Composite Score) measures the breadth of voluntary holder participation. This metric is informed by Free Adoption: a coin's holder count is meaningful as a Composite Score input only if the holders acquired the coin voluntarily. Free Adoption's satisfaction is the precondition for the metric's usefulness.

Annual Transactions and Annual Transaction Value (10% each) measure the activity rate and the economic substance moving across the network.

These metrics are informed by Freedom to Transact and by Adequate Circulation: a coin's transaction count is meaningful only if the transactions are permissionless, and a coin's transaction value is meaningful only if the supply in operation can absorb the volume natively rather than through substitutes.

Active Developers (10%) measures the technical foundation on which continued operation depends. This metric is informed by Decentralized Governance: a coin's developer count is meaningful as a Composite Score input only if the developers are distributed across multiple contributing parties rather than concentrated in a single entity.

Scarcity and Stable Pricing do not map to the four Composite Score metrics. Scarcity is evaluated at the protocol level as a binary admission condition; the metric system does not measure supply schedules because the protocol-level evaluation has already established that admitted coins satisfy the Pillar. Stable Pricing is reported developmentally as part of the Coin Report's price-volatility profile; the metric system does not measure it because the framework treats stability as informational rather than gating.

Combined Pillar-and-metric structure is the framework's analytical discipline. The Pillars produce a tracked set of coins whose monetary properties admit valuation. The metrics produce, for each admitted coin, the analytical outputs the framework's allocation discipline operates on.

X. The Admission Checklist

A practitioner who has worked through the Pillars has the admission checklist. The procedure: identify a candidate Layer-1 Coin that has passed the architectural test from Chapter 2.

Examine the candidate against each of the five gating Pillars in turn. Scarcity: is the supply schedule protocol-enforced and non-discretionary? Free Adoption: is the holder base voluntarily assembled? Decentralized Governance: do validator distribution, holder distribution, upgrade history, and institutional arrangements converge on distribution rather than concentration? Freedom to Transact: is the base layer permissionless? Adequate Circulation: does the supply in operation support transactional purposes without forcing fractional-reserve substitutes?

A candidate that satisfies all five gating Pillars is eligible for admission to the Foundation's tracked set. The Foundation's admission review then evaluates the candidate's documentation, conducts the chain-level verification the admission requires, and admits or does not admit. Pillar satisfaction is the necessary condition; the Foundation's review is the institutional check that satisfaction has been verified through the procedures the framework's reproducibility discipline requires.

A candidate that fails any one of the five gating Pillars is not admitted. Such a candidate may have analytical interest for other purposes, and a practitioner who finds such purposes may operate under disciplines the framework does not develop. The framework's tracked set contains coins that satisfy all five gating Pillars; coins outside the set are coins the framework does not value, and the framework's analytical outputs do not extend to them.

Admission of a coin to the tracked set is the outcome of the architectural test, the five gating Pillars, and the Foundation's verification combined. That combination produces the operationally maintained set against which the framework operates.

A coin in the set has demonstrated the architectural properties and the monetary properties the framework requires. A coin outside the set has not, and the framework offers no analytical claim about it.

Five gates. One developmental Pillar. One tracked set. The discipline is what the Pillars enforce.

4. THE FOUNDATION'S TRACKED SET

The twelve coins the framework values, and DGD

I. What the Tracked Set Is

The Foundation's tracked set is the operationally maintained list of Layer-1 Coins for which the Foundation produces and publishes quarterly Coin Reports. Each coin on the list has passed the architectural test from Chapter 2 and the five gating Pillars from Chapter 3, and each has been verified through the Foundation's admission review. Coin Reports document the four-metric measurements that produce the Composite Score, the Crypto Fair Value, the Fair Coin Price, and the Margin of Safety the practitioner applies.

This set is finite. At the time of this writing, it contains twelve Layer-1 Coins. The Foundation additionally supports Digital Gold (DGD), its own coin, which is structurally always at its Crypto Fair Value at each level by design and is therefore presented as a separate item rather than as a thirteenth tracked coin. The combined object the practitioner works with is twelve coins plus DGD. Never thirteen. This presentation is canonical across the framework's documentation.

The twelve, in canonical order:

1. Cardano (ADA)
2. Digital Cash (DASH)
3. DigiByte (DGB)
4. MultiversX (EGLD)

5. Internet Computer (ICP)
6. Near Protocol (NEAR)
7. Ravencoin (RVN)
8. Chia (XCH)
9. eCash (XEC)
10. Monero (XMR)
11. Nano (XNO)
12. ZClassic (ZCL)

Plus Digital Gold (DGD), the Foundation's own coin.

Bitcoin is not on the list. The reason was developed in Chapter 1: Bitcoin's role is the December 2024 calibration of the Benchmark, and the contribution is permanent and historically singular. Bitcoin is one Layer-1 cryptocurrency among others, measured against the Benchmark on identical terms with every other Layer-1 Coin, but its admission to the tracked set is not what advances the framework's purpose. CFV CoinFunds the Foundation supports do not hold Bitcoin. Ethereum, similarly, is not on the list; Ethereum appears in the framework's worked examples as a near-CFV calibration test, not as a tracked coin.

What follows develops the twelve tracked coins by architectural family rather than coin by coin. The twelve cluster naturally into three families: proof-of-work-derived architectures, proof-of-stake architectures, and novel-resource consensus architectures. Each family section discusses the shared properties that bring the family's coins together, then names what is distinctive about each coin within the family. The full Coin Reports the Foundation publishes provide the quarterly measurements the framework's arithmetic operates on; this

chapter establishes which coins exist on the tracked set and what each contributes that the others do not. Digital Gold is treated separately at the end, because its role is structurally different from any tracked coin's.

II. The Proof-of-Work-Derived Family

Six coins on the tracked set derive their consensus from Bitcoin's proof-of-work tradition, though each has modified the architecture in operationally significant ways: Digital Cash, DigiByte, Ravencoin, eCash, Monero, and ZClassic. Each operates its own blockchain, supports its own native wallet, and is the protocol's native coin; the architectural test is satisfied for all six. Each has been assembled voluntarily by its holder base across years of operational history, and Free Adoption is satisfied across the family. The family is unified by its inheritance from Bitcoin's consensus security model and by the protocol-enforced supply caps that satisfy Scarcity at the protocol level.

What distinguishes the six is what each has done with the proof-of-work base.

Digital Cash (DASH) launched in January 2014, originally as XCoin, then renamed to Darkcoin, and finally rebranded to Dash in March 2015. Digital Cash is its canonical name in the framework's documentation; DASH is the ticker; the network's promotional documentation often uses Dash. Its architectural distinction is the two-tier consensus structure. A proof-of-work base layer operates the mining architecture; a second layer of masternodes provides InstantSend transaction finality (one to two seconds, an operational property that distinguishes Digital Cash from base-layer Bitcoin's ten-minute average), optional CoinJoin-based privacy (PrivateSend), and effective elimination

of fifty-one-percent attack risk (ChainLocks). The masternode tier requires 1,000 DASH collateral per masternode, creating a meaningful operator class while preserving the broader holder base's independence. The 18.9 million DASH supply cap follows a halving schedule similar to Bitcoin's. Decentralized Governance is satisfied through the masternode distribution, the proof-of-work mining distribution, and the on-chain treasury governance procedures by which masternode votes determine the allocation of treasury proceeds across development proposals. Digital Cash's role in the tracked set is the case study in two-tier consensus and the operational utility that architecture enables for medium-of-exchange use at point of sale.

DigiByte (DGB) launched in January 2014, founded by Jared Tate. Its architectural distinction is five-algorithm proof-of-work: SHA-256, Scrypt, Skein, Qubit, and Odocrypt mining algorithms operate simultaneously, distributing mining across multiple hardware categories. Each algorithm contributes a fraction of total block production; no single algorithm dominates the validation, and no single class of mining hardware can monopolize the chain. The 21 billion DGB supply cap is cryptographically enforced. Block times average fifteen seconds; operational capacity is approximately 1,066 transactions per second at the base layer, an order of magnitude faster than Bitcoin. Decentralized Governance is satisfied through the multi-algorithm validator distribution, among the most varied profiles in the tracked set, and through the development community structure that has maintained the protocol across its eleven-year operational history without foundation funding or institutional capture. DigiByte demonstrates architectural decentralization through algorithmic diversity in a way no single-algorithm protocol can match.

Ravencoin (RVN) launched in January 2018 from a Bitcoin codebase fork. The protocol was launched without a pre-mine, without an initial coin offering, and without founder allocations: an architectural-distribution property that distinguishes Ravencoin from the majority of post-Bitcoin Layer-1 Coins. Its Bitcoin-derived codebase preserves proof-of-work consensus and the supply-cap discipline; Ravencoin's modifications focus on the asset-issuance layer the protocol adds. The 21 billion RVN supply cap satisfies Scarcity at the protocol level. Free Adoption is satisfied with unusual purity here: the launch-distribution properties (no pre-mine, no ICO) produce a holder base whose voluntary character is structurally cleaner than protocols whose early distribution concentrated coin in identifiable insider classes. The asset-issuance layer permits named assets to be created on the Ravencoin chain; those assets are tokens, not coins, and outside the framework's analytical scope, but Ravencoin itself satisfies the Pillar regardless of what its issuance layer hosts. Ravencoin's role in the tracked set is the case study in launch-distribution discipline.

eCash (XEC) forked from Bitcoin Cash in 2020, operating under the original Nakamoto consensus architecture inherited from Bitcoin. Its architectural distinction is the combination of the Nakamoto consensus base with the Avalanche pre-consensus layer that provides instant-finality transaction processing alongside the base proof-of-work layer. This dual-layer construction produces transaction finality patterns appropriate to medium-of-exchange use while preserving the security properties Bitcoin's proof-of-work establishes. The 21 trillion XEC supply satisfies Scarcity through the architecturally identical halving schedule Bitcoin uses, scaled to the larger nominal supply (eCash denominates the supply in smaller units, producing the larger nominal figure without altering the relative

scarcity Bitcoin's architecture establishes). Avalanche pre-consensus reduces effective settlement times to under one second for the substantial majority of transactions. eCash demonstrates that the security properties of Bitcoin's original consensus are compatible with the operational improvements the framework's Adequate Circulation evaluation rewards.

Monero (XMR) launched in April 2014. Its architectural distinction is cryptographic privacy at the protocol level. Three constructions combine to produce transactions whose sender, recipient, and amount are not derivable from the public ledger: ring signatures (which obscure the sender's address by including the sender among a set of decoy signatures), stealth addresses (which generate a unique one-time address for each transaction), and confidential transactions (which obscure the transaction amount). Monero's emission schedule is bounded with a tail emission of 0.6 XMR per block after the main emission completes; Scarcity is satisfied through the tail emission's predictable schedule and the absence of discretionary issuance authority. Decentralized Governance is satisfied through proof-of-work mining distribution (with the RandomX algorithm explicitly designed to resist ASIC mining specialization, producing a more distributed validator base than ASIC-dominated protocols), through the development community structure, and through the protocol's history of upgrades ratified by genuine network consensus. Freedom to Transact is satisfied here in the strongest form among coins in the tracked set: cryptographic privacy resists the transaction-graph analysis that public-ledger coins admit, supporting the Pillar's satisfaction beyond protocol-level permissionlessness alone. Monero's role in the tracked set is the case study in cryptographic privacy at the protocol level.

ZClassic (ZCL) launched in November 2016 as a fork of Zcash that removed the founder’s reward allocation Zcash imposed. Its architectural distinction is the combination of zero-knowledge cryptographic privacy (inherited from Zcash) with the launch-distribution discipline that the founder-reward removal supplies. ZClassic operates the zk-SNARK construction Zcash pioneered, supporting shielded transactions whose sender, recipient, and amount are cryptographically concealed. Its supply is bounded with the halving schedule inherited from Zcash; Scarcity is satisfied at the protocol level. Free Adoption is satisfied through the voluntarily assembled holder base with the additional property that the founder-reward removal eliminates the structural insider class that Zcash’s distribution included. Decentralized Governance is satisfied through proof-of-work mining distribution and through development history that has proceeded without the foundation-controlled allocation that founder-reward protocols admit. ZClassic pairs Monero’s privacy commitment with Ravencoin’s launch-distribution discipline, demonstrating that strong cryptographic privacy can be integrated into a Layer-1 Coin without the founder-reward structure that compromises the community’s claim on the issued supply.

III. The Proof-of-Stake Family

Three coins on the tracked set operate proof-of-stake consensus architectures: Cardano, MultiversX, and Near Protocol. Each operates its own blockchain, supports its own native wallet, and is the protocol’s native coin. Across the family, Decentralized Governance is satisfied through validator distribution under staking-and-delegation structures that distribute consensus participation broadly across the holder base, and Adoption has been assembled voluntarily.

What distinguishes the three is the architectural specifics of how proof-of-stake has been implemented and what each protocol does at higher operational capacity than the proof-of-work family typically supports.

Cardano (ADA) launched in September 2017, founded by Charles Hoskinson, an Ethereum co-founder who departed in 2014 over disagreements about Ethereum's governance posture. Cardano's architectural distinction is its commitment to peer-reviewed academic publication of the protocol's design. The Ouroboros family of proof-of-stake consensus protocols, which Cardano operates under, has been specified through publications in established cryptography venues, an unusual operational property in the Layer-1 Coin space. Validation infrastructure consists of stake-pool operators distributed across thousands of independent operators globally; the development organization, Input Output, transitioned the protocol to full on-chain governance through the 2025 Plomin hard fork, addressing the centralized-development concerns that had characterized the protocol's prior operational history. Cardano's 45 billion ADA maximum supply is among the largest hard caps in the Layer-1 Coin universe, but the cap is cryptographically enforced and satisfies Scarcity. Operational capacity is approximately 250 transactions per second at the base layer, with higher throughput available through the Hydra layer-2 architecture. Cardano's role in the tracked set is the case study in protocol-development discipline: the peer-reviewed specification, the on-chain governance transition, and the academic engagement with the cryptography establish a development posture that distinguishes Cardano from protocols developed through less formal procedures.

MultiversX (EGLD) launched in July 2020, originally as Elrond Network, and rebranded to MultiversX in November

2022. Its architectural distinction is sharded proof-of-stake consensus that produces high-throughput transaction processing while maintaining base-layer settlement security. MultiversX operates an adaptive state sharding architecture that partitions the validator set and the state across multiple shards, each processing transactions in parallel and synchronizing through cross-shard atomic execution. EGLD is bounded by a deflationary issuance schedule that approaches a fixed cap of 31.4 million; Scarcity is satisfied at the protocol level. Operational capacity reaches the thousands of transactions per second at the protocol level, with transaction costs that approach zero on a per-transaction basis. That capacity is what makes MultiversX suitable for use cases that older Layer-1 Coin architectures cannot support at competitive economics: high-frequency commerce, micropayment settlement, and application-layer infrastructure that requires base-layer throughput. MultiversX demonstrates that base-layer throughput in the thousands of transactions per second is achievable without sacrificing the security properties Bitcoin's consensus provides at substantially lower throughput.

Near Protocol (NEAR) launched in April 2020. Its architectural distinction is the combination of sharded proof-of-stake consensus with a usability-focused account-naming convention. Where most Layer-1 Coins identify addresses through long alphanumeric strings, Near's account names operate as human-readable identifiers, an operational property that distinguishes the protocol's user experience from the address-string conventions of older Layer-1 Coins. NEAR's 1 billion initial supply is bounded with a 5% annual issuance schedule that the framework's Scarcity evaluation has accepted as satisfying the Pillar through the protocol-enforced rules. Protocol development is maintained by Near Foundation and the broader development community that has formed around

the protocol; the institutional arrangement distributes decision-making across multiple contributing parties. Near's role in the tracked set is the case study in user-experience-oriented Layer-1 Coin design: account-naming convention and developer-experience emphasis demonstrate that monetary-grade Layer-1 architecture is compatible with the usability properties that broader adoption requires.

IV. The Novel-Resource Consensus Family

Three coins on the tracked set operate consensus architectures that depart from both proof-of-work computation and proof-of-stake capital lock-up: Chia, Internet Computer, and Nano. Each uses a different resource as its consensus mechanism, and each demonstrates that monetary-grade Layer-1 consensus is achievable through architectural choices that older protocols did not consider. Across the family, the architectural test is satisfied, Free Adoption is satisfied, and Decentralized Governance is satisfied through validator distributions specific to each protocol's resource model.

Chia (XCH) launched in March 2021, founded by Bram Cohen, the creator of BitTorrent. Its architectural distinction is the proof-of-space-and-time consensus, which uses storage capacity rather than computational hashing as the consensus resource. Validators commit storage capacity to the protocol through plot files generated and stored on commodity hard drives; the protocol's consensus assigns block production rights based on the proof-of-space the validator can produce in response to challenges, combined with a proof-of-time function that prevents grinding attacks. XCH supply is bounded with a long emission tail; Scarcity is satisfied at the protocol level through the issuance schedule the protocol enforces. Decentralized Governance is satisfied through

validator distribution that the proof-of-space architecture produces, in which validation is accessible to any participant with commodity hardware rather than concentrated in specialized mining facilities. The protocol's transaction architecture admits permissionless settlement and supports application-layer programmability through the Chialisp smart-coin model, a constraint-based programming approach that differs structurally from the contract-account models of older smart-contract platforms. Commercial integrations have matured: the Arizona SB 1649 named-asset list includes Chia, and the asset-issuance capabilities the protocol supports have produced commercial usage in tokenization contexts. Chia demonstrates that monetary-grade Layer-1 consensus is achievable through resources other than the energy-intensive computational hashing of proof-of-work and the capital-locked staking of proof-of-stake.

Internet Computer (ICP) launched in May 2021, developed by the DFINITY Foundation under the leadership of Dominic Williams. Its architectural distinction is the chain-key cryptography architecture that permits the protocol to host both cryptocurrency operations and broader computational workloads at the protocol level. Internet Computer operates under a hierarchical structure of subnets, each running an independent consensus, with the subnets coordinated through chain-key cryptographic procedures that produce a single network-level state. ICP supply at 469.2 million is bounded with an inflationary issuance pattern that the framework's Scarcity evaluation has accepted as satisfying the Pillar through the protocol-enforced rules governing the issuance, the staking-and-burning mechanisms that offset issuance against utility-driven coin destruction, and the absence of discretionary supply expansion outside the protocol's specified procedures.

Decentralized Governance is satisfied through the network nervous system, an on-chain governance structure that admits any neuron-holding participant to vote on protocol proposals, with the neuron-staking-and-aging dynamics producing a governance distribution that has matured across the protocol's operational history. Operational capacity is substantially higher than first-generation Layer-1 Coins, and the architecture's chain-key construction admits transaction patterns that older architectures cannot support. Computational workloads the protocol hosts include applications that operate as if on a public cloud but settle on-chain at protocol-level cost. Internet Computer extends Layer-1 Coin capability beyond settlement to include the computational workloads that traditional applications run on centralized cloud infrastructure.

Nano (XNO) launched in 2015, originally as RaiBlocks. Its architectural distinction is the feeless transaction architecture using a directed acyclic graph (DAG) ledger structure rather than a linear blockchain. Each account in the protocol operates its own chain; transactions are signed by the account holder and confirmed by the network through a delegated proof-of-stake voting procedure that requires no transaction fees. XNO's 133.2 million supply is fully issued, with no further issuance scheduled; Scarcity is satisfied at the protocol level in the strictest form, with no inflation and no discretionary issuance authority. Decentralized Governance is satisfied through the principal-representative voting structure, in which account holders delegate voting weight to representatives who participate in the consensus procedure. Freedom to Transact is satisfied at the base layer; the feeless property is operationally distinctive, producing a transaction-cost profile that no other coin in the tracked set matches. Sub-second confirmation times combined with the absence of transaction fees produce a circulation pattern that reflects genuine commercial usage

rather than speculative holding. Nano demonstrates that monetary-grade Layer-1 Coins can operate without transaction fees while preserving the consensus security that protocol-level operation requires.

V. Digital Gold (DGD)

Digital Gold is the Foundation's own coin. Its architectural distinction is structural rather than consensus-based: Digital Gold is engineered to be at its Crypto Fair Value at each level by design. The protocol's four metrics (Adoption, Annual Transactions, Annual Transaction Value, Active Developers) move with the protocol's growth, and the issuance, distribution, and validation architectures are calibrated so that the Composite Score the protocol produces against the Benchmark equals the Composite Score the protocol's measured metrics warrant. Fair Coin Price the framework computes from the protocol's Composite Score equals the price at which the protocol's coin is supported, and the Margin of Safety against that price is structurally zero by design.

This architectural choice is what distinguishes Digital Gold from the twelve tracked coins. Tracked coins exist independently of the framework, and the framework measures them against the Benchmark to produce Margins of Safety that vary with market price. Digital Gold exists as the Foundation's own coin and is calibrated so that the Foundation's commercial commitments to the coin's structural value are integrated into the protocol's design rather than asserted around it.

Digital Gold satisfies the architectural test (own blockchain, own native wallet, own native coin) and the five gating Pillars. Technical details of the protocol, the calibration procedures the Foundation operates under, the legal structure that

supports the structural-CFV property, and the seven structural distinctions that distinguish Digital Gold's legal posture under the Howey test are developed in *Cryptocurrency Analysis* Chapter 19 and in the Digital Gold White Paper.

Digital Gold's role in the framework is the practitioner's structurally-stable position. Where the twelve tracked coins admit Margins of Safety that vary with measurement date and market conditions, Digital Gold provides a position whose value relationship to the Benchmark is engineered to be stable. Presentation of Digital Gold as a separate item (twelve coins plus DGD, never thirteen) reflects the structural distinction at the level of the framework's documentation.

VI. The Set the Practitioner Holds

These twelve coins plus Digital Gold are the operational object the framework's allocation discipline operates on. A practitioner who has worked through the architectural test, the Six Pillars, and the present chapter has the tracked set. Allocation decisions, developed in Chapters 10 and 11, operate on this set. Worked Coin Reports in Chapter 8 develop five of the twelve in operational depth, demonstrating the framework's arithmetic in application. Remaining coins in the tracked set are documented in the Foundation's quarterly Coin Reports, which the practitioner consults for the measurement inputs the framework's arithmetic requires.

This set is finite. It is canonical. It is twelve plus DGD. The framework values these coins, and the framework does not extend its analytical claims to coins outside the set.

5. BITCOIN AS CALIBRATION SOURCE

Why Bitcoin produced the Benchmark, and why the Benchmark no longer depends on Bitcoin

I. The Calibration Question

The Benchmark requires a reference. Some specific cryptocurrency, measured at some specific moment, must supply the four-metric values against which every subsequent Layer-1 Coin is evaluated. That reference is what makes the framework's arithmetic possible: the Composite Score is a candidate coin's metrics divided by the reference's metrics, weighted and summed. Without a reference, there is no Composite Score, no Crypto Fair Value, and no Fair Coin Price.

Reference choice is not neutral. Different reference cryptocurrencies, measured at different moments, would produce different Benchmark values and therefore different valuations for every coin the framework evaluates. The choice has to be defensible: the reference must be a Layer-1 Coin whose December measurements can be made through verifiable channels, whose monetary properties have been examined long enough to be characterized, and whose institutional infrastructure had matured to the point that the measurement could be conducted with confidence at the calibration date.

Bitcoin, in December 2024, was the only cryptocurrency that satisfied all three conditions. This chapter develops why, develops what the calibration produced, and develops why the

Benchmark, once calibrated, is independent of Bitcoin from the calibration moment forward.

II. Why Bitcoin and Not Gold

Gold is the historical reference against which monetary cryptocurrency has been compared throughout the asset class's existence. That comparison is not without force. Gold possesses the monetary properties the Six Pillars identify: bounded supply (the earth's accessible gold reserves are finite and grow slowly through extraction), voluntary adoption (gold's monetary use is not imposed by any central authority), distributed governance (no single party controls gold's supply or its acceptance), permissionless settlement (gold transactions require no intermediary's approval), and adequate circulation (the gold supply has historically supported the commercial activity of the holders who used it).

Gold is, however, not a cryptocurrency. The four-metric methodology the framework operates depends on metrics that exist for cryptocurrencies and that do not exist for gold. Gold has no Adoption count in the sense the framework measures Adoption: gold's holders are not enumerable through chain-level inspection, exchange-custody attestations, or cross-validated holder surveys, because gold has no chain and no custody-attestation infrastructure equivalent to what regulated digital-asset exchanges provide.

Gold has no Annual Transaction count: gold transactions are not recorded on a public ledger, and gold's volume estimates are produced through industry surveys whose methodology cannot be cross-validated. Gold has no Active Developer count: gold has no protocol to develop.

The framework values cryptocurrencies, and its metrics are cryptocurrency metrics. Gold's monetary properties are evaluable, but not against the four-metric methodology. Gold is therefore not a candidate for the calibration reference.

III. Why Bitcoin and Not Ethereum

Ethereum is the second-largest cryptocurrency by market capitalization and operates the most extensive smart-contract ecosystem in the Layer-1 Coin space. Its candidacy as the calibration reference is not facially absurd.

That candidacy fails on the four-metric measurement. Ethereum's Adoption count, at the December 2024 calibration moment, was not measurable through verifiable channels at the confidence the four-metric methodology requires. Ethereum's exchange-custody distribution had not yet matured to the regulated infrastructure that produced Bitcoin's spot-ETF approvals; principal entity-adjusted holder counts for Ethereum diverged across analytics providers by margins substantially larger than the divergences for Bitcoin at the same date. Ethereum's Annual Transactions count is complicated by the protocol's smart-contract architecture, which produces transaction patterns (failed transactions, contract-internal calls, MEV-related churn) that the four-metric measurement must adjust for in ways that introduce methodological uncertainty Bitcoin's simpler transaction model does not.

A deeper objection is operational. The calibration reference is the coin whose metrics define the four-metric Benchmark values. The Composite Score formula then operates by computing each subsequent coin's metric-ratios against those Benchmark values.

Had Ethereum been the calibration reference, Ethereum would by construction produce a Composite Score of 1.0 at the calibration date, and Bitcoin's Composite Score against the Ethereum-derived Benchmark would have been less than 1.0 (because Bitcoin's adoption count, transaction volume, and developer count were smaller than Ethereum's at the calibration date on some of the four metrics, even though Bitcoin's market capitalization and holder concentration exceeded Ethereum's on others). The framework's outputs would have positioned Bitcoin as below-Benchmark on the four-metric weighted aggregate. This is not the framework's actual reading on Bitcoin, but it would have been the framework's reading had Ethereum been the calibration reference.

Reference choice shapes the outputs. The framework chose Bitcoin not because Bitcoin's properties are superior to Ethereum's, but because Bitcoin's measurement infrastructure had matured to the point that the calibration could be made with confidence. Measurement infrastructure was what the calibration required, and Bitcoin had it at the December 2024 moment.

IV. Why Bitcoin and Not a Basket

A basket of cryptocurrencies, measured collectively at the calibration date, could in principle have supplied the Benchmark values. The basket approach is what equity-index methodologies typically use: the S&P 500 is a basket, not a single company.

A basket-Benchmark would have produced an aggregate calibration whose values are insulated from the specific properties of any one cryptocurrency.

That basket approach fails for the framework's purpose. The framework's task is to value individual Layer-1 Coins against a permanent reference, not to value the cryptocurrency category against itself. A basket-Benchmark would, by construction, produce Composite Scores of approximately 1.0 for the average member of the basket and Composite Scores meaningfully above or below 1.0 only for outliers. The discriminating power the framework requires (the capacity to identify coins whose fundamentals substantially exceed their market pricing, and coins whose fundamentals substantially fall short of their market pricing) would have been blunted by the averaging the basket performs.

A basket approach also requires defining the basket. A basket of five cryptocurrencies would produce different Benchmark values from a basket of ten or twenty. Selection of basket members is itself a methodological choice that the framework would have to defend, with no objective external standard against which the defense could be conducted. The single-reference choice, by contrast, has an objective external standard: Bitcoin was the cryptocurrency whose institutional measurement infrastructure had matured to the point that the calibration could be performed with confidence, and that measurement was the operational criterion.

V. The December 2024 Moment

Bitcoin's December 2024 state is the specific moment from which the Benchmark was calibrated. That state had specific properties distinguishing it from any earlier moment in Bitcoin's history.

Bitcoin's price at the calibration moment was approximately \$100,000 per coin. The price level mattered not because of its

symbolic significance but because it reflected the institutional capital that had concentrated in Bitcoin through the regulated infrastructure that admitted it. Spot exchange-traded products that had received SEC approval in January 2024 had accumulated holdings approaching one million BTC by year-end. Public corporations with treasury accumulation strategies had built positions in the hundreds of thousands of BTC. Institutional capital was operationally measurable through audited custody attestations and through the issuers' regulatory filings, in a way that earlier-cycle institutional Bitcoin holdings had not been.

Bitcoin's market capitalization at the calibration moment was approximately \$1.983 trillion. That market capitalization is the calibration's multiplier in the Crypto Fair Value formula: each subsequent coin's Composite Score is multiplied by this figure to produce the Crypto Fair Value. Choice of this specific market-capitalization figure as the multiplier is what scales the framework's outputs to the cryptocurrency category's economic substance at the calibration moment.

Bitcoin's circulating supply at the calibration moment was approximately 19.83 million BTC. That supply is the calibration's contextual divisor: each candidate coin's Fair Coin Price is the Crypto Fair Value divided by the candidate's own circulating supply, and the relationship between Crypto Fair Value (a market-capitalization quantity) and Fair Coin Price (a per-coin quantity) is mediated by the candidate's supply. Bitcoin's supply at the calibration moment is not used directly in any subsequent coin's Fair Coin Price calculation, but it provides the contextual figure against which the framework's documentation describes the calibration.

Four metric values at the calibration moment:

- **Adoption: approximately 80 million unique global holders.** Measured through the five-source triangulation methodology Appendix A specifies: on-chain address counts after entity clustering, exchange-custody contributions from regulated custodians, institutional holdings from issuer disclosures, industry survey research, and secondary-layer end-user counts from custodial-wallet provider attestations.

- **Annual Transactions: approximately 6.09 billion.** The trailing-twelve-month count of transactions processed across Bitcoin's base layer and the Lightning Network (the principal layer-2 infrastructure for Bitcoin), with the non-economic-activity adjustments the four-metric methodology specifies.

- **Annual Transaction Value: approximately \$13.49 trillion.** The dollar value of economic transfers processed across Bitcoin and Lightning across the trailing twelve months, entity-adjusted to exclude exchange internal transfers and other non-commercial flows.

- **Active Developers: approximately 905 unique contributors.** The count of unique contributors to Bitcoin's essential repositories across the trailing twelve months, produced through the three-layer methodology the framework operates: the principal protocol implementation, the extended ecosystem repositories, and the application-layer developer layer, with cross-validation against the Electric Capital developer-tracking dataset where available.

These six values together (the four metrics plus the market-capitalization multiplier and the contextual circulating supply) constitute the Benchmark. The values are what subsequent measurements are made against.

VI. The Permanence Commitment

Once calibrated, the Benchmark does not change. Not when Bitcoin's price moves. Not when Bitcoin's adoption count moves. Not when exchange-traded-fund flows reverse. Not when sovereign allocations expand or contract. Not when subsequent cryptocurrencies surpass Bitcoin on any of the four metrics or on all of them.

Permanence is what distinguishes the Benchmark from rolling averages, periodic recalibrations, and basket replacements, all of which appear in equity-index methodologies but none of which would produce the operational reference the framework requires. The argument for permanence has been developed in *Cryptocurrency Analysis* Chapter 2 at the depth that document warrants; this chapter states the argument in the compressed form a Volume 2 reader requires.

The kilogram in Sèvres is the operational analogy. From 1889 until 2019, a specific platinum-iridium cylinder stored at the International Bureau of Weights and Measures defined the kilogram. That cylinder was the standard. Every mass measurement everywhere on Earth was traceable to it. The cylinder itself was measured against the standard the cylinder defined; across its operational life, the cylinder drifted by approximately fifty micrograms relative to its official sister copies, and the drift was reported as a property of the cylinder, not as a revision of the definition. The cylinder did not exempt itself from the test it produced.

The Benchmark operates on the same principle. Bitcoin produced the Benchmark through its December 2024 state. From the calibration moment forward, the Benchmark is independent of Bitcoin, and Bitcoin is measured against the

Benchmark on identical terms with every other Layer-1 Coin the framework evaluates. Measuring Bitcoin against its own calibration produces, at later dates, the Composite Score that quantifies how Bitcoin has performed against the standard it produced.

October 2025 is the framework's first sustained demonstration of the principle in action. Bitcoin's price at the October 2025 peak was approximately \$126,000. Bitcoin's market capitalization at the peak was approximately \$2.50 trillion. The four metrics at the peak measurement date produced a Composite Score against the Benchmark that, multiplied by the \$1.983 trillion market-capitalization anchor, generated a Crypto Fair Value substantially below Bitcoin's market capitalization at the peak. The framework reported the reading: Bitcoin was overvalued against the standard Bitcoin itself had produced ten months earlier. That reading was not a critique of Bitcoin and was not a revision of the Benchmark; it was the framework's measurement reporting what the framework's measurements reported.

Fifty micrograms of drift across 130 years, reported as a property of the cylinder rather than as a revision of the standard. Bitcoin's measured fundamentals, against the Benchmark Bitcoin produced, exhibit drift on similar terms. The drift is reported. The Benchmark is not revised.

VII. Why Not a Rolling Average

A rolling-average alternative would update the Benchmark periodically, smoothing the calibration across a trailing window of Bitcoin's metrics rather than freezing the calibration at a single moment.

That approach has surface appeal: it would keep the Benchmark current with the cryptocurrency category's evolving institutional state, and it would avoid the problem of the Benchmark becoming stale as the category matures.

A rolling-average approach fails operationally. A Benchmark that updates produces Composite Scores that depend on when the Composite Score is calculated rather than only on the candidate coin's measurements. Two analysts applying the framework to the same coin at the same measurement date, using different rolling-window definitions, would produce different Composite Scores. The reproduction discipline the framework's later chapters develop would be impossible: no fixed standard against which the practitioner's measurements could be verified, because the standard itself would be moving.

A rolling average also produces a structural problem with the Margin of Safety calculation. The Margin of Safety is the percentage by which the framework's computed Fair Coin Price exceeds the candidate coin's current market price. If the Fair Coin Price changes because the Benchmark has updated rather than because the candidate's fundamentals have changed, the Margin of Safety is not a measurement of fundamental undervaluation. The discipline the framework requires (a Margin of Safety that reflects the gap between the candidate's measured fundamentals and the candidate's market price) is what the permanent Benchmark supplies and what a rolling-average Benchmark cannot.

VIII. Why Not Periodic Recalibration

A periodic-recalibration alternative would hold the Benchmark fixed across multi-year periods and then update it through a discrete recalibration event when the cryptocurrency category's

institutional state had evolved sufficiently. That recalibration would produce a new Benchmark with new values, and the framework's outputs would migrate to the new Benchmark prospectively.

Periodic recalibration is less operationally damaging than the rolling average, but it introduces its own problems. A periodic recalibration is a discretionary act by the framework's producer.

The producer must decide when the recalibration is warranted, must defend the timing of the recalibration against the alternative timings the framework could have chosen, and must explain why the new calibration is more authoritative than the prior one. That discretion is what the framework's commitment to a fixed standard is designed to avoid. The Benchmark's authority rests on its insulation from discretion; reintroducing discretion through periodic recalibration undermines the authority the permanence supplies.

The Foundation's protocol-version architecture addresses the legitimate question that periodic-recalibration approaches attempt to solve.

The Benchmark is permanent within a specified protocol version. Subsequent protocol versions may, through the Foundation's governance procedures, modify the calibration approach for the prospective period the new version covers. A current protocol version's Benchmark applies to all measurements taken under that version; a future version's Benchmark would apply prospectively to the period the future version covers.

That architecture preserves the permanence within each version while admitting the methodology to evolve across versions.

IX. The Reader's Working Understanding

A practitioner who has worked through this chapter has the calibration. Bitcoin's December 2024 state produced the six Benchmark values: 80 million holders, 6.09 billion annual transactions, \$13.49 trillion annual transaction value, 905 active developers, \$1.983 trillion market-capitalization anchor, and 19.83 million circulating supply. Those values are fixed. The framework's arithmetic operates against them from December 31, 2024 forward.

A practitioner working through this book will encounter the Benchmark values repeatedly in the framework's arithmetic. The four metrics enter the Composite Score formula directly. The market-capitalization anchor multiplies the Composite Score to produce the Crypto Fair Value. The circulating-supply figure is contextual to the Benchmark's calibration but does not enter subsequent coins' Fair Coin Price calculations. The Margin of Safety operates on the Fair Coin Price the framework computes and on the candidate coin's current market price.

What the calibration supplies is the reference against which the analytical work proceeds. Bitcoin produced it. The Benchmark holds it. The framework operates on it. Bitcoin's role from the calibration moment forward is to be measured against the standard it produced, on terms identical to every other Layer-1 Coin the framework evaluates.

6. THE CRYPTO FAIR VALUE FORMULA

The three equations the framework operates on

I. What a Formula Does and Does Not Do

The Crypto Fair Value formula converts a candidate coin's four measured metrics into three outputs: a Composite Score (dimensionless), a Crypto Fair Value (a market-capitalization figure in United States dollars), and a Fair Coin Price (a per-coin figure in United States dollars). This conversion is arithmetic. Given the candidate's four-metric values and the Benchmark's four-metric values, the formula produces the three outputs deterministically.

What the formula does not produce is a forecast. The Fair Coin Price is what the candidate's measured fundamentals warrant against the Benchmark; it is not a projection of where the market price will move. Market prices move based on factors the formula does not measure: institutional flows, narrative cycles, macroeconomic conditions, regulatory developments, and the actions of individual large holders. The formula's output is the reference value against which the practitioner evaluates the current market price, not a forecast of where the market price will arrive.

The formula does not produce certainty either. The four-metric measurements carry uncertainty propagated from the underlying measurement methodology. Propagated uncertainty in the Composite Score is approximately fourteen percent under the framework's measurement-grade tolerance; propagated uncertainty in the Crypto Fair Value and Fair Coin Price is approximately one percent under the publication-grade

tolerance applied to a documented Coin Report. The framework operates with these uncertainties stated explicitly; the practitioner integrates the uncertainty band into the Margin of Safety discipline.

What the formula does produce is a reproducible analytical output. Two practitioners applying the formula to the same candidate at the same measurement date, using the same documented inputs, produce the same output to the publication-grade tolerance. Reproducibility is what makes the formula a framework rather than an opinion. This chapter develops the three equations, the weighting that produces the Composite Score, and two worked numerical examples.

II. The Three Equations

This formula consists of three equations applied in sequence.

Equation 1: The Composite Score.

$$S = (0.70 \times R_A) + (0.10 \times R_T) + (0.10 \times R_V) + (0.10 \times R_D)$$

where:

- R_A is the Adoption ratio, the candidate's Adoption count divided by the Benchmark's Adoption count (80 million)
- R_T is the Annual Transactions ratio, the candidate's Annual Transactions count divided by the Benchmark's Annual Transactions count (6.09 billion)
- R_V is the Annual Transaction Value ratio, the candidate's Annual Transaction Value divided by the Benchmark's Annual Transaction Value (\$13.49 trillion)

- R_D is the Active Developers ratio, the candidate's Active Developers count divided by the Benchmark's Active Developers count (905)

The Composite Score is dimensionless. A Composite Score of 1.0 indicates parity with the Benchmark on the weighted aggregate of the four metrics. Bitcoin's Composite Score at the December 2024 calibration moment is 1.0 by construction.

Equation 2: The Crypto Fair Value.

$$CFV = S \times \$1.983 \text{ trillion}$$

The Crypto Fair Value is the Composite Score multiplied by the Benchmark's market-capitalization anchor. Its output is a market-capitalization figure denominated in United States dollars. A coin whose Composite Score is 0.5 has a Crypto Fair Value of approximately \$991 billion; a coin whose Composite Score is 0.01 has a Crypto Fair Value of approximately \$19.83 billion.

This figure is the framework's estimate of the candidate coin's intrinsic market capitalization based on its measured fundamentals against the Benchmark. It is not a forecast of where the candidate's market capitalization will move.

Equation 3: The Fair Coin Price.

$$FCP = CFV / S_C$$

where S_C is the candidate coin's circulating supply at the measurement date.

The Fair Coin Price is the Crypto Fair Value divided by the candidate's circulating supply. Its output is a per-coin figure denominated in United States dollars.

This Fair Coin Price is the operational basis on which the Margin of Safety is computed against the current market price.

These three equations apply in sequence: compute the four-metric ratios, weight and sum them to produce the Composite Score, multiply by the market-capitalization anchor to produce the Crypto Fair Value, divide by circulating supply to produce the Fair Coin Price. That sequence is reproducible. Given the inputs, the outputs follow deterministically.

III. Why These Four Metrics

Four metrics drive the formula: Adoption, Annual Transactions, Annual Transaction Value, and Active Developers. This selection is not arbitrary. Each metric captures a distinct dimension of network monetary function, and the four together constitute what a Layer-1 cryptocurrency does as money. A framework that valued cryptocurrencies on the basis of fewer metrics would miss dimensions monetary function requires; a framework that valued cryptocurrencies on the basis of more metrics would introduce redundancies that double-count what the four already capture.

Adoption captures the breadth of voluntary holder participation. A monetary instrument's value as money depends on the population of holders willing to accept and hold it. A coin with one million holders functions as money for a small population; a coin with eighty million functions as money for a substantially larger one. Adoption count measures the scope of monetary acceptance the network has achieved.

Annual Transactions captures the activity rate at which the network functions as a medium of exchange. A coin whose supply is held but never moved functions less as money than a coin whose supply circulates among holders. Transaction count

measures the operational use of the coin for the medium-of-exchange function.

Annual Transaction Value captures the economic substance moving across the network. A coin can produce a high transaction count through small, repetitive transactions that do not reflect serious commercial activity. The transaction-value figure distinguishes between networks whose activity reflects substantial economic substance and networks whose activity reflects micropayment or test patterns of limited economic weight.

Active Developers captures the technical foundation on which continued operation depends. A coin whose protocol is no longer maintained by an active developer community will, over time, lose its capacity to adapt to operational requirements, security vulnerabilities, and the technical evolution of the broader cryptocurrency space. Developer count measures the protocol's continuing capacity to function as money across the multi-year horizons monetary instruments require.

Together these four describe what a Layer-1 cryptocurrency does as money. A coin's measured values against the four metrics, weighted and summed against the Benchmark's values, produce the Composite Score that quantifies the candidate's network monetary function relative to the reference Bitcoin established in December 2024.

IV. The Weighting Defended

The four metrics enter the Composite Score with weights of 0.70, 0.10, 0.10, and 0.10. Adoption carries seventy percent of the weight; each of the three remaining metrics carries ten percent.

This weighting is the framework's analytical commitment about which dimension of network monetary function matters most.

Adoption carries the heaviest weight because the holder base is the single most consequential dimension of network monetary function. A monetary instrument with substantial transactional activity but few holders is not money for many people; the activity may reflect a small population conducting frequent transactions among themselves. A monetary instrument with many holders has, by the fact of the broad holder base, demonstrated that the population willing to acquire and hold the instrument is substantial. Breadth of the holder base is what makes a cryptocurrency available to function as money across a population large enough to support monetary function operationally.

A seventy-percent weight is defensible against alternative weightings. A weight substantially below seventy percent would understate the convergence that exists between Adoption and the other three metrics: networks with many holders typically exhibit high transaction counts, high transaction values, and large developer ecosystems, and a low Adoption weight would over-weight metrics that correlate with Adoption rather than measuring something independently. A weight substantially above seventy percent would leave the other three metrics with insufficient signal to discriminate between coins of similar Adoption count. The seventy-and-thirty split sits at the architecturally defensible point.

Ten-percent weights on Annual Transactions, Annual Transaction Value, and Active Developers reflect the framework's position that each of the three matters but that none alone captures network monetary function. Annual Transactions without Annual Transaction Value would over-weight micropayment-pattern networks; Annual Transaction

Value without Annual Transactions would over-weight networks whose activity reflects a small number of large transfers; Active Developers without the transactional metrics would over-weight protocols whose technical maturity has not yet translated into operational usage. The three metrics together cross-validate each other while individually contributing the dimensions Adoption alone does not measure.

A 70/10/10/10 weighting is the framework's commitment, defended in *Cryptocurrency Analysis* Chapter 4 at the depth the question warrants. Sensitivity analysis the Foundation publishes documents how the Composite Score varies with weight perturbations.

A practitioner who finds the weighting analytically problematic can apply the formula with alternative weights and observe how the outputs change; the framework's reproduction discipline supports the alternative analysis. Published outputs operate on the 70/10/10/10 weighting because the weighting is the framework's analytical position.

V. A Worked Example: Digital Cash (DASH)

A practitioner who has worked through the three equations and the weighting rationale needs to see the formula in operation against real data. The first worked example applies the formula to Digital Cash (DASH), one of the twelve tracked coins, using measurement inputs the Foundation has published.

Measurement inputs (Q1 2026 measurement date, Digital Cash):

- Adoption: approximately 1.43 million unique holders
- Annual Transactions: approximately 27.2 million (base layer plus InstantSend)

- Annual Transaction Value: approximately \$4.16 billion
- Active Developers: approximately 87 unique contributors

Step 1: Compute the four-metric ratios.

$$R_A = 1,430,000 / 80,000,000 = 0.01788$$

$$R_T = 27,200,000 / 6,090,000,000 = 0.004467$$

$$R_V = \$4,160,000,000 / \$13,490,000,000,000 = 0.0003084$$

$$R_D = 87 / 905 = 0.09613$$

Step 2: Compute the Composite Score.

$$S = (0.70 \times 0.01788) + (0.10 \times 0.004467) + (0.10 \times 0.0003084) + (0.10 \times 0.09613)$$

$$S = 0.01252 + 0.0004467 + 0.00003084 + 0.009613$$

$$S = 0.02261$$

Digital Cash's Composite Score against the Benchmark is approximately 0.0226. That score is substantially below 1.0, reflecting the gap between Digital Cash's measured fundamentals and Bitcoin's December 2024 fundamentals. That gap is what the framework's arithmetic captures and what the subsequent steps translate into a Fair Coin Price.

Step 3: Compute the Crypto Fair Value.

$$CFV = 0.02261 \times \$1,983,000,000,000 = \$44,833,000,000$$

Digital Cash's Crypto Fair Value is approximately \$44.8 billion. This figure is the framework's estimate of Digital Cash's intrinsic market capitalization based on the candidate's measured fundamentals against the Benchmark.

Step 4: Compute the Fair Coin Price.

Digital Cash's circulating supply at the measurement date is approximately 12.05 million DASH.

$$FCP = \$44,833,000,000 / 12,050,000 = \$3,721$$

Digital Cash's Fair Coin Price at the Q1 2026 measurement date is approximately \$3,721. This figure is the framework's estimate of what one DASH is worth, based on Digital Cash's measured fundamentals against the Benchmark.

Step 5: Read the Margin of Safety against the current market price.

Digital Cash's market price at the measurement date is approximately \$27. The Margin of Safety, developed in detail in Chapter 7, is the percentage by which the Fair Coin Price exceeds the market price, divided by the Fair Coin Price.

$$MoS = (\$3,721 - \$27) / \$3,721 = 0.9928 = 99.28\%$$

Digital Cash's Margin of Safety at the Q1 2026 measurement date is approximately 99.3%. This reading indicates that the market price is approximately 0.7% of the framework's computed Fair Coin Price.\

The reading does not project where the market price will move; it reports the gap between the measured fundamentals and the current market.

VI. A Second Worked Example: Cardano (ADA)

One worked example demonstrates the arithmetic. A second worked example demonstrates that the arithmetic produces meaningfully different readings across coins with different fundamental profiles.

Cardano is the natural second choice because Cardano's measured fundamentals differ from Digital Cash's on every metric.

Measurement inputs (Q1 2026 measurement date, Cardano):

- Adoption: approximately 4.85 million unique holders
- Annual Transactions: approximately 142 million
- Annual Transaction Value: approximately \$98.4 billion
- Active Developers: approximately 312 unique contributors

Step 1: Compute the four-metric ratios.

$$R_A = 4,850,000 / 80,000,000 = 0.06063$$

$$R_T = 142,000,000 / 6,090,000,000 = 0.02332$$

$$R_V = \$98,400,000,000 / \$13,490,000,000,000 = 0.007294$$

$$R_D = 312 / 905 = 0.3448$$

Step 2: Compute the Composite Score.

$$S = (0.70 \times 0.06063) + (0.10 \times 0.02332) + (0.10 \times 0.007294) + (0.10 \times 0.3448)$$

$$S = 0.04244 + 0.002332 + 0.0007294 + 0.03448$$

$$S = 0.07998$$

Cardano's Composite Score against the Benchmark is approximately 0.0800. Compared to Digital Cash's 0.0226, Cardano scores approximately 3.5 times higher, driven principally by the substantially larger holder base and the substantially larger developer count, with transactional activity also higher in absolute terms.

Step 3: Compute the Crypto Fair Value.

$$CFV = 0.07998 \times \$1,983,000,000,000 = \$158,600,000,000$$

Cardano's Crypto Fair Value is approximately \$158.6 billion.

Step 4: Compute the Fair Coin Price.

Cardano's circulating supply at the measurement date is approximately 35.6 billion ADA.

$$FCP = \$158,600,000,000 / 35,600,000,000 = \$4.46$$

Cardano's Fair Coin Price at the Q1 2026 measurement date is approximately \$4.46.

Step 5: Read the Margin of Safety against the current market price.

Cardano's market price at the measurement date is approximately \$0.62.

$$MoS = (\$4.46 - \$0.62) / \$4.46 = 0.8610 = 86.10\%$$

Cardano's Margin of Safety at the Q1 2026 measurement date is approximately 86.1%. The reading is substantial but lower than Digital Cash's, reflecting both Cardano's substantially higher Composite Score and the differing relationship between the candidates' market prices and their respective Fair Coin Prices.

Two examples, two distinct fundamental profiles, two distinct Margin of Safety readings produced by the same arithmetic operating on different measured inputs. That is what the formula does, and the difference between the two readings is what the practitioner uses to discriminate between candidates when the allocation discipline operates.

VII. Reading the Output

These worked examples produce specific numbers. A practitioner needs to know what those numbers mean and what they do not mean.

Digital Cash's Composite Score of 0.0226 means its measured fundamentals across the four metrics, weighted and aggregated, are approximately 2.26% of Bitcoin's December 2024 fundamentals. That figure is small because Bitcoin's Adoption count is approximately fifty-six times Digital Cash's, and Adoption carries seventy percent of the Composite Score weight. The figure is the framework's measurement of Digital Cash relative to the Benchmark; it is not a measurement of Digital Cash's absolute monetary capacity. Cardano's Composite Score of 0.0800 reflects its larger Adoption count, larger transactional activity, and larger developer ecosystem against the same Benchmark.

A Crypto Fair Value figure (\$44.8 billion for Digital Cash, \$158.6 billion for Cardano) is the framework's estimate of the candidate coin's intrinsic market capitalization based on its measured fundamentals. The figures are substantially larger than the candidates' current market capitalizations. The gap is what the Margin of Safety quantifies.

A Fair Coin Price figure (\$3,721 for Digital Cash, \$4.46 for Cardano) is the per-coin expression of the Crypto Fair Value divided by the candidate's circulating supply. The figures can be compared to current market prices to produce the Margin of Safety. Fair Coin Prices are not projections of where market prices will move; they are per-coin figures the measured fundamentals warrant.

A Margin of Safety figure (99.3% for Digital Cash, 86.1% for Cardano) means the framework computes the current market price as a small fraction of the per-coin figure the measured fundamentals warrant. Both readings are substantial. The framework's allocation discipline operates on readings of these magnitudes through thresholds, trajectory considerations, and conviction-weighted position sizing, at the depth a practitioner deploying capital requires.

VIII. The Practitioner's Working Procedure

A practitioner who applies the formula to a candidate coin in the tracked set executes the following procedure:

1. Obtain the four-metric measurements for the candidate at the measurement date. The Foundation's Coin Reports supply the measurements for the twelve tracked coins quarterly. A practitioner can additionally produce independent measurements through the reproduction protocol Chapter 13 develops.
2. Compute the four-metric ratios by dividing each candidate metric by the corresponding Benchmark metric.
3. Weight the four ratios with the 70/10/10/10 weighting and sum them to produce the Composite Score.
4. Multiply the Composite Score by \$1.983 trillion to produce the Crypto Fair Value.
5. Divide the Crypto Fair Value by the candidate's circulating supply at the measurement date to produce the Fair Coin Price.
6. Compute the Margin of Safety by subtracting the candidate's current market price from the Fair Coin Price and dividing the difference by the Fair Coin Price.

That procedure is reproducible. Two practitioners executing it on the same candidate at the same measurement date, using the same measurement inputs, will produce the same outputs to the publication-grade tolerance. The framework's outputs are not opinions; they are arithmetic. That arithmetic operates on the four metrics, the Benchmark values, and the candidate's circulating supply. Given the inputs, the outputs follow.

The framework supplies the formula. The Foundation publishes the inputs quarterly for the twelve tracked coins. The practitioner combines the formula and the inputs to produce the outputs. The Margin of Safety the outputs produce is the input to the allocation discipline the framework operates under, at the depth a practitioner deploying capital requires.

Three equations, four-metric ratios, one Composite Score, one Crypto Fair Value, one Fair Coin Price, one Margin of Safety. The arithmetic is the framework. The application is the practitioner's.

7. THE MARGIN OF SAFETY

Graham's central concept, applied to bearer assets

I. Why This Chapter Exists

Graham named the Margin of Safety as the single concept on which all sound investment rests. He developed it in the 1934 edition of *Security Analysis* with David Dodd. He returned to it in the 1949 *Intelligent Investor*. In the 1973 revision, he made it Chapter 20, the final chapter, the book's conceptual conclusion. "The function of the margin of safety is, in essence, that of rendering unnecessary an accurate estimate of the future." That sentence is Graham's, and it carries the discipline this chapter develops.

This chapter is the framework's central commitment. The architectural test established what makes a cryptocurrency a Layer-1 Coin. The Six Pillars established which Layer-1 Coins the Foundation admits. The Benchmark established the reference. The formula established how the framework converts the four measured metrics into a Fair Coin Price. What this chapter on the Margin of Safety establishes is what the practitioner does with that Fair Coin Price when the practitioner has capital to deploy.

What this chapter asks the practitioner to do is precise: act on a calculated number when the market disagrees with the number. Graham was clear that this is hard. The market price exists; it is what other holders are willing to transact at; it is, in some operational sense, the consensus valuation the holder base has produced. The Fair Coin Price the framework computes is a different number, produced by a different procedure, against a

different reference. When the two numbers disagree, the practitioner has to choose which to act on. Graham's commitment, and the framework's commitment, is that the practitioner acts on the Fair Coin Price, and the cushion between the Fair Coin Price and the market price is what protects the practitioner from being wrong.

This chapter develops the protection. It does not promise the protection will hold every time. Tail events occur; analytical procedures miss things; the future contains conditions no measurement can anticipate. What the chapter offers is the discipline that produces defensible positions when the discipline is followed, and the framework against which the practitioner can examine what went wrong when the position fails.

II. The Engineering Origin

The Margin of Safety did not begin in finance. It began in engineering. A bridge designed to support a maximum load of ten tons is built to support twenty or thirty. A vessel rated for a certain pressure is built to withstand much more. The reason is that engineering specifications are estimates; the actual load a structure will encounter cannot be known in advance with the precision the design phase admits. The Margin of Safety is what closes the gap between the estimate and the reality.

Graham took this directly. In the 1973 *Intelligent Investor* he wrote: "Just as the engineer who designs a bridge specifies the maximum load it will safely support, so the investor who buys a stock at a price well below its appraised value provides himself with a margin of safety against the contingencies of the future and against the inevitable errors of his own analysis." The translation is exact.

The engineer specifies a load; the investor specifies a price. The engineer overbuilds against the load; the investor underpays against the value. A cushion absorbs the gap between what was estimated and what occurs.

That engineering analogy carries the central insight: safety comes from the difference, not from the estimate. A bridge engineer who builds exactly to the rated load has produced a structure whose safety depends on the rating being correct. A bridge engineer who builds for triple the rated load has produced a structure that survives ratings that turn out to be wrong. The same logic applies to the investor. An investor who pays the appraised value of a position has produced a holding whose safety depends on the appraisal being correct. An investor who pays half the appraised value has produced a holding that survives appraisals that turn out to be wrong.

III. Graham's Translation to Equity Valuation

Graham translated the engineering principle into equity valuation by computing an intrinsic value for a stock and then requiring a substantial discount to that intrinsic value before recommending purchase. The intrinsic-value calculation in Graham's work used the company's earnings, its book value, its dividend record, and its balance-sheet position to produce a defensible estimate of what the company was worth as a productive enterprise. The market price was compared to the intrinsic value, and the Margin of Safety was the discount the market price embedded.

Graham's recommendation was that the investor purchase only at substantial discounts. The exact threshold varied across his work, but the operational rule was approximately a one-third discount to intrinsic value: buy at sixty-six cents on the dollar

or less. A position taken at this discount had a meaningful cushion against the intrinsic-value estimate being wrong, against the company's business deteriorating, against the market price moving against the position in the short term while the intrinsic value remained defensible.

That discount produced two protections. First was capital preservation: a position taken at a substantial discount could lose ground to market volatility without putting the principal at risk, because the intrinsic value supported a recovery to which the position would naturally migrate. Second was the return profile: a position taken at sixty-six cents on the dollar, held until the market priced the position at intrinsic value, produced a fifty-percent gain on the original investment, independent of any further appreciation in the intrinsic value itself.

Graham was explicit that the Margin of Safety did not require predictive accuracy. The discipline did not depend on knowing when the market would re-rate the position. It depended on the position having a structural property, the discount to intrinsic value, that produced returns over time as the market re-rated, without specifying when the re-rating would occur. That is the discipline the framework translates to cryptocurrency.

IV. The Bearer-Asset Translation

A Layer-1 Coin is not an equity claim. Translating Graham's Margin of Safety to a Layer-1 Coin requires translating intrinsic value itself, because intrinsic value as Graham used the term (the discounted present value of a company's future cash flows, the company's break-up value, the asset coverage on the balance sheet) does not apply to an instrument that produces

no cash flows, has no break-up value, and operates no balance sheet.

The framework's intrinsic-value translation is the Fair Coin Price, computed as the Crypto Fair Value divided by the candidate's circulating supply. That Fair Coin Price is the per-coin figure the framework's measurement of the candidate's four fundamental metrics, weighted against the Benchmark, produces. Denominated in United States dollars, as Graham's intrinsic value was, it serves the same analytical role: the reference against which the market price is compared, with the gap between the two quantifying the Margin of Safety.

This translation is honest about what it preserves and what it changes. The framework preserves Graham's commitment that an intrinsic value can be computed from the asset's fundamental properties, that the intrinsic value differs from the market price in measurable ways, and that the difference produces the cushion the practitioner deploys capital against. What the framework changes is what counts as the fundamental properties. For equities, fundamentals are cash flows, earnings, balance sheets, and the productive capacity of a business. For Layer-1 Coins, fundamentals are the four monetary-function metrics: Adoption, Annual Transactions, Annual Transaction Value, and Active Developers, weighted against the Benchmark.

Without the bearer-asset translation, Graham's framework either fails to apply to Layer-1 Coins (the position Paul Krugman and others have argued) or applies only by force, with intrinsic value asserted rather than derived. The framework's translation derives intrinsic value from the cryptocurrency-specific fundamentals the asset class actually produces. That translation is what enables the rest of the discipline to operate.

V. The Calculation

A Margin of Safety for a candidate Layer-1 Coin is computed as follows:

$$MoS = (FCP - P_C) / FCP$$

where FCP is the framework's computed Fair Coin Price and P_C is the candidate's current market price. The output is a percentage, expressing the gap between the Fair Coin Price and the market price as a fraction of the Fair Coin Price.

A positive Margin of Safety indicates that the framework computes a Fair Coin Price above the current market price. The candidate is, by the framework's measurement, undervalued relative to its measured fundamentals. A practitioner who takes a position at the market price is acquiring the coin at a discount to what the framework computes the coin is worth.

A negative Margin of Safety indicates that the framework computes a Fair Coin Price below the current market price. The candidate is, by the framework's measurement, overvalued relative to its measured fundamentals. A practitioner who holds the coin at the market price is holding an asset whose price exceeds what the framework's measurements warrant.

A Margin of Safety of zero indicates that the Fair Coin Price equals the market price. That candidate is at the framework's computed valuation; the position carries no cushion against analytical error and produces no excess return relative to the framework's expectations.

Computation is arithmetic. Two practitioners computing the Margin of Safety for the same candidate at the same measurement date, using the same Fair Coin Price and the

same market price, produce the same output. Reproducibility is what makes the Margin of Safety a framework discipline rather than an analyst opinion.

VI. The Two Thresholds

The framework operates under two operational thresholds for the Margin of Safety: thirty-five percent and fifty percent. A candidate whose Margin of Safety exceeds the applicable threshold is operationally actionable under the framework's allocation discipline. A candidate whose Margin of Safety falls below the threshold is not.

These thresholds differ by measurement quality. A candidate whose four-metric measurements are fully documented, meaning each measurement satisfies the cross-provider validation requirement, the Electric Capital convergence requirement, and the reproduction-protocol documentation specification the framework requires, operates against the thirty-five percent threshold. A candidate whose measurements include one or more inputs that do not satisfy the cross-validation or convergence requirements operates against the fifty percent threshold.

A thirty-five percent threshold reflects Graham's one-third-discount rule, scaled marginally upward to account for the additional measurement uncertainty Layer-1 Coin fundamentals carry relative to equity fundamentals. A fifty percent threshold reflects the additional cushion required when one or more measurement inputs are not fully validated; the wider cushion absorbs the larger measurement-uncertainty band the inadequately validated inputs introduce.

Thresholds are operational. They do not promise outcomes; they specify the discipline. A position taken at a Margin of

Safety below the applicable threshold is not a position the framework supports. A position taken at a Margin of Safety above the applicable threshold is operationally defensible under the discipline. A practitioner who applies the discipline takes positions at the threshold or above; positions below the threshold are operating outside the discipline, and the framework offers no defense for them.

VII. The Trajectory Dimension

A point-in-time Margin of Safety captures the position at the measurement date. That point-in-time figure is the snapshot. The trajectory is the sequence of point-in-time figures across the trailing four quarters, and the trajectory informs how the practitioner reads the snapshot.

A coin whose Margin of Safety has widened across the trailing four quarters (rising trajectory) has fundamentals that are improving against the market price or a market price that is falling against the fundamentals. Either condition supports the position's conviction: the framework's reading is moving in the practitioner's favor, and the trajectory analysis does not produce evidence that the reading should be discounted.

A coin whose Margin of Safety has narrowed across the trailing four quarters (falling trajectory) has fundamentals that are deteriorating against the market price or a market price that is rising against the fundamentals. That narrowing is not, by itself, a reason to reduce the position; the narrowing may reflect the framework's reading converging with the market, which is the outcome the practitioner is positioned for. Narrowing through convergence is the framework's reading being vindicated by the market. Narrowing through deterioration is a different condition that the practitioner

identifies through the trajectory analysis and through examination of the underlying metrics that produced the deterioration.

A coin whose Margin of Safety has been flat across the trailing four quarters (flat trajectory) has fundamentals and a market price that are moving in coordination. The framework's reading is stable; the practitioner's position is operating at the same conviction level it carried at the last measurement.

Trajectory and point-in-time figure together produce the conviction category. A coin with a Margin of Safety above the applicable threshold and a rising trajectory is high conviction. A coin with a Margin of Safety above the applicable threshold and a flat trajectory is acceptable conviction. A coin with any condition under a falling trajectory is low conviction. A coin with a Margin of Safety below the applicable threshold is below-threshold, regardless of trajectory.

These conviction categories drive the allocation discipline the framework operates under. They are the framework's translation of the point-in-time Margin of Safety into the operational reading the practitioner applies when sizing positions.

VIII. Negative Margins of Safety

A negative Margin of Safety is the framework's reading that a coin is overvalued. Such a reading is a required output of a discriminating framework; a framework that produced only positive Margins of Safety would not be discriminating, because the framework would be incapable of identifying the conditions under which the asset class is mispriced upward.

Bitcoin's October 2025 peak is the framework's first sustained demonstration of the negative-margin reading on a major coin. The Composite Score Bitcoin produced against the Benchmark at the peak measurement date, multiplied by the \$1.983 trillion market-capitalization anchor, generated a Crypto Fair Value substantially below Bitcoin's market capitalization at the peak. The Fair Coin Price the framework computed was substantially below Bitcoin's market price. The Margin of Safety against the market price was negative.

Bitcoin at approximately \$126,000, with a market capitalization of approximately \$2.50 trillion, was overvalued against the standard Bitcoin itself had produced ten months earlier. That reading is not a critique of Bitcoin and is not a revision of the Benchmark. It is what the framework's measurements reported when applied at the peak. As Chapter 5 developed in the kilogram-in-Sèvres analogy, drift relative to a standard is reported as a property of the measured object, not as a revision of the standard. Bitcoin's drift against the Benchmark, at the October 2025 peak, was reported as a property of Bitcoin against the standard Bitcoin produced.

The framework's response to a negative Margin of Safety reading on a coin in the tracked set is operationally explicit. A coin with a negative Margin of Safety is not actionable for new positions under the framework's allocation discipline; the position would be acquired at a market price that exceeds the framework's computed value, providing no cushion against analytical error. A coin with a negative Margin of Safety that is already held under the framework is a candidate for reduction, with the reduction timed against the framework's trajectory analysis and against the operational considerations the framework's discipline includes.

Bitcoin is not in the framework's tracked set, and the CFV CoinFunds do not hold Bitcoin. October 2025's reading on Bitcoin therefore does not require an exit decision under the framework's allocation discipline; the framework's allocation discipline never produced a Bitcoin position. The reading is documented because it demonstrates the framework's capacity to discriminate, not because it requires action under the discipline.

IX. The Voting Machine and the Weighing Machine

Graham observed in *The Intelligent Investor* that “in the short run, the market is a voting machine; in the long run, it is a weighing machine.” That observation is more than a rhetorical flourish; it names the structural property the Margin of Safety discipline relies on.

A voting machine is the market's short-term price-discovery mechanism. Prices in the short run reflect the aggregate of buyer and seller decisions in the current window, weighted by the capital each participant deploys, influenced by narrative cycles, institutional flows, and the technical conditions of the market microstructure. Voting machines are responsive to information that does not fundamentally affect the asset, to sentiment shifts that do not reflect underlying value, and to the actions of large participants whose decisions move prices without communicating analytical content. The voting machine produces prices, and prices are what the market quotes at any given moment, but those prices are not the same as the values the asset's fundamentals warrant.

A weighing machine is the market's long-term price-discovery mechanism. Prices in the long run reflect the fundamentals the asset produces over time, the holder base's accumulated

understanding of those fundamentals, and the convergence of capital allocation toward the values the fundamentals warrant. Weighing machines are slower than voting machines; they operate across years and cycles rather than across hours and days; and they produce prices that approximate the values a disciplined framework can defend.

The Margin of Safety discipline operates on the gap between the voting machine and the weighing machine. A practitioner who computes a Fair Coin Price against the framework's measurements has produced a weighing-machine estimate of the candidate's value. The market price the candidate trades at is a voting-machine output. The Margin of Safety is the gap between the two. A practitioner who takes positions at substantial Margins of Safety is positioning to benefit from the convergence the weighing machine eventually produces, without requiring that the convergence occur on any specific schedule.

Graham did not promise that the convergence would always occur. The market sometimes votes for years against what the weighing machine ultimately produces. The Margin of Safety provides the cushion the practitioner needs to survive the years of disagreement. A position taken at a thirty-five percent or greater Margin of Safety can absorb substantial market-price movement without putting the principal at structural risk, because the underlying fundamentals support the position even as the market disputes them.

X. The Discipline of Holding Through Divergence

The hardest part of the Margin of Safety discipline is not the calculation; it is the holding. A position taken at a substantial Margin of Safety has, by definition, been taken against the

market's contemporary judgment. The market believes the asset is worth less than the framework computes; the market continues to express this belief at the position's continued price level; the position's near-term performance reflects the market's expressed belief rather than the framework's computed value.

A practitioner holding such a position experiences the divergence as a discipline cost. The framework's computed Fair Coin Price has not been validated by the market; the position's value, marked to market, does not reflect the framework's reading; the practitioner's reporting to fiduciaries, partners, or family members may have to explain a position that is not, in the near term, producing the returns the practitioner anticipated when the position was taken.

What is being asked of the practitioner deserves naming. The practitioner is asked to hold a position that, by the framework's calculation, is undervalued, while the market expresses sustained disagreement. The discipline says: the cushion is what protects the position; the convergence is what produces the return; the timing of the convergence is not what the discipline depends on. The discipline does not ask the practitioner to ignore the market; it asks the practitioner to recognize that the market's near-term verdict and the framework's measured reading are different outputs, and the framework's reading is what the practitioner deployed capital against.

Trajectory analysis is the practitioner's protection against false discipline. A position whose Margin of Safety remains substantial while the underlying fundamentals deteriorate is not a position the discipline supports. The framework's discipline rests on the fundamentals supporting the Fair Coin Price the calculation produced; if the fundamentals deteriorate, the Fair Coin Price deteriorates with them, and the Margin of Safety

the position carried at acquisition is no longer the Margin of Safety the position carries at the present measurement. Trajectory analysis tells the practitioner when this is occurring, and the framework's allocation discipline includes the procedures for reducing or exiting positions whose trajectories have turned.

Discipline of holding through divergence is therefore not blind. It is informed by the trajectory analysis, validated by the quarterly Coin Reports the Foundation publishes, and disciplined by the operational procedures the framework documents. What the discipline asks is patience with the market while remaining honest about the fundamentals.

XI. What the Practitioner Has

A practitioner who has worked through this chapter has the discipline that converts the Fair Coin Price into an operational position. Margin of Safety calculation produces a percentage. Thirty-five percent and fifty percent thresholds determine whether the percentage is operationally actionable. Trajectory across the trailing four quarters produces the conviction category. Conviction category drives the position-sizing the framework's allocation discipline applies.

I have written this chapter in my own voice at the moments where the framework asks the practitioner to make a commitment, because the commitment cannot be made by the framework's analytical voice alone. That commitment is the practitioner's, and the practitioner makes it against a discipline I am responsible for. The framework I have built operates under the Margin of Safety discipline. The CFV CoinFunds the Foundation supports operate under the discipline. The Coin Reports the Foundation publishes report the discipline's

outputs quarterly. The discipline is what the framework offers, and the discipline is what the practitioner accepts when proceeding past this chapter.

A Margin of Safety of thirty-five percent or greater on fully documented coins. A Margin of Safety of fifty percent or greater on coins with any unreliable input. A trajectory that does not contradict the point-in-time reading. A position sized against conviction. A discipline of holding while the market and the framework converge.

That is the discipline. The rest of the book applies it.

8. WORKED COIN REPORTS

Five tracked coins, run through the framework end to end

I. What This Chapter Does

By this point the framework has been developed at the architectural, conceptual, and arithmetic levels. This chapter applies the framework to five specific tracked coins and produces the analytical outputs the framework operates on: the four-metric measurements, the Composite Score, the Crypto Fair Value, the Fair Coin Price, the Margin of Safety, and the trajectory across the trailing four quarters.

Five coins are presented: Digital Cash, DigiByte, Chia, eCash, and Cardano. That selection is not a recommendation list. These coins were chosen because they represent five different architectural categories within the tracked set: a two-tier proof-of-work-plus-masternode protocol (Digital Cash), a five-algorithm proof-of-work protocol (DigiByte), a proof-of-space-and-time protocol (Chia), an optimized Nakamoto-consensus protocol with pre-consensus finality (eCash), and a peer-reviewed proof-of-stake protocol (Cardano). Each produces a different Composite Score and a different Margin of Safety reading, and the variation across the five demonstrates that the framework discriminates rather than producing uniform outputs.

Measurement inputs are Foundation-published Q1 2026 measurements. Market prices are Q1 2026 averages. Practitioners applying the framework at a later measurement date will produce different outputs; the framework's arithmetic operates on the measurement inputs available at the date of

application. Reports below illustrate the framework's application as of the measurement date stated; the procedure is what generalizes.

Each Coin Report below follows the same structure: protocol identification, the four-metric measurements, the Composite Score computation, the Crypto Fair Value, the Fair Coin Price, the market price, the Margin of Safety, the trajectory, the conviction category, and a working-conclusion paragraph that summarizes what the framework reports about the coin at the measurement date.

II. Coin Report: Digital Cash (DASH)

Protocol. Digital Cash operates a two-tier consensus architecture: a proof-of-work base layer secured by the X11 mining algorithm and a masternode tier providing InstantSend transaction finality, optional PrivateSend cryptographic privacy through CoinJoin, and ChainLocks fifty-one-percent attack resistance. Masternode operation requires 1,000 DASH collateral per masternode. Launched in January 2014, the protocol has operated continuously since.

Four metric measurements (Q1 2026):

- Adoption: 1.43 million unique holders (entity-adjusted)
- Annual Transactions: 27.2 million (base layer plus InstantSend, trailing twelve months)
- Annual Transaction Value: \$4.16 billion (trailing twelve months, entity-adjusted)
- Active Developers: 87 unique contributors (trailing twelve months, three-layer methodology)

Composite Score computation.

$$R_A = 1,430,000 / 80,000,000 = 0.01788$$

$$R_T = 27,200,000 / 6,090,000,000 = 0.004467$$

$$R_V = 4,160,000,000 / 13,490,000,000,000 = 0.0003084$$

$$R_D = 87 / 905 = 0.09613$$

$$S = (0.70 \times 0.01788) + (0.10 \times 0.004467) + (0.10 \times 0.0003084) \\ + (0.10 \times 0.09613) = 0.02261$$

$$\textbf{Crypto Fair Value. } 0.02261 \times \$1,983,000,000,000 = \\ \$44,833,000,000$$

$$\textbf{Fair Coin Price. } \$44,833,000,000 / 12,050,000 = \$3,721$$

Market price (Q1 2026 average). Approximately \$27.

$$\textbf{Margin of Safety. } (\$3,721 - \$27) / \$3,721 = 99.28\%$$

Trajectory (trailing four quarters). Margin of Safety has been widening across the trailing four quarters as the market price has remained near multi-year lows while the underlying metrics have shown modest improvement. Q2 2025: 98.6%. Q3 2025: 98.9%. Q4 2025: 99.1%. Q1 2026: 99.3%. The trajectory is rising.

Conviction category. Margin of Safety substantially above the 35% threshold; trajectory rising; measurement inputs fully documented across the four metrics under the framework's reproduction protocol. This coin falls into the high-conviction category.

Working conclusion. Digital Cash, at the Q1 2026 measurement date, presents the framework with a substantial gap between the market's voting-machine pricing and the framework's weighing-machine reading.

Two-tier consensus architecture supports operational properties (InstantSend finality, optional cryptographic privacy, ChainLocks attack resistance) that distinguish Digital Cash from base-layer Bitcoin and from most other Layer-1 Coins in the tracked set. Metric measurements support the framework's reading: the holder base has been assembled voluntarily over the protocol's eleven-year history, the transaction throughput reflects active commercial usage rather than holding-only behavior, and the developer count reflects a sustained development community without foundation funding or institutional capture. High conviction is the framework's reading on Digital Cash at the Q1 2026 measurement date.

III. Coin Report: DigiByte (DGB)

Protocol. DigiByte operates a five-algorithm proof-of-work consensus mechanism: SHA-256, Scrypt, Skein, Qubit, and Odocrypt mining algorithms run simultaneously, with each algorithm contributing a fraction of block production. This architecture resists single-algorithm validator concentration and admits validation across multiple hardware categories. Launched in January 2014, the protocol has operated continuously since.

Four metric measurements (Q1 2026):

- Adoption: 0.91 million unique holders (entity-adjusted)
- Annual Transactions: 12.4 million (trailing twelve months)
- Annual Transaction Value: \$1.87 billion (trailing twelve months, entity-adjusted)
- Active Developers: 64 unique contributors (trailing twelve months)

Composite Score computation.

$$R_A = 910,000 / 80,000,000 = 0.01138$$

$$R_T = 12,400,000 / 6,090,000,000 = 0.002036$$

$$R_V = 1,870,000,000 / 13,490,000,000,000 = 0.0001386$$

$$R_D = 64 / 905 = 0.07072$$

$$S = (0.70 \times 0.01138) + (0.10 \times 0.002036) + (0.10 \times 0.0001386) + (0.10 \times 0.07072) = 0.01528$$

$$\text{Crypto Fair Value. } 0.01528 \times \$1,983,000,000,000 = \$30,300,000,000$$

$$\text{Fair Coin Price. } \$30,300,000,000 / 17,710,000,000 = \$1.71$$

Market price (Q1 2026 average). Approximately \$0.011.

$$\text{Margin of Safety. } (\$1.71 - \$0.011) / \$1.71 = 99.36\%$$

Trajectory. Margin of Safety has been substantially flat across the trailing four quarters, with the market price and the underlying metrics moving in approximate coordination. Q2 2025: 99.2%. Q3 2025: 99.3%. Q4 2025: 99.4%. Q1 2026: 99.4%. The trajectory is flat to modestly rising.

Conviction category. Margin of Safety substantially above the 35% threshold; trajectory flat to modestly rising; measurement inputs fully documented under the framework's reproduction protocol. This coin falls into the acceptable-to-high conviction category, with the conviction modestly tempered by the flat trajectory.

Working conclusion. DigiByte, at the Q1 2026 measurement date, presents the framework with substantial structural decentralization through the five-algorithm consensus architecture and a holder base assembled voluntarily across the

protocol's eleven-year operational history. Base-layer transaction capacity (approximately 1,066 transactions per second) substantially exceeds Bitcoin's, and the protocol has not produced the wrapped-representation infrastructure that indicates Adequate Circulation strain. Metric measurements are lower in absolute terms than Digital Cash's, reflecting the smaller commercial usage pattern, but the framework's arithmetic produces a Margin of Safety in the same range as Digital Cash. Acceptable-to-high conviction characterizes the reading; the flat trajectory is what tempers the assessment.

IV. Coin Report: Chia (XCH)

Protocol. Chia operates a proof-of-space-and-time consensus architecture, using committed storage capacity rather than computational hashing as the consensus resource. Validators plot files on commodity storage hardware; the protocol's consensus assigns block production rights based on the proof-of-space the validator produces in response to challenges, combined with a proof-of-time function that prevents grinding attacks. Application-layer programmability is supported through the Chialisp smart-coin programming model. Launched in March 2021, the protocol has operated continuously since.

Four metric measurements (Q1 2026):

- Adoption: 0.42 million unique holders (entity-adjusted)
- Annual Transactions: 8.6 million (trailing twelve months)
- Annual Transaction Value: \$1.32 billion (trailing twelve months, entity-adjusted)
- Active Developers: 142 unique contributors (trailing twelve months)

Composite Score computation.

$$R_A = 420,000 / 80,000,000 = 0.005250$$

$$R_T = 8,600,000 / 6,090,000,000 = 0.001412$$

$$R_V = 1,320,000,000 / 13,490,000,000,000 = 0.0000978$$

$$R_D = 142 / 905 = 0.1569$$

$$S = (0.70 \times 0.005250) + (0.10 \times 0.001412) + (0.10 \times 0.0000978) + (0.10 \times 0.1569) = 0.01952$$

$$\text{Crypto Fair Value. } 0.01952 \times \$1,983,000,000,000 = \$38,710,000,000$$

$$\text{Fair Coin Price. } \$38,710,000,000 / 22,830,000 = \$1,695$$

Market price (Q1 2026 average). Approximately \$19.

$$\text{Margin of Safety. } (\$1,695 - \$19) / \$1,695 = 98.88\%$$

Trajectory. Margin of Safety has been rising across the trailing four quarters as the developer count has expanded and the protocol's commercial integrations have matured. Q2 2025: 98.4%. Q3 2025: 98.6%. Q4 2025: 98.7%. Q1 2026: 98.9%. The trajectory is rising.

Conviction category. Margin of Safety substantially above the 35% threshold; trajectory rising; measurement inputs fully documented under the framework's reproduction protocol. Active Developers metric is notably high relative to the holder base, reflecting Chia's substantial development ecosystem around the Chialisp smart-coin model. This coin falls into the high-conviction category.

Working conclusion. Chia, at the Q1 2026 measurement date, presents the framework with a developer-rich protocol whose commercial integrations have begun to mature without the

holder base having yet caught up to the development substance. Arizona SB 1649 named-asset list inclusion is one indicator of the protocol's commercial maturation; broader asset-issuance use of the Chialisp programming model in tokenization contexts is another. Proof-of-space-and-time consensus architecture is structurally distinctive and demonstrates that monetary-grade Layer-1 consensus is achievable through resources other than computational hashing or capital-locked staking. High conviction with rising trajectory: that is what the framework reports on Chia at the Q1 2026 measurement date.

V. Coin Report: eCash (XEC)

Protocol. eCash operates a dual-layer architecture: a proof-of-work base layer inherited from Bitcoin Cash (and ultimately from Bitcoin's Nakamoto consensus) and an Avalanche pre-consensus layer that produces sub-second transaction finality alongside the base proof-of-work settlement. This combination supports transaction finality patterns appropriate to medium-of-exchange use while preserving the security properties Bitcoin's proof-of-work establishes. Forked from Bitcoin Cash in 2020, the protocol has operated continuously since.

Four metric measurements (Q1 2026):

- Adoption: 1.18 million unique holders (entity-adjusted)
- Annual Transactions: 89.4 million (trailing twelve months)
- Annual Transaction Value: \$2.31 billion (trailing twelve months, entity-adjusted)
- Active Developers: 51 unique contributors (trailing twelve months)

Composite Score computation.

$$R_A = 1,180,000 / 80,000,000 = 0.01475$$

$$R_T = 89,400,000 / 6,090,000,000 = 0.01468$$

$$R_V = 2,310,000,000 / 13,490,000,000,000 = 0.0001713$$

$$R_D = 51 / 905 = 0.05635$$

$$S = (0.70 \times 0.01475) + (0.10 \times 0.01468) + (0.10 \times 0.0001713) + (0.10 \times 0.05635) = 0.01744$$

$$\text{Crypto Fair Value. } 0.01744 \times \$1,983,000,000,000 = \$34,580,000,000$$

$$\text{Fair Coin Price. } \$34,580,000,000 / 19,820,000,000,000 = \$0.001745$$

Market price (Q1 2026 average). Approximately \$0.000033.

$$\text{Margin of Safety. } (\$0.001745 - \$0.000033) / \$0.001745 = 98.11\%$$

Trajectory. Margin of Safety has been substantially flat across the trailing four quarters, with the high transaction count reflecting the protocol's design emphasis on transactional usage but the holder base and the developer count remaining stable. Q2 2025: 97.9%. Q3 2025: 98.0%. Q4 2025: 98.1%. Q1 2026: 98.1%. The trajectory is flat.

Conviction category. Margin of Safety substantially above the 35% threshold; trajectory flat; measurement inputs fully documented under the framework's reproduction protocol. Annual Transactions count is notably high relative to the holder base, reflecting the protocol's optimization for transactional throughput.

This coin falls into the acceptable-conviction category, with the conviction tempered by the flat trajectory.

Working conclusion. eCash, at the Q1 2026 measurement date, presents the framework with a Nakamoto-consensus-optimized protocol whose transaction count substantially exceeds what the holder base alone would generate, reflecting the protocol's commercial usage in transactional applications. Avalanche pre-consensus layer produces operational properties (sub-second finality) that distinguish eCash from base-layer Bitcoin and from most other Layer-1 Coins. Acceptable conviction with flat trajectory is the framework's reading at the Q1 2026 measurement date.

VI. Coin Report: Cardano (ADA)

Protocol. Cardano operates the Ouroboros family of proof-of-stake consensus protocols, with the protocol's design developed through peer-reviewed academic publication in established cryptography venues. Validation is distributed across thousands of independent stake-pool operators globally. The protocol transitioned to full on-chain governance through the 2025 Plomin hard fork. Application-layer programmability is supported through the Plutus and Aiken smart-contract languages, with base-layer throughput supplemented by the Hydra layer-2 architecture. Launched in September 2017, the protocol has operated continuously since.

Four metric measurements (Q1 2026):

- Adoption: 4.21 million unique holders (entity-adjusted)
- Annual Transactions: 91.7 million (trailing twelve months)
- Annual Transaction Value: \$87.4 billion (trailing twelve months, entity-adjusted)

- Active Developers: 312 unique contributors (trailing twelve months)

Composite Score computation.

$$R_A = 4,210,000 / 80,000,000 = 0.05263$$

$$R_T = 91,700,000 / 6,090,000,000 = 0.01506$$

$$R_V = 87,400,000,000 / 13,490,000,000,000 = 0.006479$$

$$R_D = 312 / 905 = 0.3448$$

$$S = (0.70 \times 0.05263) + (0.10 \times 0.01506) + (0.10 \times 0.006479) + (0.10 \times 0.3448) = 0.07346$$

Crypto Fair Value. $0.07346 \times \$1,983,000,000,000 = \$145,710,000,000$

Fair Coin Price. $\$145,710,000,000 / 35,690,000,000 = \4.08

Market price (Q1 2026 average). Approximately \$0.62.

Margin of Safety. $(\$4.08 - \$0.62) / \$4.08 = 84.80\%$

Trajectory. Margin of Safety has been substantially rising across the trailing four quarters, with the developer count, the holder base, and the transaction volume each contributing to the Composite Score expansion. The Plomin governance transition in 2025 produced measurable improvements in Decentralized Governance evaluation that the four-metric calculation captures through the developer-count expansion the transition drove. Q2 2025: 76.4%. Q3 2025: 79.1%. Q4 2025: 82.3%. Q1 2026: 84.8%. The trajectory is rising substantially.

Conviction category. Margin of Safety substantially above the 35% threshold; trajectory rising substantially; measurement inputs fully documented under the framework's reproduction

protocol. Active Developers count is the largest in the tracked set and reflects the substantial development ecosystem the protocol's peer-reviewed publication discipline has produced over the protocol's eight-year operational history. This coin falls into the high-conviction category.

Working conclusion. Cardano, at the Q1 2026 measurement date, presents the framework with the largest developer ecosystem in the tracked set, a holder base that has expanded substantially across the protocol's operational history, and a governance transition (Plomin) that addresses the centralized-development concerns that had characterized the protocol's earlier years. Peer-reviewed publication discipline distinguishes the protocol's development posture from protocols developed through less formal procedures. Margin of Safety, while substantial, is lower in percentage terms than the other four coins in this chapter, reflecting Cardano's higher market price relative to its Fair Coin Price. High conviction with substantially rising trajectory characterizes the framework's reading on Cardano.

VII. What the Five Reports Demonstrate

These five Coin Reports demonstrate the framework's discrimination in operation. Five different architectural categories produce five different Composite Scores, Crypto Fair Values, and Fair Coin Prices. Each coin's Margin of Safety reflects the specific gap between its measured fundamentals and its current market price; the Margins range from 84.8% (Cardano) to 99.4% (DigiByte). Trajectories vary across the five: Digital Cash and Chia show rising trajectories, Cardano shows substantially rising trajectory, DigiByte shows flat-to-modestly-rising, and eCash shows flat.

Variation is what a discriminating framework produces. A framework that reported the same Margin of Safety for every tracked coin would not be discriminating, and a framework whose outputs did not vary with the coin's measured fundamentals would not be analytical. These five reports illustrate that the framework's arithmetic operates on the coin-specific measurements and produces coin-specific outputs.

These five Coin Reports also illustrate the conviction-category structure. High conviction (Digital Cash, Chia, Cardano), acceptable-to-high conviction (DigiByte), acceptable conviction (eCash): the categories distinguish coins whose readings support different allocation postures. Remaining seven tracked coins, plus DGD, are documented in the Foundation's quarterly Coin Reports the practitioner consults for the corresponding measurements at the same measurement date.

A Coin Report is the framework's per-coin documentation. Each report presents the four-metric measurements, the formula's outputs, the trajectory, the conviction category, and the working conclusion the framework's analysis supports. A practitioner who consults a Coin Report at any measurement date receives the framework's analytical reading on the coin at that date. The framework's allocation discipline operates on the readings the Coin Reports produce.

9. CALIBRATION TESTS

The framework applied to Bitcoin, Ethereum, and XRP

I. Why the Framework Must Test Itself

A framework that values cryptocurrencies must produce outputs the practitioner can trust. That trust depends on the framework's calibration being correct, which requires the framework to demonstrate that its arithmetic produces defensible outputs when applied to coins whose properties are known. Coin Reports on the tracked set demonstrate the framework's outputs on coins the framework values. Calibration tests demonstrate the framework's outputs on coins the framework does not include in the tracked set, specifically to verify that the discriminating power the framework claims is actually present.

This chapter applies the framework to four off-tracked-set readings: Bitcoin at the December 2024 calibration moment, Bitcoin at the October 2025 peak, Ethereum at Q1 2026, and XRP at Q1 2026. Each application produces a Composite Score, a Crypto Fair Value, a Fair Coin Price, and a Margin of Safety against the prevailing market price. The four readings together illustrate the framework's behavior across the full range of measurement outcomes: parity with the Benchmark (Bitcoin at calibration), substantial overvaluation against the Benchmark (Bitcoin at peak), approximate parity at a non-calibration date (Ethereum), and the discriminating output against a coin with structural failure modes (XRP).

Four tests serve as the framework's accountability to its own claims. The framework asserts that the Benchmark is

permanent, that Bitcoin is measured against it on identical terms with every other coin, that the formula discriminates rather than producing uniform outputs, and that the procedure is reproducible. These four tests demonstrate each assertion through the framework's actual application to coins whose properties illustrate what the framework is supposed to detect.

II. Bitcoin at December 2024: The Calibration Moment

Bitcoin's December 2024 state produced the Benchmark. Applying the framework to Bitcoin at the calibration moment is a logical-consistency check: the framework's output should be a Composite Score of exactly 1.0, a Crypto Fair Value of exactly \$1.983 trillion, and a Fair Coin Price that, divided by Bitcoin's December 2024 circulating supply, produces a Fair Coin Price approximately equal to the December 2024 market price of approximately \$100,000 per BTC.

Four metric measurements (December 31, 2024):

- Adoption: 80 million unique holders (Benchmark)
- Annual Transactions: 6.09 billion (Benchmark)
- Annual Transaction Value: \$13.49 trillion (Benchmark)
- Active Developers: 905 unique contributors (Benchmark)

Composite Score computation.

$$R_A = 80,000,000 / 80,000,000 = 1.0000$$

$$R_T = 6,090,000,000 / 6,090,000,000 = 1.0000$$

$$R_V = 13,490,000,000,000 / 13,490,000,000,000 = 1.0000$$

$$R_D = 905 / 905 = 1.0000$$

$$S = (0.70 \times 1.0000) + (0.10 \times 1.0000) + (0.10 \times 1.0000) + (0.10 \times 1.0000) = 1.0000$$

Crypto Fair Value. $1.0000 \times \$1,983,000,000,000 = \$1,983,000,000,000$

Fair Coin Price. $\$1,983,000,000,000 / 19,830,000 = \$100,000$

Market price (December 31, 2024). Approximately \$100,000.

Margin of Safety. $(\$100,000 - \$100,000) / \$100,000 = 0.00\%$

This output is exactly what the calibration anchors the framework to produce. Bitcoin at the calibration moment is at the framework's Fair Coin Price by construction, with a Margin of Safety of zero. The framework's arithmetic, applied to the calibration source at the calibration date, produces the calibration values; this is the logical-consistency property the calibration requires.

This test is not interesting in the sense that the output is predetermined by the calibration procedure. It is consequential in the sense that any framework applied to its own calibration source should produce the calibration values, and frameworks that do not produce their own calibration values are not internally consistent. The framework passes the test.

III. Bitcoin at October 2025: The Peak

Bitcoin's October 2025 peak is the framework's first sustained demonstration of the negative-margin reading. That reading is what the framework reported at the peak measurement date, applying the same arithmetic to Bitcoin that the framework applies to every coin it evaluates.

Four metric measurements (October 31, 2025):

- Adoption: approximately 92 million unique holders (entity-adjusted, trailing twelve months)
- Annual Transactions: approximately 6.31 billion (base layer plus Lightning, trailing twelve months)
- Annual Transaction Value: approximately \$14.82 trillion (trailing twelve months, entity-adjusted)
- Active Developers: approximately 887 unique contributors (trailing twelve months)

These four metrics had moved modestly from the December 2024 calibration values. Adoption had increased by approximately 15%; Annual Transactions by approximately 4%; Annual Transaction Value by approximately 10%; Active Developers had declined by approximately 2%.

Composite Score computation.

$$R_A = 92,000,000 / 80,000,000 = 1.1500$$

$$R_T = 6,310,000,000 / 6,090,000,000 = 1.0361$$

$$R_V = 14,820,000,000,000 / 13,490,000,000,000 = 1.0986$$

$$R_D = 887 / 905 = 0.9801$$

$$S = (0.70 \times 1.1500) + (0.10 \times 1.0361) + (0.10 \times 1.0986) + (0.10 \times 0.9801) = 1.1168$$

$$\text{Crypto Fair Value. } 1.1168 \times \$1,983,000,000,000 = \$2,214,600,000,000$$

$$\text{Fair Coin Price. } \$2,214,600,000,000 / 19,870,000 = \$111,454$$

Market price (October 31, 2025 average). Approximately \$126,000.

Margin of Safety. $(\$111,454 - \$126,000) / \$111,454 = -13.05\%$

Margin of Safety is negative. The framework reports that Bitcoin's market price at the October 2025 peak exceeded the framework's computed Fair Coin Price by approximately 13%. Bitcoin was overvalued against the Benchmark Bitcoin itself had produced ten months earlier.

That reading is what the framework's discriminating power produces. The framework's arithmetic operates on Bitcoin's measured fundamentals against the Benchmark's reference values; when the measured fundamentals improve more slowly than the market price rises, the Composite Score's expansion lags the market capitalization's expansion, and the Margin of Safety turns negative. October 2025 conditions produced this exact pattern: institutional inflows had driven the market price higher faster than the underlying fundamentals expanded, and the framework's calculation reported the gap.

The framework's allocation discipline, applied to Bitcoin at the peak, would have called for not taking new positions and for reducing existing positions if any had been held. CFV CoinFunds did not hold Bitcoin at the peak, by structural decision documented in Chapter 4, and the reading produced no exit decision for the CoinFunds. That reading is documented because it demonstrates the framework's capacity to discriminate against the calibration source itself, on identical terms with every other coin the framework evaluates.

IV. Ethereum at Q1 2026

Ethereum is the second-largest cryptocurrency by market capitalization and the most extensive smart-contract Layer-1 protocol. Ethereum is not on the Foundation's tracked set, for reasons developed in *Cryptocurrency Analysis* Chapter 10 at the

depth that document warrants. Application of the framework to Ethereum at the Q1 2026 measurement date is a calibration test rather than a Coin Report.

Four metric measurements (Q1 2026):

- Adoption: approximately 110 million unique holders (entity-adjusted, trailing twelve months)
- Annual Transactions: approximately 14.2 billion (trailing twelve months, after MEV and failed-transaction adjustments)
- Annual Transaction Value: approximately \$9.7 trillion (trailing twelve months, entity-adjusted)
- Active Developers: approximately 1,840 unique contributors (trailing twelve months)

Composite Score computation.

$$R_A = 110,000,000 / 80,000,000 = 1.3750$$

$$R_T = 14,200,000,000 / 6,090,000,000 = 2.3317$$

$$R_V = 9,700,000,000,000 / 13,490,000,000,000 = 0.7190$$

$$R_D = 1,840 / 905 = 2.0331$$

$$S = (0.70 \times 1.3750) + (0.10 \times 2.3317) + (0.10 \times 0.7190) + (0.10 \times 2.0331) = 1.4709$$

$$\text{Crypto Fair Value. } 1.4709 \times \$1,983,000,000,000 = \$2,916,000,000,000$$

$$\text{Fair Coin Price. } \$2,916,000,000,000 / 121,300,000 = \$24,040$$

Market price (Q1 2026 average). Approximately \$3,820.

$$\text{Margin of Safety. } (\$24,040 - \$3,820) / \$24,040 = 84.11\%$$

Margin of Safety is substantial and positive. The framework's Q1 2026 reading on Ethereum is that the market price was approximately 16% of the framework's computed Fair Coin Price.

That reading requires interpretation, because Ethereum is not on the Foundation's tracked set, and the framework's reading on Ethereum is not an actionable allocation signal under the framework's discipline. It is, however, an accountability test: the framework's reading on Ethereum should not be obviously wrong, and the reading should be consistent with what an honest evaluation of Ethereum's measured fundamentals against the Benchmark would produce.

That reading is defensible on those terms. Ethereum's Adoption count exceeds Bitcoin's, its Annual Transactions count exceeds Bitcoin's by more than a factor of two, its Active Developers count exceeds Bitcoin's by more than a factor of two, and only its Annual Transaction Value is below Bitcoin's. A Composite Score of 1.47 reflects the metric values as measured; the Crypto Fair Value of approximately \$2.92 trillion is what the metrics warrant; and the Fair Coin Price of approximately \$24,040 is the per-coin expression. A market price of approximately \$3,820 reflects the market's voting-machine output, which the framework reports as substantially below the weighing-machine reading.

Ethereum's absence from the tracked set is a Pillar-evaluation outcome rather than a metric-based exclusion. Decentralized Governance considerations (the Ethereum Foundation's influence on protocol upgrades, the staking-concentration patterns under proof-of-stake, the institutional capture risks the protocol has faced) drove the Pillar evaluation toward non-admission. The metric-based reading produces a substantial Margin of Safety, but the Pillar evaluation excludes the coin

from the framework's actionable tracked set. That reading is documented because it demonstrates the framework's discrimination structure: the four-metric reading and the Pillar admission are distinct procedures that produce distinct outputs, and a coin can present a substantial four-metric reading without being admitted to the tracked set.

V. XRP at Q1 2026

XRP is the native coin of the Ripple protocol. The Ripple protocol was developed by Ripple Labs (now Ripple), a venture-backed corporation that retains substantial holdings of XRP and substantial influence over the protocol's continued development. Application of the framework to XRP is a calibration test that illustrates how the framework discriminates against coins with structural Pillar failure modes.

Four metric measurements (Q1 2026):

- Adoption: approximately 4.85 million unique holders (entity-adjusted; the methodology adjusts for the Ripple-controlled escrow positions that would inflate the raw holder count)
- Annual Transactions: approximately 712 million (trailing twelve months, after correction for the on-ledger transactions that do not represent economic activity)
- Annual Transaction Value: approximately \$187 billion (trailing twelve months, entity-adjusted to exclude exchange-internal transfers and Ripple-controlled flows)
- Active Developers: approximately 38 unique contributors (trailing twelve months; the count is heavily concentrated in Ripple-employed developers, which the framework's reading does not adjust for under the four-metric methodology but which the Pillar evaluation does adjust for)

Composite Score computation.

$$R_A = 4,850,000 / 80,000,000 = 0.06063$$

$$R_T = 712,000,000 / 6,090,000,000 = 0.1169$$

$$R_V = 187,000,000,000 / 13,490,000,000,000 = 0.01386$$

$$R_D = 38 / 905 = 0.04199$$

$$S = (0.70 \times 0.06063) + (0.10 \times 0.1169) + (0.10 \times 0.01386) + (0.10 \times 0.04199) = 0.05968$$

$$\text{Crypto Fair Value. } 0.05968 \times \$1,983,000,000,000 = \$118,340,000,000$$

$$\text{Fair Coin Price. } \$118,340,000,000 / 59,400,000,000 = \$1.99$$

Market price (Q1 2026 average). Approximately \$2.18.

$$\text{Margin of Safety. } (\$1.99 - \$2.18) / \$1.99 = -9.55\%$$

Margin of Safety is negative. The framework reports that XRP's market price at the Q1 2026 measurement date exceeded the framework's computed Fair Coin Price by approximately 10%.

That reading is consistent with the framework's Pillar evaluation of XRP. The Pillar failure modes are explicit: Decentralized Governance fails through Ripple's continued control over the protocol's development and through the holder-concentration pattern that the Ripple-controlled escrow positions produce; Free Adoption is borderline, with substantial portions of the holder base concentrated in entities that operate within Ripple's commercial ecosystem rather than as fully independent voluntary holders; and the Active Developers count, while modest in absolute terms, is structurally concentrated in Ripple-employed contributors

rather than distributed across an independent development community.

A four-metric calculation does not capture these Pillar considerations directly. The metrics measure what they measure; the Pillar evaluation interprets what the metrics measure in light of structural properties the four-metric calculation does not encode. XRP's Composite Score against the Benchmark produces a Margin of Safety reading that is negative at the Q1 2026 market price, and the Pillar evaluation produces a non-admission determination. The two outputs are distinct procedures, and both produce results consistent with the framework's overall reading on XRP: a coin the framework does not admit and a coin whose Q1 2026 market price exceeds the framework's computed value.

VI. What the Four Tests Demonstrate

These four calibration tests demonstrate the framework's behavior across the range of outputs the framework can produce.

A Bitcoin December 2024 test produced a Composite Score of exactly 1.0 and a Margin of Safety of exactly zero, demonstrating logical consistency between the calibration procedure and the framework's arithmetic. The framework, applied to its own calibration source at the calibration date, produces the calibration values; this is the property a calibrated framework must possess.

A Bitcoin October 2025 test produced a Composite Score of approximately 1.12 and a Margin of Safety of approximately negative 13%, demonstrating the framework's discriminating power against the calibration source itself. The framework reports that Bitcoin, at the October 2025 peak, was overvalued

against the Benchmark Bitcoin had produced ten months earlier. Discriminating power is the property a framework must possess if its negative-margin readings are to be credible.

An Ethereum Q1 2026 test produced a Composite Score of approximately 1.47 and a Margin of Safety of approximately 84%, demonstrating the framework's evaluation of a Layer-1 Coin whose measured fundamentals substantially exceed the Benchmark on three of the four metrics. That reading is defensible on the metric calculation alone; the coin's non-admission to the tracked set reflects the Pillar evaluation, which is a distinct procedure that operates on structural properties the four-metric calculation does not encode.

An XRP Q1 2026 test produced a Composite Score of approximately 0.06 and a Margin of Safety of approximately negative 10%, demonstrating the framework's evaluation of a coin whose Pillar failure modes produce a non-admission determination and whose market price exceeds the framework's computed value at the measurement date. Two outputs converge on the framework's overall reading: XRP is not admitted to the tracked set, and the market price at the measurement date is above the framework's computed Fair Coin Price.

These four tests together establish that the framework's outputs vary with the coin's measured fundamentals, that the calibration is internally consistent, that the framework discriminates against the calibration source itself when the source is overvalued, and that the four-metric calculation and the Pillar evaluation are distinct procedures that produce complementary outputs. The framework is accountable to these properties.

This chapter has demonstrated the accountability through the actual application of the framework's arithmetic to four specific coins at four specific measurement dates.

Four readings together are the framework auditing itself. Two practitioners running the framework against the same four coins at the same measurement dates, using the same documented inputs, will produce the same four readings. Reproducibility is what makes the framework's outputs trustworthy. Accountability is what the framework's adversarial reviewers can verify against the framework's published procedures.

10. ALLOCATION UNDER FIDUCIARY CONSTRAINT

The framework applied by the institutional allocator

I. The Defensibility Constraint

An institutional allocator does not allocate capital that belongs to them. The capital belongs to a fund's investors, a family's beneficiaries, an endowment's perpetual mission, a pension's eventual retirees, a sovereign-wealth fund's national constituency, or a state treasury's residents. An allocator's relationship to the capital is fiduciary. That relationship is what makes the defensibility constraint operationally binding.

Every position the institutional allocator takes must be defensible to the reporting constituency. Defense may be exercised in an investment-committee meeting, in a quarterly board presentation, in a regulatory filing, in a public-records request from a constituent, or in litigation following a reporting period in which the position underperformed. Defense is not a formality; it is the structural property that distinguishes fiduciary capital allocation from personal capital allocation. A position that cannot be defended in the appropriate forum is a position that cannot, operationally, be held.

This chapter is written to the allocator who has accepted this constraint and is asking what the framework offers them. The framework offers a methodology whose outputs were designed to be defensible at the institutional standard the allocator operates under. Coin Reports document the four-metric measurements through cross-validated procedures the reporting constituency can independently verify.

Margin of Safety thresholds operationalize Graham's central concept in a form the constituency recognizes from equity-investing decades. A reproduction protocol produces the audit trail that survives adversarial review. Falsification commitments state the conditions under which positions are expected to be reduced or exited; those conditions are documented in advance, not asserted after the fact.

This is not a black box. The methodology is published. Inputs are published. Arithmetic is published. Outputs follow from the inputs through procedures the constituency can verify. Defensibility is structural to the methodology, not asserted around it. That is what makes the framework usable by the institutional allocator.

II. The Conviction-Weighted Allocation Schema

The framework's allocation discipline operates on the conviction categories Chapter 7 developed. A coin in the framework's tracked set produces, at any measurement date, a Composite Score, a Margin of Safety, a trajectory across the trailing four quarters, and a conviction category that integrates the point-in-time reading with the trajectory.

An institutional allocator's position-sizing operates against the conviction category. The framework specifies the operational schema as follows:

High-conviction coins (Margin of Safety above the applicable threshold; trajectory rising): position weight in the upper portion of the allocation range the allocator's mandate admits. The allocator's specific mandate dictates the absolute scale; a venture allocator may operate ranges substantially larger than a pension allocator may.

Acceptable-conviction coins (Margin of Safety above the applicable threshold; trajectory flat or modestly rising): position weight in the middle portion of the range.

Low-conviction coins (Margin of Safety above the applicable threshold; trajectory falling): position weight in the lower portion of the range, with reduction or exit triggered by the trajectory analysis if the falling trajectory reflects deteriorating fundamentals rather than convergence with the framework's reading.

Below-threshold coins (Margin of Safety below the applicable threshold regardless of trajectory): no position. The framework's discipline does not admit positions in coins whose Margin of Safety falls below the threshold the measurement-quality classification establishes.

A conviction-weighted schema is not a portfolio-construction algorithm. The allocator's mandate, the constituency's risk tolerance, the broader portfolio's composition, and the operational constraints the allocator faces (concentration limits, liquidity requirements, tax considerations) all shape the absolute position sizes the framework's conviction categories support. The schema names the operational discipline that translates the framework's per-coin readings into the allocator's per-coin positions.

III. The Allocation Range I Recommend

I have been asked, in the years since the framework's first publication, what allocation range I recommend for the institutional allocator's cryptocurrency exposure. That question has two parts: what portion of the broader portfolio the allocator commits to cryptocurrency as an asset class, and how

the cryptocurrency allocation is distributed across the framework's tracked set.

On the first part, the framework does not produce a specific answer. Asset-class allocation depends on the allocator's mandate, the constituency's posture toward digital assets, the macroeconomic conditions the allocator is positioning against, and the operational considerations specific to the allocator's institution. What the framework does produce is the analytical infrastructure that supports the allocation decision: the Coin Reports, the Margin of Safety thresholds, the reproducibility protocol, and the falsification commitments. The allocator decides what portion of the portfolio is appropriate; the framework supplies the inputs the allocator applies.

For institutions with no prior cryptocurrency exposure considering an initial allocation, I recommend the allocation begin modestly relative to the broader portfolio, with the size scaled to the constituency's comfort and the institution's capacity to defend the position through cycles. An initial allocation in the low single-digit percentage range is operationally defensible under the framework's discipline; the allocation can be expanded as the framework's outputs accumulate operational record and as the constituency's posture matures. The framework does not require any specific initial allocation; it supports any allocation that respects the per-coin discipline the conviction-weighted schema specifies.

On the second part, the distribution across the tracked set, the framework's discipline operates through the conviction categories the per-coin readings produce. An allocator deploying capital across the tracked set at a Q1 2026 measurement date, using the Coin Reports the Foundation publishes, would weight the high-conviction coins more heavily than the acceptable-conviction coins, would weight the

acceptable-conviction coins more heavily than the low-conviction coins, and would not take positions in below-threshold coins. Specific weights depend on the allocator's mandate and the broader portfolio considerations; the operational discipline is the conviction-weighted ordering.

IV. DGD as the Structural-Stability Position

Digital Gold (DGD), the Foundation's own coin, occupies a distinct role in the institutional allocator's working schema. DGD is structurally always at its Crypto Fair Value at each level by design; the protocol's calibration to the Benchmark is engineered rather than measured. Margin of Safety on DGD is structurally zero by design, which means the position does not carry the cushion-against-undervaluation property the tracked coins offer.

What DGD offers in compensation is structural stability. The Foundation's commitment is that DGD's value relationship to the Benchmark is engineered to be maintained, and the protocol's design and the Foundation's commercial commitments support the engineering. An allocator holding DGD is holding a position whose value relationship to the framework's reference is structurally stable; the position does not depend on the market converging to the framework's reading, because the protocol is designed so the market and the framework remain in alignment.

Within the institutional allocator's schema, DGD's role is the structurally stable component of the cryptocurrency allocation. Where the tracked coins offer the potential return that comes from the market converging to the framework's reading, DGD offers the stability that comes from the protocol's engineered alignment.

An allocator constructing a cryptocurrency allocation that balances potential return against structural stability holds both: some portion of the allocation in tracked coins at the Margin-of-Safety discipline the conviction categories support, and some portion in DGD at the structural-stability position.

DGD allocation should not dominate the cryptocurrency exposure for most institutional allocators, because the position's structural-stability property does not produce the appreciation profile the tracked coins' Margin-of-Safety positions are positioned for. The DGD allocation supplies stability within the broader cryptocurrency allocation; the broader allocation supplies the return potential the conviction-weighted schema is designed to capture. Ratio between the two depends on the allocator's mandate and the constituency's posture toward potential return relative to structural stability.

V. Documentation for the Reporting Constituency

An institutional allocator reports the cryptocurrency allocation to the reporting constituency at the cadence the institution's reporting calendar requires. The framework supports the documentation through three artifacts that the constituency receives at each reporting cycle.

First is the Coin Report for each held coin. A Coin Report documents the four-metric measurements, the Composite Score, the Crypto Fair Value, the Fair Coin Price, the Margin of Safety, and the trajectory. The constituency receives the same documentation the allocator works from. That documentation is reproducible; a constituency that wishes to verify the framework's outputs against the inputs can do so through the reproduction protocol.

Second is the conviction-category roster. A roster documents which coins held in the allocation fall into which conviction categories at the reporting date, and what the framework's reading on each coin's trajectory has been across the trailing measurement periods. The roster shows the constituency how the allocator's positions translate the framework's per-coin readings into the per-coin positions held.

Third is the falsification-tracking documentation. The framework's falsification commitments specify the conditions under which positions would be reduced or exited. Documentation states which conditions, if any, the framework's measurements have approached at the reporting date, and what the allocator's operational response would be if the conditions were satisfied. The constituency receives the same falsification framework the allocator operates under.

These three artifacts together produce reporting that supports the constituency's evaluation of the allocation. The reporting is not advocacy; it is documentation. The constituency can evaluate the allocation against the framework's published methodology and reach its own conclusions about whether the allocator's positions are appropriate. The framework's structural transparency supports this evaluation rather than substituting for it.

VI. The Rebalancing Cadence

The framework's measurements are quarterly. Coin Reports the Foundation publishes update on a quarterly cycle. An institutional allocator's rebalancing cadence should align with the measurement cycle that produces the inputs the allocator works from.

A quarterly rebalancing cadence is operationally appropriate for most institutional allocators. That cadence is frequent enough to respond to changes in the framework's readings without producing the trading-cost drag that more frequent rebalancing would impose. It is infrequent enough to avoid the noise of intra-quarter price movement that does not reflect fundamental change. The framework's readings are stable across quarters in the sense that the fundamental metrics move slowly; quarterly rebalancing captures the meaningful changes and ignores the operational noise.

Rebalancing decisions at each quarter follow from the framework's outputs. A coin whose conviction category has changed since the prior measurement is a candidate for position adjustment in the direction the category change implies.

A coin whose Margin of Safety has fallen below the applicable threshold is a candidate for exit, with the exit timed against the operational considerations the Foundation's allocation guidance documents. A coin whose Margin of Safety has expanded substantially is a candidate for position expansion, subject to the broader portfolio constraints the allocator's mandate imposes.

Discipline of the quarterly cadence is what the framework supports. An allocator who rebalances on a quarterly cycle, against the framework's published outputs, operates under a discipline whose decisions are documented, reproducible, and defensible to the reporting constituency. The constituency receives the documentation; the documentation states the readings and the responses; the responses follow from the readings through the procedures the framework specifies.

VII. Concentration Limits and Diversification Discipline

The framework's tracked set contains twelve coins plus DGD. A cryptocurrency allocation distributed across all thirteen positions produces a diversification posture different from one concentrated in a subset of the tracked set.

An institutional allocator's diversification discipline operates against the conviction-weighted ordering the framework produces. An allocation concentrated in the high-conviction coins (those whose Margin of Safety and trajectory both support substantial position weight) is consistent with the framework's discipline if the high-conviction roster is large enough to support meaningful diversification at the position-sizing the allocator's mandate admits. An allocation distributed across the full thirteen positions diversifies more broadly but at the cost of including coins whose conviction categories are lower.

I recommend the institutional allocator's cryptocurrency allocation typically include positions in at least the top high-conviction coins at any measurement date, with the exact count varying as the conviction categories shift across measurement cycles. The allocation may additionally include DGD at the structural-stability weight the allocator's schema admits. Below-threshold coins are not held under any circumstance, by the framework's operational discipline.

Concentration considerations the allocator's mandate imposes (typical institutional mandates impose per-position concentration limits that prevent any single coin from exceeding some portion of the broader portfolio) apply to the cryptocurrency allocation as they apply to any other asset class.

The framework's conviction-weighted schema operates within the broader concentration discipline rather than overriding it. A high-conviction coin whose framework-recommended weight would exceed the broader concentration limit is held at the concentration limit rather than at the framework-recommended weight; the broader portfolio discipline is the binding constraint.

VIII. The State Treasurer's Special Case

A specific institutional allocator deserves explicit mention: the state treasurer or public fiduciary operating under statutory authority. Arizona Senate Bill 1649 named that allocator and specified the framework's outputs as authoritative for the named-asset list the statute admits. State treasurers in other jurisdictions operate under different statutory authorities, but the framework's application to public-fiduciary capital follows the same operational pattern.

A state treasurer's defensibility constraint is structurally similar to the private institutional allocator's, with the addition of statutory and political-accountability layers the private allocator does not face. State treasurers' reporting constituency includes the residents of the jurisdiction, the legislative bodies that authorized the allocation, the auditors who examine the treasury's positions, and the political opposition that may scrutinize the positions in subsequent election cycles. Defensibility must extend across all these constituencies.

The framework supports the state treasurer's defensibility through the same documentation it supports the private institutional allocator's defensibility: the Coin Reports, the conviction-category roster, and the falsification-tracking documentation.

State treasurers additionally benefit from the framework's institutional-architecture documentation in *Cryptocurrency Analysis* Chapter 14 and the model legislative agenda in *Cryptocurrency Analysis* Appendix C, which develop the state-treasurer's specific operational considerations at the depth the role requires.

Arizona Senate Bill 1649 procedure is the framework's first sustained state-treasurer application. The state treasurer there operates under statutory authority that names the framework's tracked set, requires quarterly Coin Reports, and incorporates the framework's falsification commitments as statutory triggers. Other state treasurers considering similar legislation should examine the Arizona model as a working precedent and adapt the legislative architecture to their jurisdictions' statutory frameworks.

IX. The Allocator's Working Posture

An institutional allocator who has worked through this chapter has the operational schema. Conviction-weighted allocation across the tracked set; DGD at the structural-stability position; quarterly rebalancing aligned with the Coin Report measurement cycle; concentration limits applied within the cryptocurrency allocation as they apply across the broader portfolio; documentation supplied to the reporting constituency at the institution's reporting cadence; falsification tracking maintained across measurement cycles.

This schema is operational. It produces decisions the allocator can defend in the appropriate forum. It operates against the framework's published outputs, which any reviewer can verify against the published methodology.

Its outputs are not opinions; they are the institutional translation of the framework's per-coin readings into the allocator's per-coin positions, under the discipline the framework specifies.

What the framework offers the institutional allocator is the analytical infrastructure that supports the defensibility constraint. That infrastructure is the methodology, the inputs, the arithmetic, the documentation, and the reproducibility. An allocator supplies the mandate, the constituency relationship, the broader portfolio judgment, and the operational decisions that translate the framework's outputs into the institution's positions. Framework and allocator operate together; neither replaces the other.

I have written this chapter in my own voice at the points where the framework's discipline asks the allocator to commit, because the commitment is the allocator's, made under fiduciary duty, against a framework I am responsible for. The framework's structure supports defensibility; the allocator's structure supports the deployment of capital under that defensibility. Both structures meet at the position the allocator takes. That position is the work the chapter exists to inform.

11. ALLOCATION UNDER PERSONAL-CAPITAL CONSTRAINT

The framework applied by the sophisticated individual

I. The Different Discipline

A sophisticated individual does not report to a board. Capital is the individual's own; the reporting constituency is the individual themselves, in some cases a spouse, and a family or estate the capital is ultimately destined for. The operational constraint that bound the institutional allocator does not apply in the same form. A position that the institutional allocator could not hold because the investment committee would not approve it is a position the individual can hold if the individual's own discipline supports it.

Absence of the institutional constraint is not absence of all constraint. A sophisticated individual has accepted Graham's discipline across the equity-investing decades; that discipline is internalized; the individual reports to a constituency of one, but that constituency has standards. Those standards are produced by years of refined judgment, by the operational record of holding positions through cycles, and by the integration of risk management into the individual's broader financial life. The individual's discipline is not less rigorous than the institutional allocator's; it is differently structured.

A direct statement of what this chapter is. The framework's analytical outputs are the same for the individual as for the institutional allocator. Coin Reports report the same readings; Margin of Safety calculations produce the same numbers; conviction categories integrate the readings the same way.

What differs is the operational deployment of those outputs into actual positions, because the constraints differ. This chapter develops the operational schema for the individual whose discipline is personal rather than institutional.

II. What Personal Capital Admits

A personal-capital allocator can hold positions the institutional allocator cannot. Reasons are operational rather than analytical.

An institutional allocator's positions must be defensible to a reporting constituency at any moment the constituency demands the defense. An individual's positions must be defensible to the individual themselves at any moment the individual chooses to examine them. The two defenses operate on different scales.

A coin whose conviction category is high but whose narrative is unfavorable in the financial press may be impossible for the institutional allocator to hold, because the institutional defense would have to overcome the unfavorable narrative in every reporting forum. The individual, defending the position to themselves, weighs the framework's reading against the narrative without the additional burden of overcoming the narrative for skeptical reviewers who are not party to the framework's discipline.

An institutional allocator's positions must satisfy the institution's regulatory and compliance overlay. The cryptocurrency-allocation universe the institutional allocator can deploy into is constrained by the custody arrangements available to the institution, the regulatory framework the institution operates under, and the operational risk management the institution applies.

An individual operates under a different regulatory framework that, for sophisticated individuals in most jurisdictions, admits broader cryptocurrency exposure with less procedural overhead. The individual can take positions in coins, hold positions through volatility windows, and rebalance on timing the institution cannot.

Tax and estate considerations for the individual differ from the institution's. Personal capital is subject to capital-gains treatment that varies by jurisdiction, holding period, and the individual's specific tax situation. Estate planning is operationally consequential: the individual is managing capital not only for themselves but for the family or beneficiaries the estate is destined for, and the long-term horizon of the personal-capital allocation often exceeds the horizon the institutional allocator operates against.

What personal capital admits is therefore a broader allocation universe, with different timing latitude, under tax and estate considerations the institution does not face. The framework's discipline applies to both, but the operational deployment is different.

III. The Individual's Conviction-Weighted Schema

An individual's conviction-weighted schema operates against the same conviction categories as the institutional allocator's, but with different operational weights and different rebalancing latitude.

High-conviction coins can carry larger position weights than the institutional allocator's schema admits. An individual whose conviction in the framework's reading is high, who has held the discipline through prior cycles, and whose broader financial situation supports the position can take meaningful percentage

exposures to a small number of high-conviction coins. Institutional allocators' concentration limits do not apply in the same form.

Acceptable-conviction coins carry moderate position weights, providing diversification within the tracked-set allocation without dominating the high-conviction positions the schema is anchored on.

Low-conviction coins carry small or no position weights. An individual's latitude to ride out divergence between the framework's reading and the market's reading is greater than the institutional allocator's, but the latitude does not extend to positions whose trajectory the framework reads as deteriorating. Low conviction is a signal to reduce, not to hold through indefinitely.

Below-threshold coins are not held, on the same terms as the institutional allocator's schema. The framework's discipline does not admit positions below the threshold the measurement-quality classification establishes, and the discipline applies to personal capital as it applies to fiduciary capital.

Specific weights within the schema depend on the individual's broader financial situation, the cryptocurrency allocation's role within the broader portfolio, the individual's risk tolerance, and the individual's time horizon. The framework does not specify the weights; it specifies the discipline against which the weights are chosen.

IV. What I Recommend for Personal-Capital Allocators

I have been asked, in the years since the framework's first publication, what I recommend for the sophisticated individual

deploying personal capital into cryptocurrency under the framework's discipline. The recommendation has several parts.

On the cryptocurrency allocation as a fraction of the broader portfolio: I recommend approaching the cryptocurrency allocation as Graham would have approached an unfamiliar asset class: substantial enough to matter if the framework's readings prove correct, modest enough that the position can be held through prolonged divergence without producing the financial pressure that forces premature exit. For most sophisticated individuals, this is a meaningful but not dominant fraction of investable capital. The exact fraction depends on the individual's broader portfolio composition, on the individual's age and time horizon, and on the individual's emotional capacity to hold positions through periods of substantial market disagreement with the framework's reading.

On the concentration within the cryptocurrency allocation: I recommend the individual concentrate the cryptocurrency exposure in the high-conviction coins the framework identifies at the measurement date. An individual's latitude to concentrate is greater than the institutional allocator's, and the conviction-weighted schema rewards concentration when the readings support it. Spreading the cryptocurrency allocation evenly across all twelve tracked coins plus DGD produces lower expected returns than a concentrated allocation in the high-conviction coins, because the allocation includes lower-conviction positions whose readings do not support the same weight.

On the holding horizon: I recommend the individual treat the cryptocurrency allocation as a multi-year position. The framework's discipline operates on the convergence between the market and the framework's reading; that convergence operates on horizons measured in years rather than months.

An individual whose holding horizon is shorter than the framework's discipline supports is operating against the discipline rather than within it. An individual who cannot hold through a year of price movement against the framework's reading should size the allocation smaller, hold a higher proportion in DGD's structurally stable position, or both.

On DGD's role: I recommend the individual hold a portion of the cryptocurrency allocation in DGD as the structural-stability component. That portion depends on the individual's posture toward return potential relative to stability. An individual who values the structural-stability property highly holds a larger DGD position; an individual who values the return potential the tracked coins offer holds a smaller DGD position. No single correct ratio exists; the discipline is what the individual brings to the question, integrating DGD's structural property with the tracked coins' Margin-of-Safety property.

V. The Emotional Discipline

An institutional allocator faces the discipline of holding through divergence as an operational challenge with structural support: the reporting calendar, the investment committee, the documented methodology, the falsification framework. An individual faces the same discipline without the structural support. An individual's emotional discipline is what holds the position when the market disagrees with the framework's reading for prolonged periods.

Operational honesty is required here. Holding a position whose mark-to-market value has declined substantially while the framework reports the underlying fundamentals as improving is operationally hard. An individual experiences the position's underperformance directly, in the form of a portfolio value the

individual reviews periodically and that does not, at the moment of review, reflect what the framework reads. An institutional allocator can defer the emotional experience to the next investment-committee meeting; the individual cannot defer it at all.

Discipline appropriate to the individual is twofold. First is to internalize the framework's reading as the operational basis for the position, not the market price. An individual who marks the position to the framework's Fair Coin Price, while reporting the position at market for tax and accounting purposes, has separated the operational reading from the accounting reading. The framework's Fair Coin Price is what the individual deployed capital against; the market price is what the market currently offers for the position. These are different numbers, and the individual's discipline rests on the first rather than the second.

Second is to limit the frequency of position-value review. An individual who reviews the cryptocurrency allocation daily experiences the market's voting-machine output as a continuous emotional input; an individual who reviews on a quarterly cadence aligned with the framework's measurement cycle experiences the position as integrated with the framework's reading. The framework's discipline operates on quarterly readings; the individual's discipline benefits from review on the same cadence. Daily review does not produce better decisions; it produces more emotional pressure on decisions whose underlying basis updates quarterly.

VI. The Family and Estate Dimension

A personal-capital allocator is not allocating only for themselves. The capital is destined for a family, an estate,

beneficiaries, or charitable purposes the individual has designated. The allocation's long-term performance affects the constituency the estate will eventually serve.

The framework's discipline operates on horizons that often exceed the individual's remaining personal horizon. A coin whose Margin of Safety is substantial, whose trajectory is rising, and whose fundamentals support a multi-year convergence to the framework's reading may produce its returns across a period during which the individual's stewardship of the capital transitions to the next generation or to the designated beneficiaries.

An individual should document the framework's discipline as part of the estate's transition documentation. The next-generation steward or the beneficiary receiving the capital should understand the discipline under which the positions were taken, the measurement cycle the framework operates on, the conviction categories the positions reflect, and the falsification framework that would trigger exit. The framework supplies the documentation; the individual incorporates the documentation into the estate's planning so the discipline survives the stewardship transition.

Foundation quarterly Coin Reports are an asset to the individual's estate planning. The reports continue to be published regardless of which steward is receiving them, and the methodology the reports document is reproducible regardless of the steward's analytical sophistication. A successor steward inheriting cryptocurrency positions whose framework basis is documented can continue the discipline without rebuilding the analytical infrastructure from scratch.

VII. The Rebalancing Cadence for the Individual

An institutional allocator's rebalancing cadence is structured by the institution's reporting calendar. An individual's cadence is structured by the individual's choice.

Recommended cadence is quarterly, aligned with the Coin Report measurement cycle. That cadence has the same operational virtues for the individual as for the institution: frequent enough to respond to changes in the framework's readings, infrequent enough to avoid the trading-cost drag of more aggressive cadences and the emotional noise of more frequent position review.

An individual additionally has the latitude to defer rebalancing when operational considerations support deferral. A coin whose conviction category has shifted modestly from high to acceptable can be held at the existing weight if the broader portfolio is otherwise within its target ranges and if the tax or transaction-cost implications of rebalancing exceed the analytical benefit. An institutional allocator typically does not have the same latitude, because the reporting constituency expects the institutional schema to be applied rigorously at each rebalancing cycle. The individual's discretion is part of the personal-capital schema rather than a departure from discipline.

Discretion has limits. A coin whose conviction category has fallen to low or whose Margin of Safety has crossed below the applicable threshold is not held under the individual's discretion any more than under the institutional schema. The framework's discipline does not admit positions in coins whose readings do not support them, regardless of who is allocating.

VIII. What the Individual Has

A sophisticated individual who has worked through this chapter has the personal-capital application of the framework's discipline. Conviction-weighted allocation, concentrated more heavily than the institutional schema admits, in the high-conviction coins the framework identifies at the measurement date. A DGD position scaled to the individual's posture toward structural stability relative to return potential. A holding horizon measured in years. A rebalancing cadence aligned with the quarterly Coin Reports. An emotional discipline that operates on the framework's Fair Coin Price rather than on the market's voting-machine output. Documentation that supports the stewardship transition to the next generation or to the designated beneficiaries.

Discipline of this kind is the individual's, applied to capital that is the individual's, under standards the individual has internalized across the equity-investing decades. The framework supplies the analytical infrastructure; the individual supplies the operational deployment. The two operate together; neither replaces the other.

I have written this chapter in my own voice because the questions the personal-capital allocator faces are personal in a way the institutional allocator's questions are not. The framework I have built supports the individual's discipline. Foundation commitments to the Coin Reports, to the reproducibility protocol, and to the falsification commitments support the discipline across the multi-year horizons personal-capital allocations operate against. What the individual brings is the operational judgment that translates the framework's outputs into the family's positions. That judgment is what this chapter exists to inform.

12. REBALANCING, EXIT, AND HOLDING

The operational mechanics across measurement cycles

I. What This Chapter Operationalizes

The framework's analytical outputs produce per-coin readings at each quarterly measurement date. Conviction categories integrate the readings into operational classes. The allocation chapters specified how positions are taken initially and how the institutional allocator and the sophisticated individual translate the framework's outputs into deployed capital under their respective constraints. This chapter operationalizes what happens after the positions are taken: how the allocator adds to existing positions, how reductions are timed, how exits are executed, and how positions are held through the measurement cycles when no action is warranted.

Operational mechanics are not separate from the analytical framework; they are the framework expressed as practitioner conduct across measurement cycles. The quarterly cadence aligns the allocator's decisions with the framework's input cycle. Conviction categories at each measurement date update the operational status of each position. Trajectory analysis informs whether changes in the readings reflect convergence or deterioration. Margin of Safety thresholds gate which positions remain operationally actionable. This chapter develops the procedures the allocator applies at each cycle.

This chapter is written to both readers. The institutional allocator operates the procedures under fiduciary defensibility constraints; the sophisticated individual operates the procedures under personal discipline.

Operational steps are common to both; documentation and reporting differ as the allocation chapters specified.

II. The Quarterly Decision Sequence

At each quarterly measurement date, the Foundation publishes updated Coin Reports for the twelve tracked coins. An allocator's decision sequence at each cycle proceeds through four steps.

Step one: receive and review the Coin Reports. An allocator examines the updated readings for each held position: the four-metric measurements at the new measurement date, the Composite Score, the Crypto Fair Value, the Fair Coin Price, the Margin of Safety against the prevailing market price, and the trajectory across the trailing four quarters now including the new measurement.

Step two: classify each held position into the conviction categories. Classification is mechanical: high conviction if the Margin of Safety is above the applicable threshold and the trajectory is rising; acceptable conviction if the Margin of Safety is above threshold and the trajectory is flat or modestly rising; low conviction if the Margin of Safety is above threshold but the trajectory is falling; below-threshold if the Margin of Safety has crossed below the applicable threshold regardless of trajectory.

Step three: compare the new conviction classification to the prior quarter's classification. A position whose conviction category has changed since the prior measurement is a candidate for position adjustment in the direction the change implies. A position whose conviction category is unchanged is a candidate for holding at the existing weight, subject to broader portfolio considerations.

Step four: execute the operational decisions. Decisions follow from the category transitions and from the broader portfolio context. The transitions and their operational consequences are developed in Section IV.

This four-step sequence operates at each quarterly cycle. An allocator who executes the sequence consistently across cycles operates under a discipline whose decisions are documented, reproducible, and defensible. The discipline does not eliminate judgment; it structures the judgment around the framework's inputs.

III. What the Trajectory Tells the Allocator

Trajectory across the trailing four quarters is the primary signal for distinguishing convergence from deterioration. That distinction matters operationally because the two conditions warrant different responses.

A Margin of Safety that has been substantial and is narrowing presents two possibilities. First is convergence: the market is moving toward the framework's reading, and the position is approaching the conditions under which the framework's discipline produces its expected return. Such narrowing reflects the position working as the discipline intends. The allocator's response is to hold the position, monitor the narrowing across subsequent cycles, and evaluate the position for partial reduction or exit when the Margin of Safety approaches zero or turns negative.

Second is deterioration: underlying metrics are weakening, and the Fair Coin Price is falling toward the market price rather than the market price rising toward the Fair Coin Price. Such narrowing reflects the framework's reading of the coin's fundamentals declining.

The allocator's response is different: the position is a candidate for reduction even though the point-in-time Margin of Safety may still be above the applicable threshold, because the trajectory indicates the framework's reading is moving against the position rather than in its favor.

Distinction between the two possibilities is examined through the underlying metrics. A position whose Margin of Safety is narrowing because the market price has risen toward a substantially stable Fair Coin Price is converging. A position whose Margin of Safety is narrowing because the Fair Coin Price has fallen while the market price is stable or rising is deteriorating. A position where both the Fair Coin Price has fallen and the market price has fallen, with the gap narrowing or widening, requires examination of the relative magnitudes of the two movements to determine which condition is producing the trajectory.

Coin Reports document the underlying metrics at each measurement date. An allocator reviewing the trajectory examines the metric movements across the four quarters to identify which of the two conditions is producing the trajectory. That examination is the framework's discipline applied to position management.

IV. The Conviction-Transition Map

Each measurement date can produce a transition in a held position's conviction category. The framework's operational discipline specifies the response to each transition.

High to high (unchanged). No action required by the framework's discipline. The position is held at the existing weight subject to broader portfolio considerations.

If the allocator is operating below the framework-recommended weight for the position (due to prior concentration limits or to the position being a recent addition that has not yet reached target weight), the allocator may add to the position at the current measurement date.

High to acceptable. Modest reduction may be warranted, depending on the underlying-metric examination. If the transition reflects the Margin of Safety narrowing through convergence with the framework's reading, the position is held while monitoring the convergence. If the transition reflects deterioration in the underlying metrics, modest reduction is warranted, with the magnitude of the reduction proportional to the deterioration evidence.

High to low conviction (Margin of Safety above threshold, trajectory turned falling). Material reduction is warranted. The trajectory turn indicates the framework's reading is moving against the position. Reduction does not need to be complete exit; the position may be held at a reduced weight while the trajectory is examined across subsequent cycles to determine whether the falling trajectory reflects deterioration that will continue or a temporary fluctuation that will reverse.

Acceptable to high. Position addition is warranted if the broader portfolio admits expansion at the existing or expanded weight. The transition indicates the framework's reading is moving in the position's favor.

Acceptable to low. Modest reduction is warranted on the same principles as high-to-low transitions. Smaller initial position relative to high-conviction positions means the reduction magnitude is correspondingly smaller in absolute terms.

Low conviction to high. Position addition is warranted, with the addition timed against the broader portfolio considerations. A low-to-high transition is a substantial improvement in the framework's reading and supports operational confidence in the position.

Any conviction to below-threshold. Position exit is required. The framework's discipline does not admit positions whose Margin of Safety has crossed below the applicable threshold. Exit timing follows the operational considerations Section V develops.

Below-threshold to any conviction above threshold. Position re-entry is admitted, with the entry timing and weight following the same operational considerations new positions face. A position that exited under the discipline is re-entered under the discipline; the re-entry is not penalized by the prior exit.

This transition map operates against the conviction categories at each measurement date. An allocator who applies the map consistently across cycles produces position changes whose rationale is documented in the framework's quarterly outputs. The map does not produce decisions; the allocator produces decisions, informed by the map and by the broader portfolio considerations the allocator integrates.

V. The Operational Considerations Around Exit

A position whose Margin of Safety has crossed below the applicable threshold is required to be exited. That exit is not instantaneous; the operational considerations specify how the exit proceeds.

Market depth. A coin's market depth, measured by the size of the position relative to the daily trading volume the coin produces across the venues the allocator operates on, determines the time over which the exit can be executed without producing market impact that worsens the exit price. A position whose size is small relative to daily volume can be exited within a single trading day; a position whose size is substantial relative to daily volume requires execution across multiple days or weeks.

Volatility. A coin whose price exhibits substantial intra-day or intra-week volatility may produce exit prices that vary substantially across the execution window. An allocator's exit strategy may include a volume-weighted average price (VWAP) approach across the execution window, a time-weighted approach distributing the exit across a fixed period, or other algorithmic approaches the allocator's trading infrastructure supports.

Tax considerations. For the personal-capital allocator and for institutional allocators operating under tax-sensitive mandates, the timing of the exit may have material tax consequences. A position whose exit would produce a substantial capital gain may be timed against the tax year, against the recognition of offsetting losses elsewhere in the portfolio, or against the holding-period considerations that distinguish long-term from short-term capital gains.

Re-entry consideration. A position being exited because the Margin of Safety has crossed below the applicable threshold may, at a subsequent measurement date, re-enter the actionable range. An allocator does not pre-commit to re-entry at any specific subsequent date; the framework's discipline operates on the readings each cycle produces, and re-entry is admitted if

and when the readings support it. Exit does not foreclose subsequent positions in the same coin.

Documentation. Exit is documented in the allocator's reporting to the reporting constituency (for the institutional allocator) or in the allocator's personal records (for the sophisticated individual). Documentation states the Margin of Safety reading that triggered the exit, the trajectory across the cycles leading to the exit, and the operational decision the framework's discipline produced. Documentation supports the defensibility constraint the institutional allocator operates under and the personal-discipline integrity the sophisticated individual maintains.

VI. The Discipline of Holding Through Cycles

Most positions across most measurement cycles do not require operational action. A position whose conviction category is unchanged across two or more consecutive cycles, whose Margin of Safety is stable in the range the position was taken at, and whose trajectory does not contradict the original analysis is a position the discipline holds without adjustment.

Discipline of holding is operationally distinctive. The market's voting-machine output may differ substantially from the framework's reading across the holding period. A position's mark-to-market value may move against the original position cost. Financial press may report unfavorable narratives about the coin, the asset class, or the framework itself. An allocator's reporting constituency may question the position. None of these conditions, individually or collectively, warrants exit if the framework's quarterly readings remain consistent with the analysis the position was taken under.

Discipline of this kind rests on the structural property the framework's outputs supply. Four metric measurements are reproducible; the Composite Score is arithmetic; the Margin of Safety is calculated; the trajectory is documented. The framework's reading is what the practitioner positioned against, and the framework's reading at each subsequent measurement date is what the practitioner evaluates the position against. Market intermediate price movements, narrative cycles, analyst opinions, and political conditions that affect cryptocurrency policy are inputs the framework does not measure and that the framework's outputs do not depend on.

An allocator who has internalized this discipline operates without being moved by the inputs the discipline does not depend on. The allocator examines each quarterly Coin Report, applies the conviction classification, examines the trajectory, and either holds the position, adjusts it, or exits it according to what the framework's readings warrant. Decisions are documented; documentation is reproducible; discipline operates across cycles.

VII. Risks the Margin of Safety Does Not Capture

The Margin of Safety is the framework's central risk concept, but it is not the only risk dimension a serious allocator considers. Three additional risks operate alongside the Margin of Safety and warrant explicit treatment within the operational discipline: drawdown risk, correlation risk, and liquidity risk at scale. The framework does not eliminate these risks; it requires the allocator to integrate them with the framework's outputs in ways the conviction categories alone do not.

Drawdown risk is the magnitude of mark-to-market loss a position may exhibit between acquisition and convergence with

the framework's reading. A position with a substantial Margin of Safety is, by the framework's analysis, undervalued relative to its measured fundamentals. That analysis does not specify when the market will close the gap. In the interval between position acquisition and convergence, the market price can move further away from the Fair Coin Price before moving toward it. A position taken at a 70% Margin of Safety can exhibit additional 50% mark-to-market decline if the market continues to vote against the position before the weighing machine asserts itself. The framework's discipline holds the position through such drawdowns if the underlying fundamentals support the holding, but the operational consequence of drawdown is real: a position that has declined substantially affects the allocator's reporting, the constituency's posture, and the allocator's emotional discipline. Drawdown risk is sized at the portfolio level by limiting cryptocurrency exposure to a fraction of total capital the allocator can tolerate seeing decline substantially without forced action. For institutional allocators, that fraction is informed by the reporting constituency's tolerance for the volatility profile cryptocurrency exhibits. For personal-capital allocators, that fraction is informed by the individual's emotional capacity to hold positions through prolonged divergence.

Correlation risk is the property that cryptocurrency positions tend to move in coordination during stress periods, both within the cryptocurrency asset class and against the broader portfolio. A diversified cryptocurrency allocation across the framework's tracked set is not, in correlation terms, a diversified position; the twelve tracked coins plus DGD exhibit positive correlation across substantial portions of the historical record, particularly during downturns when liquidity withdraws from the asset class as a whole. Correlation against the broader portfolio is similarly real: cryptocurrency has, in recent cycles,

exhibited positive correlation with risk assets generally, particularly with high-beta equities and high-yield credit. The diversification benefit cryptocurrency contributes to a broader portfolio is therefore less than a naive analysis of cross-asset correlation would suggest. The framework's allocation discipline does not produce correlation-adjusted position sizing; the allocator must integrate the framework's per-coin readings with the correlation properties of the broader portfolio to produce position sizes that respect the portfolio's overall risk budget.

Liquidity risk at scale is the operational property that an allocator's ability to exit a position is constrained by the venues available and by the position size relative to those venues' depth. Several coins in the framework's tracked set trade on venues whose daily volume cannot absorb large exit orders without substantial price impact. For institutional allocators deploying substantial capital, that liquidity profile is operationally consequential: a position size that would be defensible under the framework's analytical discipline may be impractical under the operational discipline of exit-capacity-at-stress. The framework's Fair Coin Price assumes the candidate coin's market price reflects orderly trading at the venues the price is quoted on; an exit conducted during a stress period at scale may execute at prices substantially below the orderly market quote. Liquidity at scale is therefore an additional constraint on position size, particularly for institutional allocators whose mandate requires exit-readiness within specified time windows.

These three risks are not failures of the framework; they are dimensions the framework's analytical outputs do not measure and that the operational allocator must integrate separately.

The framework's contribution is the Margin of Safety and the conviction category. The allocator's contribution is the integration of those outputs with drawdown tolerance, correlation properties, and liquidity capacity at the operational scale the allocation is deployed at. Both contributions are required for the position to be operationally defensible.

VIII. Adding to Positions

The transition map specified the conditions under which position addition is warranted: transitions to high conviction from any lower category, and additions to high-conviction positions that have not yet reached the framework-recommended weight.

Addition mechanics follow operational considerations similar to the exit mechanics. An addition's size relative to the coin's daily trading volume determines the execution window; the volatility profile may inform the choice of VWAP or time-weighted distribution; tax considerations may favor specific timing within the broader cycle; documentation states the conviction transition or the weight gap that produced the addition.

The framework's discipline supports patient addition. A high-conviction position whose framework-recommended weight exceeds the current position weight does not need to be brought to target in a single execution. Addition can be distributed across multiple measurement cycles if the operational considerations favor distribution, with each cycle's addition documented against the framework's outputs at that cycle's measurement date. An allocator's discretion within the broader portfolio context is operational.

This discipline does not support adding to below-threshold positions. A coin whose Margin of Safety is below the applicable threshold is not held under the framework's discipline regardless of any subjective view about the coin's prospects. The threshold is what the framework's discipline operationalizes; bypassing the threshold for a particular coin compromises the discipline for every position the discipline protects.

IX. The Documentation Trail

Each cycle's decisions produce documentation that accumulates across the measurement history of the cryptocurrency allocation. That documentation serves multiple purposes.

For the institutional allocator, documentation supports the reporting constituency's evaluation of the allocation. The constituency reviews not only the current positions but the history of position changes, the rationale for each change, and the framework's readings that produced the rationale. Accumulating documentation provides the audit trail the constituency relies on across multi-year holding periods.

For the sophisticated individual, documentation supports the estate's transition discipline. A successor steward inheriting the cryptocurrency allocation reviews not only the positions held at the transition but the methodology under which positions were taken, adjusted, and exited across the prior measurement cycles. Accumulating documentation enables the successor to continue the discipline rather than rebuilding the analytical basis from scratch.

For both allocators, documentation supports learning. A measurement cycle's decision that subsequently proves to have been wrong (in the sense that subsequent readings indicate the decision should have been different) is documented alongside the readings that produced it. An allocator who examines the documentation across cycles can identify patterns in the decision-making, refine the operational application of the framework, and improve the discipline's deployment without altering the framework's underlying methodology.

The Foundation supports the documentation through the quarterly Coin Reports, the falsification-tracking documentation, and the methodology publication. An allocator supplies the per-position decision documentation and the cycle-by-cycle rationale. The two together produce the operational record the discipline accumulates.

X. The Practitioner's Working Cadence

A practitioner who has worked through this chapter has the operational cadence the framework's discipline supports. Quarterly review of the Coin Reports; conviction classification at each cycle; trajectory analysis to distinguish convergence from deterioration; conviction-transition mapping to identify the operational consequences; execution of decisions against the broader portfolio considerations; integration with the risks the Margin of Safety does not capture; documentation of decisions and rationale at each cycle; holding through cycles in which no action is warranted; patience with the convergence between the market's voting-machine output and the framework's weighing-machine reading.

That cadence is the framework's discipline in operation. The framework's analytical infrastructure supplies the inputs; the practitioner's operational discipline supplies the deployment of those inputs into actual positions across time. Neither operates alone. The two together produce the allocation the discipline supports.

Quarterly Coin Reports. Conviction classifications. Trajectory analysis. Conviction transitions. Risk integration. Position adjustments. Documentation. Holding. Eight elements are what the practitioner does at each cycle. The cycles accumulate into an operational record. The record is the discipline.

13. REPRODUCIBILITY

The framework's commitment to verifiable outputs

I. Why Reproducibility Is the Framework's Foundation

A framework whose outputs cannot be reproduced by an independent party is not a framework; it is an opinion. That distinction matters because allocators who deploy capital under analytical methodologies cannot rely on opinions held by others. They can rely on procedures that produce the same outputs when applied by anyone, anywhere, to the same inputs. Reproducibility is what makes the framework's outputs trustworthy.

The framework's reproducibility commitment is operational. Every measurement the Foundation publishes is documented through procedures any third party can execute. Every Composite Score, every Crypto Fair Value, every Fair Coin Price, and every Margin of Safety the framework reports is the deterministic output of the documented procedures applied to the documented inputs. Two practitioners working independently from the same inputs, following the same procedures, produce the same outputs to the publication-grade tolerance. This chapter develops the protocol that supports this commitment.

Reproducibility of this kind is not a self-attestation. It is the framework's accountability to adversarial review. An analyst who suspects the framework's outputs are wrong can apply the protocol to the inputs the Foundation publishes and produce their own outputs. If the analyst's outputs differ from the Foundation's, one of two conditions is true: either the analyst's

application of the protocol deviates from the documented procedure, or the Foundation's published outputs deviate from the procedure the Foundation published. In either case, the deviation is detectable, the source of the deviation is identifiable, and the framework's integrity is testable.

II. The Reproduction Protocol

This reproduction protocol consists of four operational components.

Component one: the measurement methodology. The four-metric measurements (Adoption, Annual Transactions, Annual Transaction Value, Active Developers) are produced through documented procedures the Foundation publishes in full. Each metric has a defined measurement methodology that specifies the data sources, the cross-validation requirements, the entity-adjustment procedures, the temporal-window specifications, and the convergence tests the measurement must satisfy. The methodology is published in *Cryptocurrency Analysis* Appendix A at the depth that document supplies; the Volume 2 reader has the operational summary.

Adoption methodology specifies a five-source triangulation across on-chain address counts after entity clustering, exchange-custody contributions from regulated custodians, institutional holdings from issuer regulatory disclosures, industry survey research, and secondary-layer end-user counts from custodial-wallet provider attestations. Those five sources are reconciled through documented procedures that produce a single Adoption figure. The reconciliation is reproducible by any analyst with access to the source data the methodology specifies.

Annual Transactions methodology specifies the trailing-twelve-month transaction count across the base layer and the principal layer-2 infrastructure, with documented adjustments for non-economic activity (exchange internal transfers, contract-internal calls, MEV-related churn) the four-metric calculation excludes. Those exclusions are specified procedurally; the analyst applying the methodology applies the same exclusions and produces the same transaction count to the publication-grade tolerance.

Annual Transaction Value methodology specifies the trailing-twelve-month dollar value of economic transfers, entity-adjusted to exclude exchange-internal transfers and other non-commercial flows. The methodology specifies the dollar-conversion procedure for transactions denominated in the coin's native unit, including the price-source selection and the temporal-window specification for the conversion.

Active Developers methodology specifies the three-layer count of unique contributors to the protocol's essential repositories across the trailing twelve months: the principal protocol implementation, the extended ecosystem repositories, and the application-layer developer layer, with cross-validation against the Electric Capital developer-tracking dataset where available. Those three layers are specified by repository inclusion criteria; the analyst applying the methodology examines the same repositories and produces the same developer count.

Component two: the calibration values. Benchmark values are documented and fixed: Adoption 80 million, Annual Transactions 6.09 billion, Annual Transaction Value \$13.49 trillion, Active Developers 905, market-capitalization anchor \$1.983 trillion, circulating supply 19.83 million. The values do not change. An analyst applying the framework's arithmetic uses these specific values.

Component three: the formula. Composite Score is the weighted sum of the four-metric ratios with weights 0.70, 0.10, 0.10, 0.10. Crypto Fair Value is the Composite Score multiplied by the \$1.983 trillion anchor. Fair Coin Price is the Crypto Fair Value divided by the candidate's circulating supply. Margin of Safety is the percentage gap between the Fair Coin Price and the current market price. The arithmetic is deterministic; given the inputs, the outputs follow.

Component four: the tolerance specification. The framework operates under two tolerance bands. A publication-grade tolerance applies to the documented Coin Reports the Foundation publishes; that band is approximately one percent for the Composite Score and the Crypto Fair Value, reflecting the propagated uncertainty from the underlying metric measurements operated through the documented validation procedures. A measurement-grade tolerance applies to measurements that have satisfied the metric-specific cross-validation requirements but have not yet completed the full publication-grade documentation; that band is approximately fourteen percent for the Composite Score, reflecting the larger measurement uncertainty before full validation.

These four components together constitute the reproduction protocol. An analyst applying the protocol independently to a tracked coin at a given measurement date produces a Composite Score within the publication-grade tolerance of the Foundation's published Composite Score. That match is the protocol's accountability test.

III. What Reproduction Demonstrates

A successful reproduction demonstrates several properties simultaneously.

Measurement procedures are operational. Any analyst applying the documented procedures to the documented data sources produces metric measurements consistent with the Foundation's published measurements. The procedures are not theoretical descriptions of methodology; they are operational instructions that produce specific numerical outputs.

Arithmetic is correct. The Composite Score the analyst computes from the analyst's measurements matches the Foundation's published Composite Score (within the tolerance band). The arithmetic is the simplest part of the framework, but the simplicity is structural: the arithmetic does not introduce error beyond what the input measurements carry.

Tolerance bands are honest. Propagated uncertainty from the metric measurements into the Composite Score and the Crypto Fair Value is documented and quantified. An analyst whose measurements produce a Composite Score within the tolerance band but at a different point in the band is operating within the framework's specified uncertainty; an analyst whose measurements produce a Composite Score outside the band has identified either a methodology divergence (which the analyst can document) or a methodology error (which the framework's adversarial-review posture invites the analyst to report).

Benchmark is permanent. Benchmark values the analyst uses are the same values the Foundation uses, regardless of when the analyst conducts the reproduction. Permanence is what makes reproduction across time possible: an analyst reproducing a Q1 2026 reading in 2030 uses the same Benchmark the Foundation used in Q1 2026.

Reproduction does not demonstrate that the framework's readings are predictive of market outcomes. The framework's

outputs are not forecasts; they are measurements of the gap between the framework's reading of the coin's fundamentals and the market's current price. Reproduction demonstrates that the measurements are reproducible. Whether the market eventually converges to the framework's reading is a separate question that operational results across multiple measurement cycles address; the reproducibility commitment is the precondition for the question being answerable at all.

IV. The Adversarial-Review Posture

This reproducibility is not asserted as a courtesy to interested analysts; it is structured as an invitation to adversarial review. The framework expects that capable analysts will examine the outputs, attempt to reproduce them through the documented protocol, and report deviations the analysts identify.

The Foundation maintains the channels through which deviations are reported. Analysts identifying discrepancies between their independently produced outputs and the Foundation's published outputs are invited to submit the discrepancies for review. Submissions are processed through documented procedures the Foundation publishes; those procedures specify the response timelines, the verification protocol the Foundation applies, the conditions under which the Foundation modifies its published outputs if the discrepancy is upheld, and the documentation the Foundation produces in response.

An adversarial-review posture distinguishes the framework from analytical methodologies that operate as proprietary black boxes. A proprietary methodology may produce outputs the proprietor is unwilling to subject to independent verification; the outputs are accepted on the proprietor's authority, or they

are rejected. The framework's adversarial-review posture replaces authority-based acceptance with procedure-based acceptance. Capital deployed under the framework is deployed under outputs that have survived the adversarial review the framework invites and continues to invite.

An institutional allocator's reporting constituency benefits from the adversarial-review posture directly. That constituency does not need to take the Foundation's published outputs on faith; the constituency can engage independent analysts to apply the protocol and verify the outputs. Verification is what the constituency's fiduciary discipline supports, and the framework supplies the analytical infrastructure that makes the verification operational.

V. What an Independent Verification Looks Like

The Foundation has supported several independent verifications across the framework's operational history. Those verifications followed a consistent pattern.

An institutional reviewer engages an independent analyst with the technical capacity to apply the protocol. The analyst receives the Foundation's published Coin Report for a specific coin at a specific measurement date. The analyst conducts an independent measurement of the four metrics for the same coin at the same date, applying the documented methodology to the documented data sources without relying on the Foundation's measurements as inputs. The analyst produces the four-metric values, computes the Composite Score, computes the Crypto Fair Value, computes the Fair Coin Price, and produces the Margin of Safety against the market price the analyst observes.

Analyst outputs are compared to the Foundation's published outputs. The expected condition is that the analyst's Composite Score falls within the publication-grade tolerance band of the Foundation's published Composite Score. The observed condition, across the verifications the framework has undergone, has consistently been that the analyst's Composite Score falls within the band, the analyst's Crypto Fair Value falls within the band, and the analyst's Margin of Safety reading is consistent with the Foundation's published Margin of Safety reading.

Where an analyst's outputs have differed from the Foundation's published outputs at the edges of the tolerance band, the differences have been traced to specific measurement decisions in the protocol's application: the entity-clustering boundary in the Adoption measurement, the temporal-window choice in the Annual Transactions measurement, the price-source selection in the Annual Transaction Value measurement, the developer-counting period selection. Those differences have been documented; in some cases the analyst's measurement decisions have been incorporated into the Foundation's published methodology as refinements; in other cases the differences have been retained as legitimate methodology choices within the framework's tolerance band.

Independent verifications have not produced findings that contradicted the framework's overall outputs. Reproductions have fallen within the tolerance band the framework specifies, the discriminating outputs (which coins have substantial Margins of Safety, which have negative Margins of Safety, which trajectories are rising or falling) have been consistent across verifications, and the framework's integrity has survived the reviews.

VI. The Practitioner's Verification Posture

A practitioner deploying capital under the framework does not need to conduct an independent verification before each allocation decision. The practitioner relies on the verification infrastructure the framework supports, on the institutional reviewers who have engaged independent analysts, and on the cumulative operational record the framework has produced across multiple measurement cycles.

A practitioner may, however, choose to conduct verification at specific moments. A practitioner whose allocation is approaching a position size that would warrant additional defensibility may engage an independent analyst before expanding the position.

A practitioner whose institution is initiating cryptocurrency allocation under the framework for the first time may benefit from a verification that supports the institutional reporting constituency's confidence in the framework's outputs. A practitioner whose reporting constituency raises specific questions about the framework's readings may use independent verification to produce documented responses.

The framework supports these verifications. The protocol is operational. The methodology is published. The Foundation's review procedures admit submissions of independently produced outputs and respond through documented procedures.

Verification infrastructure is part of the framework's overall methodology, not a service the framework charges for.

VII. Reproducibility Across Time

The framework's reproducibility extends across time. An analyst in 2030 examining a Q1 2026 reading uses the same protocol, the same Benchmark, the same arithmetic, and the same tolerance bands as the analyst in Q1 2026.

A 2030 analyst can verify that the Q1 2026 reading was correct given the inputs available at Q1 2026, can verify that the inputs themselves were measured according to the methodology, and can verify that the subsequent measurement cycles applied the same methodology to the evolving inputs.

Temporal reproducibility is what makes long-horizon allocation under the framework operational. An institutional allocator holding a position across multiple measurement cycles can demonstrate to the reporting constituency that the position was taken under documented procedures, has been examined under the same procedures at each subsequent cycle, and remains operative under the procedures the framework's analytical record documents.

Benchmark permanence is the structural property that makes this possible. A framework whose calibration shifted across time would produce historical outputs that could not be reproduced by analysts working under the current calibration; the framework would lose its capacity to be verified across the multi-year horizons institutional allocation operates against.

Permanence is what supports the temporal reproducibility, and the temporal reproducibility is what supports the long-horizon discipline the allocation chapters develop.

VIII. What Reproducibility Establishes

This framework's reproducibility establishes the structural property the practitioner relies on. The framework's outputs are not opinions held by the Foundation. They are the deterministic results of documented procedures applied to documented inputs, verifiable by any party with the technical capacity to apply the procedures.

Verification is invited rather than feared. The framework's institutional posture welcomes the adversarial review the reproducibility commitment makes possible. Cumulative verifications across the framework's operational history have produced outputs consistent with the framework's published outputs; the consistency is the operational record the framework accumulates.

What reproducibility establishes is the precondition for trust. An analytical methodology that cannot be independently verified cannot be trusted for the multi-year capital deployments fiduciary allocation requires; one whose verification is structural can be trusted to the extent the verifications continue to produce consistent outputs. The framework's commitment is to maintain the verifications, document the methodology, accept the adversarial reviews, and respond through documented procedures to any deviations identified. Reproducibility is the foundation. Procedures the framework documents, inputs the Foundation publishes, and outputs the formula produces are the framework's accountability to anyone who allocates capital under its discipline. That accountability is what the framework offers; the practitioner who allocates under the framework allocates under the accountability the reproducibility maintains.

14. THREE CRITICS

The framework examined through serious objections

I. Why the Critics Get a Chapter

A framework that does not engage its serious critics is not a framework; it is a sales pitch. The framework's commitment to its own methodology requires the framework to state the strongest objections honestly, respond to them through the framework's published procedures, and leave the reader to evaluate the response.

Three critics receive sustained treatment in this chapter. Lyn Alden makes the Bitcoin-maximalist case: Bitcoin is the only cryptocurrency whose monetary properties have been demonstrated at the scale and across the duration that allocation under fiduciary discipline requires; the framework's tracked set of Layer-1 Coins other than Bitcoin extends analytical attention to coins whose properties have not been similarly demonstrated, and the framework's exclusion of Bitcoin from the tracked set is the framework refusing to value the only cryptocurrency that has earned valuation. Nouriel Roubini makes the asset-class-skeptic case: cryptocurrency is not a productive asset, produces no cash flows, has no intrinsic value, and admits no valuation methodology that survives serious analytical examination; the framework's claim to value Layer-1 Coins under Graham-derived methodology is the framework asserting authority over a question the methodology cannot support. Gary Gensler makes the regulatory-skeptic case: most cryptocurrencies are securities under United States law, their issuance and trading have proceeded outside the

regulatory infrastructure that protects investors in securities markets, and an allocation methodology that operates on these instruments operates outside the regulatory framework that fiduciary capital allocation requires.

Each critic is named and quoted from public-record statements the critic has made over the years. The framework's response engages the criticism on its substance rather than dismissing it. The reader evaluates the engagement; the framework does not require the reader's agreement.

Three critics together are not a representative sample of all the framework's critics. They are three serious critics whose positions, taken together, span the spectrum of objections the framework faces. A reader who finds the framework's responses to these three persuasive will find the framework's responses to less-serious critics correspondingly persuasive. A reader who finds the framework's responses unpersuasive should not be persuaded; the framework's commitment to disclosure is what supplies the reader the basis for that judgment.

II. Lyn Alden's Bitcoin-Maximalist Case

Lyn Alden has argued across her published work that Bitcoin occupies a unique position in the cryptocurrency category, that the unique position derives from operational properties no other cryptocurrency has matched, and that allocation methodologies that extend analytical attention beyond Bitcoin extend attention to instruments whose monetary case has not been comparably demonstrated.

That case rests on three observations. Bitcoin's operational history extends approximately seventeen years at the time of this writing, a duration that no other Layer-1 Coin matches.

Bitcoin's holder distribution, while concentrated at the high end, includes the broadest geographic and demographic distribution of any cryptocurrency, with adoption across regulated institutional channels (spot exchange-traded products, public-company treasury allocations, sovereign-allocation discussions) that other Layer-1 Coins have not achieved. Bitcoin's protocol-level conservatism (the slow, careful evolution of the base layer, the resistance to feature additions that would compromise the security model, the explicit choice to support layer-2 architectures rather than expanding base-layer functionality) produces an operational stability the alternative Layer-1 architectures have not yet demonstrated across comparable durations.

The argument's force is structural. A capital allocator deploying under fiduciary discipline against multi-decade horizons requires the operational record that justifies the position. Bitcoin's record is unique. The framework's tracked set of Layer-1 Coins other than Bitcoin extends attention to coins whose operational records are shorter, whose holder distributions are smaller, and whose protocol stability has not been tested at comparable scale. From the Bitcoin-maximalist position, the framework is allocating analytical attention away from the cryptocurrency that has earned it, toward cryptocurrencies that have not.

The framework's response acknowledges the observation while disputing its conclusion. The framework agrees that Bitcoin's operational record is unique, that the institutional infrastructure that has matured around Bitcoin distinguishes it from any other cryptocurrency, and that Bitcoin's December 2024 state was the specific moment at which the four-metric measurement could be made with the confidence the calibration required. The framework's calibration choice reflects exactly this agreement.

The Benchmark is calibrated from Bitcoin precisely because Bitcoin's operational record and institutional infrastructure had matured to the point that the calibration was possible.

What the framework disputes is the conclusion that the operational record's uniqueness in 2024 forecloses analytical attention to other Layer-1 Coins in 2026 and beyond. The four-metric methodology that produced the calibration is reproducible against any Layer-1 Coin whose data can be measured through the documented procedures. The procedures do not privilege Bitcoin; they privilege the measurement infrastructure that supports verifiable inputs. As other Layer-1 Coins' measurement infrastructure has matured across the years following the calibration, the four-metric methodology has become operable for those coins. The framework's tracked set reflects the coins whose architectures address operational properties Bitcoin's design left unsolved (the base-layer throughput limit, the absence of cryptographic privacy, the proof-of-work energy intensity, the holder-concentration pattern, the limited smart-contract expressiveness) and whose Pillar evaluations admit framework valuation.

Bitcoin's exclusion from the tracked set is not a refusal to value Bitcoin; the framework measures Bitcoin against the Benchmark at each measurement date, as Chapter 9 demonstrated. That exclusion reflects the analytical purpose Bitcoin's contribution to the framework served: the calibration, which is permanent and historically singular. No further analytical work on Bitcoin advances the framework's purpose, and the framework's resources are deployed against the Layer-1 Coins whose architectures advance the technical and monetary functions Bitcoin's design did not.

A Model T framing developed in Chapter 1 supplies the framework's posture. The Model T was a remarkable achievement. The Model T is not the standard against which modern automobiles are evaluated. Standards for automobiles are what an automobile must accomplish; modern automobiles are evaluated against those standards. Bitcoin was a remarkable achievement, and the calibration the framework derived from Bitcoin's December 2024 state preserves the achievement permanently. The framework's analytical attention to Layer-1 Coins other than Bitcoin reflects the framework's evaluation of those coins against the standard Bitcoin produced, not the framework's denigration of Bitcoin itself.

Alden's position remains defensible on its own terms. A practitioner who finds Bitcoin's operational record sufficient to support concentrated allocation, who values the protocol-level conservatism the network has demonstrated, and who accepts the framework-based reading that Bitcoin at the October 2025 peak was overvalued against the Benchmark while still preferring to hold Bitcoin through cycles is operating under a discipline the framework can respect even when the framework's tracked set does not include Bitcoin. The disagreement between Alden's position and the framework's tracked-set composition is not a disagreement about Bitcoin's properties; it is a disagreement about whether analytical attention to other Layer-1 Coins is warranted. The framework's answer is yes; Alden's answer is no; the reader evaluates the answers against the operational records the two positions produce over multi-year horizons.

III. Nouriel Roubini's Asset-Class-Skeptic Case

Nouriel Roubini has argued across his published work and public speaking that cryptocurrency is not an investable asset

class, that any methodology claiming to value cryptocurrencies is fundamentally unsound, and that allocation discipline of the kind Graham developed for equity investing does not translate to instruments whose properties differ from equities in the ways cryptocurrencies do.

That case rests on three observations. Cryptocurrencies produce no cash flows; an instrument that produces no cash flows cannot be valued by discounting future cash flows to present value, and the discounting methodology is the foundation of the equity-valuation tradition Graham helped establish. Cryptocurrencies have no intrinsic value in the sense Graham used the term; intrinsic value of an equity is the productive capacity of the underlying business, and cryptocurrencies have no underlying business whose productive capacity can be evaluated. Cryptocurrencies' prices are driven by speculation, sentiment, and the actions of large holders, in ways that resemble historical asset bubbles whose collapses have produced substantial losses for the holders who participated.

This argument's force is also structural. Graham's framework operated on equities because equities had the properties the framework's methodology required: a productive enterprise that generated cash flows, balance sheets that documented the assets supporting the enterprise, and earnings records that produced the inputs the intrinsic-value calculation operated on. Cryptocurrencies lack these properties. An attempt to apply Graham's framework to cryptocurrencies must either assert intrinsic values that the underlying instruments cannot support, or redefine intrinsic value to mean something different from what Graham meant, in which case the framework's claim of continuity with Graham's tradition is compromised.

The framework's response engages the criticism at its core. The framework agrees that cryptocurrencies produce no cash flows in the equity sense, that they have no productive enterprise whose earnings can be evaluated, and that any application of Graham's framework to cryptocurrencies must address what intrinsic value means in the absence of the equity-specific inputs Graham worked from. The framework's response is the bearer-asset translation Chapter 7 developed.

Graham's intrinsic value was a per-share calculation derived from the productive capacity of the underlying business. The framework's Fair Coin Price is a per-coin calculation derived from the network monetary function the underlying protocol produces. These two calculations operate on different inputs because the two asset classes have different fundamentals, but the structural role of the calculation is the same: it produces an analytical reference against which the market price is compared, and the gap between the two is the Margin of Safety that informs the position-sizing.

Fundamentals the framework operates on are the four-metric measurements: Adoption, Annual Transactions, Annual Transaction Value, and Active Developers.

Those four metrics measure what a Layer-1 cryptocurrency does as money: the breadth of voluntary holder participation, the activity rate of medium-of-exchange function, the economic substance moving across the network, and the technical foundation that sustains continued operation. These are not cash flows, but they are the empirical properties through which a monetary instrument's function as money can be evaluated, and they are the properties an analytical methodology values when valuing monetary instruments.

The framework does not claim that the four-metric methodology produces outputs comparable to the equity-valuation methodology Graham operated. Those two methodologies operate on different asset classes with different properties, and the outputs reflect the different inputs. What the framework claims is structural continuity: the framework derives an intrinsic-value calculation from the asset's fundamental properties, compares the calculation to the market price, and produces a Margin of Safety the practitioner applies through the discipline Graham developed. That continuity is in the operational discipline, not in the specific inputs.

Roubini's deeper objection is that monetary instruments without cash flows cannot have intrinsic values derived from any properties, because intrinsic value as a concept requires the cash-flow foundation Graham built on. The framework disputes this. Gold has had intrinsic-value treatment in monetary economics for centuries without producing cash flows. Intrinsic value of gold rests on its monetary properties (durability, divisibility, fungibility, scarcity, recognizability, portability) and the holder population's collective recognition of those properties. A Layer-1 Coin's monetary properties are evaluated similarly, with the four-metric methodology supplying the empirical inputs and the Pillar evaluation supplying the structural assessment. An intrinsic-value claim is not asserted; it is derived from properties that admit measurement and against which the framework's discipline operates.

Whether Roubini finds this response persuasive is a question the framework cannot answer for him. The framework's commitment is to state the response honestly and let the response stand against Roubini's case.

A reader who finds Roubini's case stronger than the framework's response is not the reader the framework is written for; a reader who finds the framework's response addresses Roubini's objections on their substance can proceed under the framework's discipline with the engagement having been transparent.

IV. Gary Gensler's Regulatory-Skeptic Case

Gary Gensler, during his tenure as SEC Chair, argued repeatedly that most cryptocurrencies are securities under United States law, that their issuance and trading have proceeded outside the regulatory infrastructure that protects investors in securities markets, and that the regulatory framework's failure to encompass these instruments creates investor-protection gaps that allocation methodologies cannot address through analytical procedures alone.

That case rests on three observations. The Howey test, established by the Supreme Court in 1946, identifies a security as an investment of money in a common enterprise with profits expected primarily from the efforts of others. Many cryptocurrencies satisfy the Howey test on its face: holders invest money, the cryptocurrencies operate as common enterprises through the protocols' development and the holder communities' coordination, and holders expect profits from the development teams' continued work and from the broader market's adoption of the protocol. Regulatory infrastructure that applies to securities under United States law (registration requirements, disclosure obligations, broker-dealer regulation, exchange registration, anti-fraud enforcement) has not applied to most cryptocurrencies, and the gap exposes investors to risks the regulatory framework was designed to mitigate.

This argument's force is institutional. The regulatory framework exists because pre-regulatory markets produced patterns of investor harm that the framework was designed to address. An allocation methodology that operates on instruments outside the regulatory framework is operating in a market environment whose investor-protection structure has been incompletely built. A methodology may produce defensible analytical outputs, but the broader context within which the analytical outputs are applied is structurally different from the context within which Graham's framework operates for equity securities.

The framework's response begins by acknowledging the regulatory question's seriousness. Cryptocurrency's regulatory status under United States law and under the laws of other jurisdictions is unsettled in ways the equity category's regulatory status is not. That unsettled status is a structural feature of the asset class that any allocation methodology must address. The framework does not pretend the regulatory framework is settled when it is not.

What the framework disputes is the conclusion that the unsettled status forecloses defensible allocation. Several considerations bear on the framework's position.

First, the regulatory framework's actual evolution. SEC approval of Bitcoin spot exchange-traded products in January 2024 was a substantial regulatory development that admitted Bitcoin into the regulated investment-product universe under specific institutional conditions. SEC approval of Ethereum spot exchange-traded products later in 2024 extended the regulatory admission to a second cryptocurrency. Regulatory framework has been changing, and the changes have generally been in the direction of admitting cryptocurrencies into regulated infrastructure rather than excluding them.

The framework's outputs are designed to support allocation that operates within this evolving regulatory infrastructure, not to evade it.

Second, the Howey test's actual application. The framework's Pillar evaluation includes considerations that map directly to Howey factors: Decentralized Governance specifically evaluates whether the protocol's continued operation depends on the efforts of a specific identifiable party, which is the third Howey factor. A coin that fails Decentralized Governance fails the Pillar evaluation and is not admitted to the tracked set. Pillar evaluation is structurally aligned with the Howey analysis the regulatory framework operates under, even where the framework does not assert a regulatory determination it lacks the standing to make.

Third, the Foundation's institutional architecture. *Cryptocurrency Analysis* Chapter 14 documents the Foundation's posture toward regulatory engagement at the depth that volume supplies. The Foundation operates under public-benefit-corporation structure designed to support regulatory engagement rather than evade it. Coin Reports the Foundation publishes are designed to provide the documentation regulated market participants require for fiduciary defensibility. Arizona Senate Bill 1649 statutory architecture provides a working example of how the framework integrates with regulatory authority rather than operating outside it.

Fourth, what the framework explicitly does not do. The framework does not assert that the cryptocurrencies on the tracked set are not securities under United States law; the framework lacks the standing to make such determinations. The framework asserts that the cryptocurrencies on the tracked set are Layer-1 Coins under the architectural criteria the framework defines, that they satisfy the Six Pillars under the

framework's Pillar evaluation, and that they admit valuation under the framework's four-metric methodology. Whether the resulting allocation is admitted under any specific regulatory authority is a question the practitioner deploying under that authority must address through procedures appropriate to the jurisdiction. The framework supplies the analytical infrastructure that supports defensible allocation; the practitioner integrates the framework's outputs with the regulatory infrastructure applicable to the practitioner's circumstances.

Gensler's case is not refuted by the framework's response. Regulatory framework's coverage of cryptocurrency remains incomplete, the Howey-test application to specific cryptocurrencies remains contested, and the investor-protection gaps Gensler identified remain operationally significant. The framework's response is that defensible allocation under the framework's discipline can proceed within the regulatory infrastructure as it exists, with the framework's outputs supporting defensibility under the conditions the practitioner faces, and that the framework's institutional posture supports the regulatory framework's evolution toward fuller coverage rather than evading the evolution. Whether this is sufficient is the reader's judgment.

V. What Engagement With Critics Produces

This chapter has stated three critics' cases at the strength their positions deserve and engaged each on its substance. That engagement does not refute the critics. The engagement establishes that the framework has considered the strongest objections, responded through documented procedures, and accepted the reader's evaluation of the responses.

A reader who finds Alden's case more persuasive than the framework's response can hold concentrated Bitcoin positions under a discipline the framework respects without sharing. A reader who finds Roubini's case more persuasive than the framework's response can decline cryptocurrency allocation entirely under a discipline the framework respects without sharing. A reader who finds Gensler's case more persuasive than the framework's response can defer cryptocurrency allocation until the regulatory framework has evolved further under a discipline the framework respects without sharing. The framework is not written to compel agreement; it is written to support allocation by readers whose evaluation of the framework's outputs and the framework's responses produces operational confidence.

What the framework requires is honest engagement. The critics' positions are stated honestly. The framework's responses are stated honestly. The reader's evaluation produces the operational basis for whatever allocation the reader undertakes. Transparency is what the framework offers to any reader, including readers who proceed under the framework's discipline and readers who do not.

Three serious critics have been engaged at the substance their positions deserve. The framework's responses have not refuted any of them. What the responses have done is establish the engagement on the terms a serious analytical methodology owes its critics. The reader's judgment is what the chapter exists to inform.

15. FALSIFICATION CONDITIONS

The conditions under which the framework would declare itself wrong

I. Why Falsifiability Matters

Karl Popper's contribution to the philosophy of science was that a theory's scientific status depends on its being falsifiable: there must be some observable condition that, if it occurred, would establish the theory as wrong. A theory that cannot be falsified, whose advocates accommodate every observation through ad-hoc adjustments, is not a scientific theory. It may be useful for other purposes; it may produce predictions; it may achieve adherent loyalty. It is not science.

This framework is not a scientific theory in Popper's sense. It is an allocation methodology, which is a different kind of intellectual product. But the structural property Popper identified applies to the framework with operational force. A methodology whose adherents accommodate every adverse outcome through ad-hoc adjustments is not a methodology; it is an advocacy posture. A methodology whose adverse outcomes are specified in advance, whose triggers are documented, and whose responses to triggers are operationally committed is a methodology that can be evaluated, refined, and improved across its operational life.

What the framework's falsification commitments are deserves explicit statement. The framework names the conditions under which the methodology would be declared wrong, the procedures the Foundation would execute in response, and the public statements the Foundation would make.

Those commitments are in writing. They are documented in advance, not asserted after the fact. They are part of the framework's accountability to the practitioners deploying capital under it.

A framework whose proponents accept these commitments and operate by them is a framework that earns the trust the practitioner extends. A framework whose proponents avoid these commitments, soften them when challenged, or rationalize their non-occurrence is a framework that does not. This chapter is the framework's commitment, in the form Popper's discipline requires.

II. The Five Falsification Conditions

The framework operates under five falsification conditions. Each names a specific empirical pattern that, if observed across the framework's operational record, would establish that the methodology requires substantial revision or that the methodology is wrong.

Condition one: Systematic Composite Score divergence from independently produced measurements. If competent independent analysts, applying the reproduction protocol to the tracked coins across multiple measurement cycles, consistently produce Composite Scores outside the publication-grade tolerance band of the Foundation's published Composite Scores, the framework's measurement methodology is wrong. The trigger threshold is a pattern across at least three independent analysts producing systematically divergent Composite Scores on at least three tracked coins across at least two consecutive measurement cycles.

The pattern must be systematic; isolated divergences are within the framework's specified tolerance and do not satisfy the condition.

Condition two: Sustained over-prediction of Fair Coin Prices relative to subsequent market outcomes. If the framework's published Fair Coin Prices systematically exceed the market prices the tracked coins achieve across subsequent multi-year horizons, the framework's valuation methodology is wrong. The trigger threshold is a pattern across at least eight of the twelve tracked coins, where the Fair Coin Price published at a measurement date exceeds the maximum market price achieved by the coin across the following thirty-six months, and where the pattern persists for at least two consecutive measurement-cycle cohorts. The framework's claim is that the Fair Coin Price is a defensible analytical reference; if the market consistently disagrees across multi-year horizons applied to most tracked coins, the analytical reference is not defensible.

Condition three: Sustained under-prediction of Fair Coin Prices relative to subsequent market outcomes. If the framework's published Fair Coin Prices systematically fall short of the market prices the tracked coins achieve across subsequent multi-year horizons, the framework's valuation methodology is also wrong. This mirror-image condition matters because a framework that systematically undervalues is producing analytical outputs whose discriminating power is compromised. The trigger threshold is the same pattern as condition two, with the direction reversed: the Fair Coin Price published at a measurement date falls short of the minimum market price the coin sustains across the following thirty-six months, across at least eight of the twelve tracked coins, persisting for at least two consecutive cohorts.

Condition four: Pillar-failure pattern in admitted coins. If a coin admitted to the tracked set is subsequently determined to have failed one or more of the five gating Pillars at the admission date, the Pillar evaluation is wrong on that coin. A single such determination does not falsify the framework, because the Pillar evaluation operates on inputs that admit some uncertainty. The trigger threshold is a pattern across at least three admitted coins, where the Pillar failure at the admission date is established through independent analysis, and where the failures span at least two distinct Pillars. The pattern indicates the Pillar evaluation methodology is producing systematic admission errors.

Condition five: Catastrophic protocol failure of an admitted coin. If an admitted coin's protocol suffers a catastrophic failure (a fifty-one-percent attack producing reorganization of significant transaction history, a cryptographic compromise of the protocol's core security primitives, an issuance event that violates the protocol's published supply schedule, or an equivalent failure of the protocol's monetary integrity), the framework's Pillar evaluation on that coin was wrong at the admission. A single such failure does not falsify the framework, because all protocols face protocol-level risk and the Pillar evaluation cannot eliminate the risk entirely. The trigger threshold is a pattern across at least two admitted coins suffering catastrophic protocol failure within a five-year window of admission, with the failures attributable to architectural properties the Pillar evaluation should have identified.

Five conditions name specific empirical patterns. Each is observable. Each is documented in advance. Each commits the framework to specific responses if the condition is satisfied.

Those five conditions cover the framework's principal methodological commitments: the measurement methodology, the valuation accuracy in both directions, the Pillar evaluation's admission decisions, and the protocol-level evaluation that underlies the Pillar evaluation.

III. What Happens If a Condition Is Satisfied

The framework's commitments are not limited to naming the falsification conditions. The framework specifies what the Foundation does if a condition is satisfied.

Public acknowledgment. Within ninety days of the Foundation determining a falsification condition has been satisfied, the Foundation publishes a statement identifying the condition, the evidence that established the satisfaction, and the Foundation's assessment of the implications for the framework's methodology. The statement is published through the Foundation's principal communication channels and is not retracted, modified, or softened in any way.

Methodology review. The Foundation initiates a comprehensive review of the methodology element the satisfied condition implicates. That review is conducted by a panel that includes at least three external participants whose credentials and independence the Foundation documents. Panel findings are published in full, including any dissenting views or methodological alternatives the panel identifies.

Protocol-version response. If the methodology review concludes that the methodology requires revision, the revision is implemented through a documented protocol-version transition. That transition preserves the prior protocol version's outputs for the historical measurement cycles those outputs were produced under; the new protocol version applies

prospectively to subsequent measurement cycles. The protocol-version architecture is what allows methodology evolution without compromising the temporal reproducibility the framework's discipline requires.

Capital deployment implications. If the satisfied condition implies that capital deployed under the framework was deployed under flawed analysis, the Foundation states this explicitly. The Foundation does not have direct authority over capital deployed by allocators under the framework, and the framework's responses to falsification do not directly affect existing positions. The Foundation's statement provides the basis for allocators to evaluate their own positions in light of the falsification; the evaluation is the allocator's, and the Foundation does not substitute for the allocator's judgment.

Foundation governance implications. If the falsification condition implicates the Foundation's institutional governance (a measurement error attributable to procedural failures, a Pillar-evaluation error attributable to evaluation-panel composition, a protocol-failure pattern attributable to inadequate technical review), the Foundation's governance procedures are reviewed and reformed as the implicated failures require. Those reforms are documented, the procedural changes are published, and the Foundation's accountability for the underlying failure is established.

Responses are operational. They are not promises that can be deferred indefinitely; they are commitments that operate on documented timelines. Ninety-day public acknowledgment, structured methodology review, protocol-version transition, capital-deployment implications, and governance review constitute the framework's operational accountability to the conditions Popper's discipline requires.

IV. What Falsification Does Not Mean

What falsification under these conditions does not mean deserves explicit treatment, because the framework's critics are likely to characterize any adverse observation as falsification when the framework's commitments specify what counts.

Falsification does not mean a single quarter's Composite Score on a single coin produced by a single analyst falls outside the tolerance band. The framework's measurement methodology produces outputs within a specified tolerance, and isolated divergences within the tolerance, or at the tolerance edges, are operational variation rather than methodology error.

Falsification does not mean the market price of a tracked coin moves substantially against the position the framework supported. The framework's discipline operates on the gap between the framework's reading and the market's voting-machine output; the gap may persist across measurement cycles, and the persistence does not falsify the framework. Condition two specifies the pattern that would establish systematic over-prediction across the tracked set; isolated mark-to-market underperformance is the discipline operating under the conditions Graham identified.

Falsification does not mean a critic publishes an essay arguing the framework is wrong. The framework engages serious critics through documented procedures, as Chapter 14 demonstrated; the engagement does not depend on the critic's assessment of whether the engagement was successful. A critic's continued belief that the framework is wrong, in the absence of the empirical patterns conditions one through five specify, does not satisfy any falsification condition.

Falsification does not mean a regulatory action against a tracked coin or against the cryptocurrency category broadly. Regulatory actions reflect institutional decisions in particular jurisdictions, the political and policy posture of the regulator at the time of the action, and considerations the analytical framework does not measure. A regulatory action may inform the practitioner's allocation decisions, but it does not, in itself, satisfy a falsification condition that the framework's methodology specifies.

This narrowness is structural. The framework's commitments are specific, documented, and observable. That narrowness is what makes the commitments meaningful. A framework whose falsification conditions were so broad that any adverse observation would satisfy them would be a framework whose commitments were promises of failure rather than commitments to accountability.

V. What I Personally Stake

The framework is mine in the sense that I produced it, that I am the public face of it, and that my professional credibility is staked on it. What follows states explicitly what I personally stake.

If the framework satisfies one or more of the five falsification conditions, I commit publicly to the responses Section III specifies, on the timelines Section III specifies. I do not commit to defending the framework in those conditions; I commit to acknowledging the failure, conducting the methodology review, implementing the protocol-version transition or other appropriate response, and stating the capital-deployment implications honestly.

If the framework is established as wrong on a fundamental methodological dimension that the falsification conditions identify, I commit to stating publicly that the framework was wrong, that capital deployed under it was deployed under flawed analysis, and that my own analytical work on the question was inadequate.

That statement will not be conditioned on any defense of intent, any rationalization of the analytical posture, or any softening that would compromise the acknowledgment.

These commitments are made in advance because the value of the commitments depends on their being made in advance. A commitment to acknowledge failure that is made only after the failure has occurred is not a commitment; it is a description of what one would have done if the prior commitment had existed. The prior commitment is what stands now, before any falsification condition has been satisfied.

The reason for the commitment is not generous; it is operational. A practitioner deploying capital under the framework is entrusting the analytical methodology with consequential decisions. The practitioner is entitled to the accountability the framework's methodology supports.

That accountability requires the commitments stated here, because the commitments are what distinguish methodology from advocacy. A practitioner who is entrusting capital to the framework's analytical outputs is entitled to know what the framework's proponents would do if the analytical outputs proved wrong. The statement of what I would do is in writing, in advance.

VI. The Operational Posture of Falsifiability

This framework operates under continuous falsifiability rather than periodic. The five conditions are not evaluated only at scheduled review intervals; they are evaluated continuously as the framework's operational record accumulates.

The Foundation maintains the falsification-tracking documentation as part of the framework's reporting infrastructure. That documentation states which conditions have approached the trigger threshold, which conditions have not been approached, and what the operational state of each condition is at any given measurement date. The documentation is published quarterly alongside the Coin Reports and is part of the broader documentation the institutional allocator's reporting constituency receives.

A condition that has approached but not satisfied the trigger threshold receives explicit attention in the documentation. The framework does not wait for the threshold to be exceeded before reporting that the condition is being approached. Early reporting is part of the framework's commitment to honest accountability; a practitioner who is approaching the framework's discipline at a measurement date when a condition is being approached has the information to integrate the approach into the practitioner's allocation decisions.

Continuous falsifiability is what distinguishes the framework's accountability from periodic-review accountability. A framework that announces compliance with its methodology only at scheduled intervals admits the possibility that interval-end compliance has been engineered to coincide with the interval rather than reflecting genuine operational status.

Continuous reporting is what supports the practitioner's confidence that the framework's accountability is operational, not performative.

VII. What the Falsification Chapter Establishes

This chapter has named five specific conditions under which the framework would be established as wrong, has specified the operational responses to each condition's satisfaction, has distinguished what falsification means from what it does not mean, and has stated what I personally stake on the framework's accountability.

Those commitments are documented. They are observable. They are operational. They are what the framework offers the practitioner who is asked to deploy capital under the framework's discipline. A framework that cannot be falsified through specific conditions cannot be trusted across multi-year horizons; a framework whose falsifiability is specified can be trusted to the extent the framework continues to operate within the conditions the falsifiability specifies.

Trust is not a function of charisma, the Foundation's marketing, or the framework's recent operational results. Trust is a function of the falsifiability conditions remaining unsatisfied across cycles, the documentation supporting the unsatisfied status, and the framework's procedures admitting the verification of the unsatisfied status by parties external to the Foundation. Trust is built, cycle by cycle, through the framework's continued operation within the conditions it specifies. It is not asserted; it is earned.

I have written this chapter in my own voice because the commitments are mine.

If the framework is established as wrong by the conditions the chapter specifies, the acknowledgment will follow on the terms the chapter specifies, on the timelines the chapter specifies, and through the procedures the chapter specifies.

That commitment is what the framework offers the practitioner who has worked through the book. The practitioner evaluates the commitment, evaluates the framework, and decides whether to deploy under the discipline the framework supports.

Five conditions. Five responses. One framework. One commitment. The framework operates under the conditions the chapter has stated, and the framework continues to operate as long as the conditions remain unsatisfied. The work is the framework. The accountability is the chapter. The trust is the practitioner's to extend or withhold.

Acknowledgments

This framework is not the author's alone. Its development across the years preceding the December 2024 calibration, and its continued operation across the cycles that have followed, depended on the work of many.

To the Digital Gold Foundation: every member of the Foundation whose work on the Benchmark calibration, the measurement methodology, the Pillar evaluation procedures, the quarterly Coin Reports, and the verification infrastructure made the framework operational. Their contributions are the framework's institutional substance. The author's name appears on the volume; the framework's institutional capacity belongs to the Foundation.

To the independent analysts who have applied the reproduction protocol across the cycles since the framework's first publication, whose verification submissions identified methodology refinements that the protocol-version architecture has admitted in subsequent versions, and whose continued adversarial review is what makes the framework's outputs trustworthy: the framework's reproducibility commitment depends on the analysts continuing to engage it. The author thanks them for treating the framework as worth their analytical attention.

To the institutional allocators, the state treasurers, the family-office principals, the endowment officers, and the sophisticated individuals who have deployed capital under the framework's discipline across the operational cycles, whose engagement with the methodology has supported the framework's continued development, and whose questions have produced the refinements that distinguish the present volume from the first comprehensive presentation in *Cryptocurrency Analysis*: the framework was built to be useful, and their use of it has been the test that establishes whether the build succeeded.

To Arizona Senate Bill 1649's legislative architects and to the state treasurer operating the resulting statutory authority: the first sustained state-treasurer application of the framework demonstrated that the framework integrates with regulatory authority rather than operating outside it. That integration is what supports the framework's broader institutional credibility. The author thanks the legislative architects for designing the statutory architecture and the state treasurer for operating it under the conditions a public fiduciary faces.

To the critics whose serious objections are engaged in Chapter 14, and to the broader community of skeptics whose work has tested the framework's claims: the framework's accountability

depends on engagement with its serious critics rather than on dismissal of them. The author thanks the critics for treating the framework as worth criticizing.

To the practitioners of the Austrian monetary tradition whose work supplies the conceptual foundation the Six Pillars distill: Carl Menger, Ludwig von Mises, Friedrich Hayek, Murray Rothbard, and the broader scholarly community that has continued their work. The author thanks the tradition for supplying the framework the intellectual coordinates a monetary methodology requires.

To Benjamin Graham, who is not present to thank but whose discipline is what the present volume translates: this book exists because Graham's framework was rigorous enough to admit translation to an asset class he did not live to see. The translation is the author's; the discipline being translated is Graham's. The author's debt to Graham is what every page of this volume reflects.

To the readers who have worked through this book and who proceed under the framework's discipline: the framework operates because practitioners deploy capital under it. The author thanks the readers for trusting the framework with the consequential decisions capital allocation entails. The framework's continued accountability to that trust is what the present volume and the volumes to come will continue to maintain.

The framework's mistakes are the author's. Its successes are the work of all who have engaged it.

John Wright Gotts

Digital Gold Foundation

APPENDIX A: BENCHMARK SPECIFICATION

The operational specification of the Digital Gold Standard Benchmark

A.1 Purpose

This appendix supplies the technical specification of the Digital Gold Standard Benchmark at the operational depth a practitioner applying the framework requires. Body chapters introduced the Benchmark's values and the calibration moment that produced them; this appendix documents the procedures the Foundation operates under, the data sources the procedures depend on, the cross-validation requirements the framework enforces, and the protocol-version architecture that admits methodology evolution across versions.

A reader who is conducting independent verification under Chapter 13's reproduction protocol uses this appendix as the methodology reference. A reader who is reviewing the Foundation's published outputs uses this appendix as the audit-trail reference. A reader who is consulting the framework for general allocation discipline does not need the appendix's depth; the body chapters supply the operational summary the allocation discipline operates against.

A.2 The Six Benchmark Values

The Benchmark consists of six values. Four are the metric values that produce the Composite Score; the fifth is the market-capitalization anchor that converts the Composite Score to the Crypto Fair Value; the sixth is the contextual circulating supply that supports the calibration's interpretation.

Adoption (A_B): 80,000,000 unique global holders. This figure represents the entity-adjusted count of unique holders of bitcoin at the December 31, 2024 measurement date, produced through the five-source triangulation methodology Section A.4 develops.

Annual Transactions (T_B): 6,090,000,000 transactions. This figure represents the trailing-twelve-month count of Bitcoin transactions across the base layer and the Lightning Network as of December 31, 2024, with the non-economic-activity adjustments Section A.5 develops.

Annual Transaction Value (V_B): \$13,490,000,000,000. This figure represents the trailing-twelve-month United States dollar value of economic transfers across Bitcoin and Lightning as of December 31, 2024, entity-adjusted under the procedures Section A.6 develops.

Active Developers (D_B): 905 unique contributors. This figure represents the trailing-twelve-month count of unique contributors to Bitcoin's essential repositories as of December 31, 2024, produced through the three-layer methodology Section A.7 develops, with cross-validation against the Electric Capital developer-tracking dataset.

Market-capitalization anchor (M_B): \$1,983,000,000,000. This figure represents Bitcoin's market capitalization at December 31, 2024, computed as the closing price across the principal regulated exchanges multiplied by the circulating supply at that date.

Circulating supply (S_B): 19,830,000 BTC. This figure represents Bitcoin's circulating supply at December 31, 2024, computed as the protocol-issued supply less the verifiably lost

and provably burned supply identifiable at the measurement date.

These six values are fixed. They do not change as Bitcoin's measured values evolve in subsequent measurement cycles. They are the calibration reference against which subsequent measurements are conducted.

A.3 The Calibration Date

The calibration date is December 31, 2024, 23:59:59 UTC. That date is the close of the calendar year 2024, which was the operational moment at which the four-metric measurements could be conducted with the cross-validation the Benchmark requires.

Institutional infrastructure that supports the measurements (regulated exchange-traded products, audited custodians, entity-adjusted on-chain analytics, cross-validated developer accounting) had matured to the operational state the calibration could be made against.

This date is a single point. Trailing-twelve-month metrics that the Annual Transactions, Annual Transaction Value, and Active Developers values are computed against cover the period from January 1, 2024 through December 31, 2024. The point-in-time Adoption and Market-capitalization and Circulating-supply values are measured at the close of December 31, 2024.

Calibration is a single historical act. Subsequent measurements are conducted at subsequent calendar quarter-end dates and produce the per-coin readings the Foundation publishes in the quarterly Coin Reports; the calibration itself is not repeated and is not modified.

A.4 Adoption Measurement Methodology

Adoption is the entity-adjusted count of unique global holders of the coin at the measurement date. This metric is produced through five-source triangulation, with the five sources reconciled through documented procedures.

Source one: on-chain address count after entity clustering.

Raw on-chain address counts overstate holder counts substantially, because a single holder typically operates multiple addresses across wallets, exchanges, and custodial arrangements. Entity-clustering procedures apply heuristics that identify addresses controlled by the same entity (common-input clustering, change-address detection, behavioral clustering across temporal windows) to produce an entity-adjusted address count. The Foundation uses the principal entity-clustering analytics providers' outputs as inputs to the source-one count.

Source two: exchange-custody contributions from regulated custodians.

Regulated digital-asset exchanges and custodians report aggregate customer holdings under regulatory disclosure obligations. Customer counts at the principal regulated venues (Coinbase, Kraken, Gemini, Bitstamp, and equivalent venues in other jurisdictions) supply customer-count contributions that the entity-clustering procedures cannot resolve from on-chain data alone. The source-two count is the sum of unique customer counts across the venues, adjusted for cross-venue overlap through the procedures the venues' disclosures support.

Source three: institutional holdings from issuer regulatory disclosures.

Spot exchange-traded product issuers report their holdings through prospectuses, monthly holdings disclosures,

and other regulatory filings. Aggregate holdings across the issuers represent capital from holders whose individual identities are not separately disclosed but whose existence is implicit in the issuers' filings. The source-three count is derived from the issuers' assets-under-management figures, holder concentration disclosures, and prospectus-level identifications.

Source four: industry surveys. Independent research organizations conduct survey-based estimates of cryptocurrency holder populations through demographic-representative survey methodology. Survey results, while subject to the methodological limitations all survey research carries, supply a contribution to the triangulation that is not derivable from on-chain or institutional sources. The source-four count is produced by integrating the principal survey research outputs through documented procedures.

Source five: secondary-layer end-user counts. For coins with substantial layer-2 infrastructure (Lightning Network for Bitcoin, sidechain integrations for several tracked coins), the end-user counts from custodial-wallet providers operating on the secondary layer contribute to the holder count. The source-five count is derived from the layer-2 wallet-provider attestations the providers publish.

Five sources are reconciled through documented procedures that produce a single Adoption figure. Reconciliation applies entity-disambiguation procedures to identify cross-source duplications, applies confidence-weighting to the source contributions based on the documented validation each source supports, and produces the final entity-adjusted Adoption count. Specific parameters are documented in the Foundation's measurement-methodology publications; the procedures are reproducible by independent analysts with access to the source data.

Cross-validation tolerance for the Adoption measurement is approximately five percent at the publication-grade level: an independent analyst applying the procedures should produce an Adoption figure within five percent of the Foundation's published figure for the coin at the measurement date.

A.5 Annual Transactions Measurement Methodology

Annual Transactions is the trailing-twelve-month count of transactions processed across the coin's base layer and the principal layer-2 infrastructure, with adjustments for non-economic activity that the metric's monetary-function purpose requires excluding.

Base-layer transaction count. Raw on-chain transaction count for the trailing twelve months is the starting figure. The count is observable on the chain and is not subject to estimation uncertainty.

Layer-2 transaction count. For coins with substantial layer-2 infrastructure, the transaction count on the layer-2 protocols is added to the base-layer count. Layer-2 transaction counts are derivable from the protocols' observable state, with the layer-2 wallet-provider attestations supplying confirmation where the protocol's observability admits.

Non-economic-activity adjustments. Aggregate transaction count includes transactions that do not represent economic activity in the sense the metric is designed to measure. Adjustments identify and exclude: exchange-internal consolidation and rebalancing transactions, custodial-provider housekeeping transactions, transactions that resolve to the same entity through entity-clustering, transactions whose values fall below the documented dust-threshold, and contract-internal operations that do not represent transfers between

distinct holders. Those adjustments are documented in the Foundation's measurement methodology and are applied uniformly across measurement cycles to produce the adjusted Annual Transactions count.

Cross-validation tolerance for the Annual Transactions measurement is approximately two percent at the publication-grade level.

A.6 Annual Transaction Value Measurement Methodology

Annual Transaction Value is the trailing-twelve-month United States dollar value of economic transfers across the coin's base layer and the principal layer-2 infrastructure, entity-adjusted and value-converted under documented procedures.

Value computation. Each transaction in the adjusted Annual Transactions count is valued at the United States dollar price of the coin at the transaction's block-time, computed from the volume-weighted average price across the principal regulated exchanges within the relevant temporal window. Valuation methodology produces a dollar value for each transaction; the sum across the trailing twelve months produces the raw Annual Transaction Value.

Entity adjustments. Raw Annual Transaction Value is adjusted for the non-economic activity excluded from the Annual Transactions count. Exclusions are conducted at the per-transaction level: a transaction excluded from the count is also excluded from the value. The resulting adjusted Annual Transaction Value reflects the dollar value of transactions that represent economic transfers between distinct holders.

Cross-validation tolerance for the Annual Transaction Value measurement is approximately five percent at the publication-

grade level, reflecting the additional uncertainty the per-transaction price-conversion introduces.

A.7 Active Developers Measurement Methodology

Active Developers is the trailing-twelve-month count of unique contributors to the coin's essential repositories, produced through a three-layer methodology with cross-validation against the Electric Capital developer-tracking dataset.

Layer one: principal protocol implementation. Repositories that implement the protocol's core consensus mechanism, the principal client software, and the protocol-specification documentation constitute the principal protocol implementation. Contributors to these repositories across the trailing twelve months supply the layer-one count.

Layer two: extended ecosystem repositories. Repositories that supply infrastructure operating directly on the protocol, including block explorers, indexing services, the protocol's wallet implementations, and the protocol's bridge architectures, constitute the extended ecosystem. Contributors to these repositories across the trailing twelve months supply the layer-two count.

Layer three: application-layer developer layer. Repositories that implement applications operating on the protocol but that do not modify the protocol itself constitute the application layer. Contributors to these repositories across the trailing twelve months supply the layer-three count.

Cross-validation. Electric Capital developer-tracking dataset supplies an independent count of cryptocurrency developers across the trailing twelve months under a documented

methodology. The Foundation's three-layer count is cross-validated against the Electric Capital count for coins where the dataset includes the coin; the cross-validation tolerance is approximately ten percent, reflecting the methodology differences between the two counting approaches.

Cross-validation tolerance for the Active Developers measurement is approximately ten percent at the publication-grade level.

A.8 Tolerance Propagation

Tolerance bands on the four-metric measurements propagate into the Composite Score and the Crypto Fair Value through the formula's arithmetic. Propagated tolerance bands are:

Composite Score: approximately one percent at the publication-grade level. Adoption's weight of 0.70 means the Adoption measurement's five-percent tolerance propagates as approximately 3.5% of the Composite Score's central value, but the other three metrics' tolerances partially offset rather than compounding, producing a net Composite Score tolerance of approximately one percent for tracked coins under fully validated measurements.

Crypto Fair Value: approximately one percent at the publication-grade level. Market-capitalization anchor multiplier does not introduce uncertainty (the anchor is a fixed Benchmark value); the propagated tolerance is identical to the Composite Score tolerance.

Fair Coin Price: approximately one percent at the publication-grade level. Circulating-supply divisor introduces minimal additional uncertainty for coins with stable supply at the measurement date.

Margin of Safety: approximately one percent of the Fair Coin Price at the publication-grade level. Margin of Safety propagates the Fair Coin Price tolerance directly.

For measurements where one or more inputs operate at the measurement-grade level rather than the publication-grade level, the propagated tolerance is substantially wider. A fourteen-percent measurement-grade tolerance on the Composite Score is what produces the fifty-percent Margin of Safety threshold the framework's allocation discipline applies to coins with any unreliable input. The wider cushion absorbs the wider measurement uncertainty.

A.9 Protocol-Version Architecture

The framework operates under a protocol-version architecture that admits methodology evolution while preserving the temporal reproducibility historical readings require.

A protocol version specifies the methodology in operational effect for a defined period. The current protocol version's methodology is what the Foundation applies in producing the current Coin Reports; historical measurements were conducted under the protocol versions in effect at the historical measurement dates. An analyst reproducing a historical reading uses the methodology of the protocol version in effect at the historical date.

A protocol-version transition occurs through a documented procedure that names the methodology changes, justifies the changes, and specifies the transition date. That transition does not modify measurements made under prior versions; it specifies how subsequent measurements are made.

The architecture preserves the integrity of historical outputs while admitting the methodology to evolve as the framework’s operational record identifies refinements.

Benchmark values are protocol-version-invariant. The six values (Adoption, Annual Transactions, Annual Transaction Value, Active Developers, market-capitalization anchor, circulating supply) are fixed at the December 31, 2024 calibration and do not change across protocol versions. Methodology that produces measurements against the Benchmark may evolve across versions; the Benchmark itself does not.

A.10 Verification Channels

The Foundation maintains channels through which independent analysts can submit verification outputs and through which the Foundation responds to verification submissions.

Methodology publications. Full measurement methodology is published in the Foundation’s *Cryptocurrency Analysis* documentation and in the per-cycle Coin Reports. Publications are stable references that admit independent application of the methodology to the data sources the methodology specifies.

Source-data references. The Foundation publishes references to the data sources the measurements operate on, including the principal entity-clustering analytics providers, the regulated-custodian disclosure references, the institutional-holdings disclosure references, the survey-research references, and the layer-2 attestation references. Independent analysts can access the same source data the Foundation’s measurements operate on.

Submission procedures. Independent analysts identifying discrepancies between their reproduced outputs and the Foundation’s published outputs can submit the discrepancies through documented procedures.

The Foundation responds to submissions within published timelines, evaluates the discrepancies against the methodology and against the source data, and either confirms the Foundation’s published outputs, modifies the published outputs in response to the verification, or identifies methodology refinements that the protocol-version architecture admits in subsequent versions.

Verification channels are operational. The framework’s integrity depends on the verifications continuing, on the discrepancies being submitted, and on the Foundation responding through the documented procedures. Those procedures are part of the framework’s accountability to the practitioners deploying capital under it.

A.11 What the Appendix Supplies

This appendix has supplied the operational specification of the Benchmark at the depth a practitioner conducting verification or institutional review requires. Six Benchmark values are stated; the calibration date is documented; the measurement methodology for each of the four metrics is specified; the tolerance bands are quantified; the protocol-version architecture is named; the verification channels are identified.

A reader who has worked through this appendix has the technical reference the framework’s reproduction protocol operates against. A reader conducting verification under Chapter 13’s reproduction protocol applies this appendix as the methodology specification.

A reader reviewing the Foundation's outputs uses this appendix as the audit-trail reference. A reader's confidence in the framework's outputs depends on the appendix's specifications being operational, the verifications being conducted, and the framework continuing to operate within the methodology the appendix documents.

APPENDIX B: REPRODUCTION PROTOCOL

The procedure for independent verification of Coin Reports

B.1 Purpose

This appendix supplies the step-by-step procedure an independent analyst applies to verify a published Coin Report. That procedure operationalizes the reproducibility commitment Chapter 13 specified: any party with the technical capacity to apply the procedure can produce, from the documented data sources, outputs consistent with the Foundation's published outputs within the tolerance bands the framework specifies.

This protocol is operational rather than aspirational. It names the data sources, specifies the application sequence, identifies the comparison procedure, and documents the submission channel for discrepancies. An analyst conducting verification works through the protocol sequentially; the protocol produces a complete reproduction of any Coin Report at the publication-grade tolerance.

B.2 Prerequisites

An analyst conducting verification requires the following inputs.

The Coin Report being verified. The Foundation publishes Coin Reports on a quarterly cycle; the analyst identifies the specific Coin Report (coin name, measurement date) that the verification will operate on.

The framework’s published methodology. Measurement methodology for each of the four metrics is specified in Appendix A and at greater depth in *Cryptocurrency Analysis*. An analyst applies the methodology in effect at the measurement date being verified, which the Foundation identifies through the protocol-version reference on the Coin Report.

Access to the data sources. Methodology specifies data sources for each metric. An analyst accesses the same data sources the Foundation operates against, through the public references the methodology documents or through equivalent providers the methodology admits.

Technical capacity. Verification requires capacity for on-chain analytics (address clustering, transaction extraction, value computation), regulatory-filing review (institutional disclosure parsing, custodian attestation review), survey-research analysis (where applicable), and developer-repository accounting (commit history extraction, contributor identification, repository categorization). That capacity is typical of cryptocurrency-analytics teams, accounting firms with digital-asset specialization, or independent analysts with the relevant technical background.

B.3 The Four Metric Reproductions

Verification reproduces each of the four-metric measurements through documented procedures.

B.3.1 Adoption Reproduction

Step one. Obtain the entity-clustered address count for the coin at the measurement date from the principal entity-clustering analytics providers the methodology references. Principal providers in the current protocol version are

Chainalysis, Glassnode, and Coin Metrics; the analyst may use one provider or multiple, with multi-provider use producing the analyst's source-one count through documented reconciliation.

Step two. Obtain the exchange-custody contribution counts from regulated custodians' published disclosures. Custodian aggregate customer counts at the measurement date supply the source-two count. Cross-venue overlap is adjusted through the procedures the methodology specifies.

Step three. Obtain the institutional-holdings counts from spot exchange-traded product issuers' regulatory filings at the measurement date. Aggregate assets-under-management figures, the prospectus-level holder identifications, and the holder concentration disclosures produce the source-three count.

Step four. Obtain the industry survey research outputs that cover the coin at the measurement date or at the nearest available reference date. Principal providers in the current protocol version include the Cambridge Centre for Alternative Finance, the Federal Reserve's Survey of Consumer Finances digital-asset module, and equivalent research organizations in other jurisdictions.

Step five. For coins with substantial layer-2 infrastructure, obtain the secondary-layer end-user counts from the layer-2 wallet-provider attestations.

Step six. Reconcile the five sources through the documented procedures. Reconciliation applies entity-disambiguation to identify cross-source duplications, applies confidence-weighting based on the validation each source supports, and produces the analyst's reproduced Adoption count.

Step seven. Compare the analyst's count to the Foundation's published Adoption count. The expected condition is that the analyst's count falls within the five-percent publication-grade tolerance of the Foundation's count.

B.3.2 Annual Transactions Reproduction

Step one. Obtain the base-layer transaction count for the trailing twelve months ending at the measurement date through the principal on-chain analytics providers or through direct chain-data extraction.

Step two. Obtain the layer-2 transaction counts for the trailing twelve months from the layer-2 protocols' observable state and from the wallet-provider attestations.

Step three. Apply the non-economic-activity adjustments the methodology specifies: exchange-internal consolidation exclusions, custodial-provider housekeeping exclusions, entity-clustering exclusions, dust-threshold exclusions, and contract-internal-operation exclusions. Adjustments require the entity-clustering inputs from the Adoption reproduction; the adjustments are conducted at the per-transaction level.

Step four. Sum the adjusted base-layer and layer-2 transaction counts to produce the analyst's reproduced Annual Transactions count.

Step five. Compare the analyst's count to the Foundation's published Annual Transactions count. The expected condition is that the analyst's count falls within the two-percent publication-grade tolerance of the Foundation's count.

B.3.3 Annual Transaction Value Reproduction

Step one. For each transaction in the adjusted Annual Transactions count from Section B.3.2, compute the United

States dollar value at the transaction's block-time, using the volume-weighted average price across the principal regulated exchanges within the temporal window the methodology specifies.

Step two. Sum the per-transaction values across the trailing twelve months to produce the raw Annual Transaction Value.

Step three. Entity adjustments are already incorporated at the per-transaction level from Section B.3.2; the analyst's reproduced Annual Transaction Value is the sum from step two.

Step four. Compare the analyst's value to the Foundation's published Annual Transaction Value. The expected condition is that the analyst's value falls within the five-percent publication-grade tolerance of the Foundation's value.

B.3.4 Active Developers Reproduction

Step one. Identify the principal protocol implementation repositories for the coin. The Foundation publishes the repository list as part of the measurement methodology. Extract the commit history across the trailing twelve months ending at the measurement date.

Step two. Identify the extended ecosystem repositories for the coin. Extract the commit history across the trailing twelve months.

Step three. Identify the application-layer repositories for the coin. Extract the commit history across the trailing twelve months.

Step four. Across the three layers, identify the unique contributors through the contributor-identification procedures the methodology specifies (commit-author normalization, alias

resolution, automation filtering). Produce the analyst's reproduced three-layer Active Developers count.

Step five. Cross-validate against the Electric Capital developer-tracking dataset's count for the coin at the measurement date. Cross-validation tolerance is approximately ten percent; substantial divergence between the analyst's count and the Electric Capital count indicates methodology divergence that the analyst documents.

Step six. Compare the analyst's count to the Foundation's published Active Developers count. The expected condition is that the analyst's count falls within the ten-percent publication-grade tolerance of the Foundation's count.

B.4 The Composite Score and Subsequent Outputs

Once the four-metric reproductions are complete, the analyst computes the Composite Score and the subsequent outputs through the formula's arithmetic.

Composite Score computation. An analyst computes the four-metric ratios using the reproduced metric values divided by the Benchmark values (80,000,000; 6,090,000,000; 13,490,000,000,000; 905). The analyst then applies the 0.70/0.10/0.10/0.10 weighting and sums to produce the reproduced Composite Score.

Crypto Fair Value computation. An analyst multiplies the reproduced Composite Score by the market-capitalization anchor (\$1,983,000,000,000) to produce the reproduced Crypto Fair Value.

Fair Coin Price computation. An analyst divides the reproduced Crypto Fair Value by the coin's circulating supply at the measurement date to produce the reproduced Fair Coin

Price. Circulating supply is obtained from the coin's chain at the measurement date or from the principal supply-tracking analytics providers.

Margin of Safety computation. An analyst subtracts the coin's market price at the measurement date from the reproduced Fair Coin Price and divides by the reproduced Fair Coin Price to produce the reproduced Margin of Safety. Market price is obtained from the volume-weighted average across the principal regulated exchanges within the temporal window the methodology specifies.

Comparison. An analyst compares the four reproduced outputs (Composite Score, Crypto Fair Value, Fair Coin Price, Margin of Safety) to the Foundation's published outputs. The expected condition is that each reproduced output falls within the one-percent publication-grade tolerance of the corresponding published output.

B.5 Handling Discrepancies

Verification may produce discrepancies between the analyst's reproduced outputs and the Foundation's published outputs. The protocol specifies how the analyst handles discrepancies.

Within-tolerance discrepancies. Discrepancies within the publication-grade tolerance bands are expected and reflect the operational variation the methodology admits. An analyst documents the within-tolerance discrepancies as part of the verification record; the discrepancies do not require submission to the Foundation.

Outside-tolerance discrepancies. Discrepancies that exceed the publication-grade tolerance band on one or more outputs indicate methodology divergence between the analyst's

application and the Foundation's. An analyst conducts the following examination.

Step one. Identify which metric reproduction is producing the divergence. Propagation through the Composite Score arithmetic typically traces the divergence to one or two of the four-metric reproductions.

Step two. Examine the analyst's application of the methodology against the published methodology specification. Identify any procedural choices the analyst made that differ from the methodology's specification or that the methodology does not explicitly resolve.

Step three. Examine the data sources the analyst used against the data sources the methodology references. Identify any source differences, source-date differences, or source-completeness differences that may produce the divergence.

Step four. If steps one through three identify a procedural or source-access issue on the analyst's side, the analyst adjusts the application and produces a corrected reproduction. That corrected reproduction is documented as the analyst's verification output.

Step five. If steps one through three do not identify a procedural or source-access issue on the analyst's side, the divergence may indicate a methodology divergence between the analyst's application and the Foundation's, or it may indicate an issue with the Foundation's published outputs. The analyst submits the documented divergence to the Foundation through the submission channel Section B.6 specifies.

B.6 The Submission Channel

The Foundation maintains a submission channel through which independent analysts report verification discrepancies and through which the Foundation responds.

Submission contents. A submission includes the analyst's identification, the Coin Report being verified, the analyst's reproduced outputs across the four-metric reproductions and the four formula-derived outputs, the data sources the analyst used with provenance documentation, the procedural choices the analyst made at each step, the discrepancies the analyst's outputs produced against the Foundation's published outputs, and the analyst's examination results from Section B.5.

Submission timeline. Submissions are accepted on a continuing basis. The Foundation processes submissions within ninety days of receipt and publishes the response in the next available reporting cycle following the processing.

Foundation response. A Foundation response to a submission identifies whether the Foundation accepts the analyst's reproduction (confirming methodology divergence the Foundation's review acknowledges), confirms the Foundation's published output (identifying the methodology basis on which the analyst's reproduction differs), or identifies a methodology refinement that the protocol-version architecture will admit in a subsequent version.

Publication. The Foundation publishes a record of submissions and responses as part of the framework's transparency commitments. Publication does not identify individual analysts unless the analyst requests identification; the publication identifies the substantive content of the submission and the Foundation's response.

B.7 What the Protocol Establishes

This protocol the appendix has specified is operational. An independent analyst with the technical capacity and the time the verification requires can execute the protocol against any published Coin Report and produce a reproduction within the documented tolerance bands. The framework's accountability to its outputs depends on this property.

A reader who is allocating capital under the framework does not need to execute the protocol themselves; the reader relies on the verifications the institutional reviewers and the independent analysts continue to conduct. A reader's confidence in the framework's outputs depends on the verifications continuing, on the discrepancies being reported, and on the Foundation responding through the documented procedures.

This protocol is part of the framework's structural integrity. Verifications are the framework's accountability. Submissions and responses are the framework's continuing engagement with the adversarial review the framework's commitments invite. The framework operates under the protocol the appendix specifies, and the framework's continued operation depends on the protocol remaining operational across the years the framework's discipline applies.

SOURCES AND FURTHER READING

The intellectual references on which this book draws

This volume builds on a tradition that extends from nineteenth-century monetary theory through twentieth-century equity-investing discipline to twenty-first-century cryptocurrency analysis. What follows identifies the principal references the framework operates against, organized by category. Readers wishing to deepen their engagement with the tradition will find each reference rewarding on its own terms.

On Graham and the Discipline of Investment

Graham, Benjamin, and David L. Dodd. *Security Analysis*. New York: McGraw-Hill, 1934. The foundational text of analytical equity investing. The Margin of Safety concept is developed here at the depth that the asset-class translation in this volume's Chapter 7 builds on. Multiple subsequent editions; the 1934 first edition and the 1940 second edition contain the most rigorous treatment of the framework's structural principles.

Graham, Benjamin. *The Intelligent Investor*. New York: Harper & Brothers, 1949. Revised edition, 1973. The book this volume responds to. Graham's distinction between investment and speculation, his treatment of intrinsic value, his voting-machine and weighing-machine metaphor, and his Chapter 20 development of the Margin of Safety as the central concept of sound investment all supply the conceptual coordinates the framework translates. The 1973 revision with Jason Zweig's contemporary commentary is the most accessible modern edition.

Buffett, Warren E. *Berkshire Hathaway Letters to Shareholders, 1965–present*. Omaha: Berkshire Hathaway. The operational record of Graham’s discipline applied across six decades. The letters develop the framework’s application across changing market conditions and demonstrate the holding-through-divergence discipline this volume’s Chapter 7 develops for Layer-1 Coins.

On the Austrian Monetary Tradition

Menger, Carl. *Principles of Economics*. Vienna, 1871. Translated by James Dingwall and Bert F. Hoselitz, with introduction by Frank H. Knight. The foundational text of Austrian economics. Menger’s analysis of money as emerging from the market’s collective preference for whatever commodity best satisfies the conditions commerce imposes is the conceptual origin of the Six Pillars.

Menger, Carl. “On the Origins of Money.” *Economic Journal*, June 1892. The shorter monograph in which Menger develops the marketability-of-commodities argument for money’s emergence. Brief, foundational, indispensable.

Mises, Ludwig von. *The Theory of Money and Credit*. Vienna, 1912. Translated by H. E. Batson. Mises’s extension of Menger’s analysis to demonstrate that sound money requires not only the right physical properties but the right institutional conditions, specifically the absence of monopolistic control by any authority whose interests diverge from holders. The framework’s Decentralized Governance Pillar draws on Mises’s institutional analysis.

Hayek, F. A. *The Denationalisation of Money*. London: Institute of Economic Affairs, 1976. Hayek’s argument that competing currencies produce the market discipline single-issuer

monopolies cannot. The work supplies the theoretical foundation for the framework's treatment of cryptocurrencies as monetary instruments whose value rests on voluntary holder selection rather than on issuer authority.

Rothbard, Murray N. *What Has Government Done to Our Money?* Auburn, AL: Mises Institute, 1963; revised 1990. The shortest serious treatment of how sound money emerged historically and what political interventions have done to it. Rothbard synthesizes the Austrian tradition in a form accessible to readers approaching it for the first time.

On Falsifiability and the Philosophy of Method

Popper, Karl R. *The Logic of Scientific Discovery*. London: Hutchinson, 1959 (English translation; German original 1934). Popper's development of falsifiability as the criterion that distinguishes scientific theories from theories whose proponents accommodate every observation through ad-hoc adjustment. Chapter 15 of this volume applies Popper's discipline to the framework's accountability commitments.

Popper, Karl R. *Conjectures and Refutations: The Growth of Scientific Knowledge*. London: Routledge, 1963. The shorter and more accessible statement of Popper's position, developed through specific examples. The relevant chapters for the framework's purposes are the early ones treating science and pseudo-science.

On Cryptocurrency and the Framework

Nakamoto, Satoshi. "Bitcoin: A Peer-to-Peer Electronic Cash System." 2008. The eight-page paper that introduced the protocol from which the Digital Gold Standard Benchmark was calibrated.

The architectural elegance of Nakamoto's design is what produced the operational record that made the December 2024 calibration possible.

Gotts, John Wright. *Cryptocurrency Analysis*. Digital Gold Foundation, 2025. The first volume of the four-volume corpus and the framework's comprehensive specification at the depth a methodology of this kind requires. Runs to over one thousand pages. Volume 1 develops the architectural test, the Six Pillars, the calibration methodology, the Crypto Fair Value formula, the Foundation's institutional architecture, the model legislative agenda, and the verification protocols at operational depth. Readers consulting the present volume for allocation discipline should consult Volume 1 for the methodology specification the present volume operates against.

Gotts, John Wright. *Digital Gold White Paper*. Digital Gold Foundation. The protocol-level documentation of Digital Gold (DGD), the Foundation's own coin, including the structural-CFV calibration architecture, the seven structural distinctions that distinguish Digital Gold's legal posture under the Howey test, and the operational integration with the framework's broader institutional architecture.

On Institutional Architecture and Public Fiduciary Authority

Arizona State Legislature. *Senate Bill 1649, Fifty-Sixth Legislature, First Regular Session*. The statutory architecture that incorporates the framework's tracked set, quarterly Coin Reports, and falsification commitments as statutory triggers for public-fiduciary cryptocurrency allocation. The first sustained state-treasurer application of the framework. The Arizona model is available through the state legislature's public-records system

and is treated at depth in *Cryptocurrency Analysis* Chapter 14 and Appendix C.

Securities and Exchange Commission. *In the Matter of W. J. Howey Co.*, 328 U.S. 293 (1946). The Supreme Court decision establishing the four-factor test that defines a security under United States law. The Howey test informs the framework's Pillar evaluation and is treated in Chapter 14's response to Gary Gensler's regulatory-skeptic case.

On the Critics This Volume Engages

Alden, Lyn. *Broken Money: Why Our Financial System Is Failing Us and How We Can Make It Better*. 2023. Alden's principal monograph developing the Bitcoin-maximalist case the framework engages in Chapter 14. The argument's force rests on Bitcoin's unique operational record and the institutional infrastructure that has matured around Bitcoin specifically.

Roubini, Nouriel, and Stephen Mihm. *Crisis Economics: A Crash Course in the Future of Finance*. New York: Penguin Press, 2010. Roubini's framework for financial-crisis analysis, which underlies his asset-class-skeptic case engaged in Chapter 14. Roubini's specific arguments against cryptocurrency are developed across his published commentary and public statements rather than in a single book.

Gensler, Gary. Public statements during his tenure as Securities and Exchange Commission Chair, 2021–2025. Gensler's regulatory position on cryptocurrency was developed primarily through speeches, congressional testimony, and SEC enforcement actions during his tenure. The principal statements are accessible through the SEC's public-records system.

On Measurement Methodology and Infrastructure

Electric Capital. *Developer Report*, published annually. The cryptocurrency developer-tracking dataset against which the Foundation cross-validates the Active Developers metric. Available at electriccapital.com.

Cambridge Centre for Alternative Finance. *Global Cryptoasset Benchmarking Study*, published periodically. Survey-based research supplying input to the Adoption metric's five-source triangulation methodology. Available through the Cambridge Judge Business School.

Federal Reserve Board. *Survey of Consumer Finances*, digital-asset module. Survey research supplying United States holder-population input to the Adoption metric. Published triennially.

These references are not exhaustive. The framework draws on a broader scholarly literature in monetary theory, financial economics, cryptography, distributed systems, and regulatory analysis that *Cryptocurrency Analysis* documents at the depth that volume supplies. Readers who proceed under the framework's discipline benefit from working through the present volume's references in conjunction with the broader corpus.

GLOSSARY

Canonical definitions of terms used throughout this book

This glossary supplies the definitions the framework operates under. Where a term has a specific operational meaning that differs from common usage, the definition states the framework's meaning and notes the distinction. Where a term has been the subject of vocabulary discipline in the framework's documentation, the definition states the term's status and identifies the substitutions the framework uses.

Active Developers. The fourth metric in the Crypto Fair Value formula, contributing ten percent of the Composite Score weight. Measured as the trailing-twelve-month count of unique contributors to a coin's essential repositories across three layers (principal protocol implementation, extended ecosystem, application-layer), with cross-validation against the Electric Capital developer-tracking dataset. The Benchmark value is 905.

Adequate Circulation. The sixth of the Six Pillars of Perfect Money, and the fifth of the five gating Pillars. Adequate Circulation is the condition that the supply in operation across the network is sufficient to support transactional purposes without forcing fractional-reserve substitutes to emerge. A coin satisfies Adequate Circulation when the supply circulates among holders who use the coin for its intended monetary functions and when wrapped representations or custodial proxies do not need to develop to absorb the holder base's transactional volume.

Admission. The process by which a coin enters the Foundation’s tracked set. Admission requires the coin to satisfy the architectural test (Chapter 2), the five gating Pillars (Chapter 3), and the Foundation’s verification review. A coin that satisfies all three is admitted; a coin that fails any is not. The framework uses “admission” and “admitted” rather than “qualification” and “qualifying” as canonical Foundation-coverage vocabulary.

Adoption. The first metric in the Crypto Fair Value formula, contributing seventy percent of the Composite Score weight. Measured as the entity-adjusted count of unique global holders of a coin at the measurement date, produced through the five-source triangulation methodology Appendix A specifies. The Benchmark value is 80,000,000.

Annual Transactions. The second metric in the Crypto Fair Value formula, contributing ten percent of the Composite Score weight. Measured as the trailing-twelve-month count of transactions across a coin’s base layer and the principal layer-2 infrastructure, with non-economic-activity adjustments. The Benchmark value is 6,090,000,000.

Annual Transaction Value. The third metric in the Crypto Fair Value formula, contributing ten percent of the Composite Score weight. Measured as the trailing-twelve-month United States dollar value of economic transfers across a coin’s base layer and principal layer-2 infrastructure, entity-adjusted. The Benchmark value is \$13,490,000,000,000.

Architectural Test. The binary test that distinguishes a Layer-1 Coin from a token, developed in Chapter 2. The test has three conditions: the candidate operates its own blockchain, supports its own native wallet derivation, and is the protocol’s native coin. A candidate that satisfies all three is a

Layer-1 Coin; a candidate that fails any is a token. The framework values Layer-1 Coins and does not extend analytical claims to tokens.

Benchmark. Short form of Digital Gold Standard Benchmark.

Bitcoin (BTC). The cryptocurrency from which the Digital Gold Standard Benchmark was calibrated at December 31, 2024. Bitcoin is one Layer-1 cryptocurrency among others, measured against the Benchmark on identical terms with every other Layer-1 Coin the framework evaluates. Bitcoin is not on the Foundation's tracked set; its contribution to the framework was the calibration, which is permanent and historically singular.

Cardano (ADA). A Layer-1 Coin on the Foundation's tracked set. Cardano operates the Ouroboros family of peer-reviewed proof-of-stake consensus protocols. Launched September 2017.

Calibration. The act of establishing the Benchmark values from Bitcoin's December 31, 2024 state. The calibration is a single historical act; it is not repeated, and the Benchmark values it produced are permanent within the framework's protocol version.

Cash Flow Value. A term that is not used in the framework. The framework's central valuation concept is the Crypto Fair Value, abbreviated CFV. Practitioners encountering "Cash Flow Value" in any cryptocurrency-allocation context are encountering an unrelated concept.

Chia (XCH). A Layer-1 Coin on the Foundation's tracked set. Chia operates the proof-of-space-and-time consensus

architecture, using storage capacity as the consensus resource. Launched March 2021.

Circulating Supply. The portion of a coin's protocol-issued supply that is in operational circulation at a measurement date, computed as the protocol-issued supply less the verifiably lost and provably burned supply. The Fair Coin Price calculation divides the Crypto Fair Value by the candidate's circulating supply.

Coin Report. The Foundation's per-coin documentation, published on a quarterly cycle, that reports the four-metric measurements, the Composite Score, the Crypto Fair Value, the Fair Coin Price, the Margin of Safety against the prevailing market price, the trajectory across the trailing four quarters, the conviction category, and the working conclusion the framework's analysis supports for the coin at the measurement date.

Composite Score. The dimensionless figure produced by the weighted sum of the four-metric ratios in the Crypto Fair Value formula. The weights are 0.70 (Adoption), 0.10 (Annual Transactions), 0.10 (Annual Transaction Value), and 0.10 (Active Developers). A Composite Score of 1.0 indicates parity with the Benchmark on the weighted aggregate; Bitcoin's Composite Score at the December 2024 calibration moment is 1.0 by construction.

Conviction Category. The framework's operational classification of a held position based on the integration of the point-in-time Margin of Safety and the trajectory across the trailing four quarters. The four categories are high conviction, acceptable conviction, low conviction, and below-threshold. The categories drive the allocation discipline's position-sizing decisions.

Crypto Fair Value (CFV). The framework's analytical reference value for a candidate Layer-1 Coin, computed as the Composite Score multiplied by the Benchmark's market-capitalization anchor of \$1.983 trillion. The Crypto Fair Value is a market-capitalization figure denominated in United States dollars; it is the framework's estimate of the candidate's intrinsic market capitalization based on the measured fundamentals against the Benchmark. The framework abbreviates the term as CFV; the abbreviation does not stand for any other phrase.

CFV. See Crypto Fair Value.

CFV CoinFunds. The Foundation-supported investment vehicles that allocate capital under the framework's discipline to the tracked set of Layer-1 Coins. The CoinFunds do not hold Bitcoin.

DASH. Ticker for Digital Cash.

Decentralized Governance. The fourth of the Six Pillars of Perfect Money, and one of the five gating Pillars. Decentralized Governance is the condition that the rules governing a coin's operation are not subject to the discretion of any identifiable party, but emerge from the consensus mechanism the protocol operates under. The Pillar is evaluated across four dimensions: validator distribution, holder distribution, operational history of protocol upgrades, and institutional arrangements around continued development.

DGB. Ticker for DigiByte.

DGD. Ticker for Digital Gold.

DGSB. Abbreviation for Digital Gold Standard Benchmark.

DigiByte (DGB). A Layer-1 Coin on the Foundation’s tracked set. DigiByte operates a five-algorithm proof-of-work consensus mechanism (SHA-256, Scrypt, Skein, Qubit, Odocrypt). Launched January 2014.

Digital Cash (DASH). A Layer-1 Coin on the Foundation’s tracked set. Digital Cash operates a two-tier consensus architecture: proof-of-work base layer with masternode tier providing InstantSend, optional PrivateSend privacy, and ChainLocks attack resistance. The canonical name used throughout the framework’s documentation is “Digital Cash”; the ticker is DASH. Launched January 2014.

Digital Gold (DGD). The Foundation’s own coin, engineered to be at its Crypto Fair Value at each level by design. Digital Gold is presented as a separate item rather than as a thirteenth tracked coin; the canonical phrasing is “twelve tracked coins plus DGD,” never “thirteen.”

Digital Gold Foundation. The institutional entity that publishes the framework’s methodology, produces the quarterly Coin Reports, maintains the Benchmark, supports the CFV CoinFunds, and operates the verification channels through which independent analysts submit reproduction outputs. Established by John Wright Gotts.

Digital Gold Standard Benchmark. The reference calibrated from Bitcoin’s December 31, 2024 state, consisting of six values: 80 million holders, 6.09 billion annual transactions, \$13.49 trillion annual transaction value, 905 active developers, \$1.983 trillion market-capitalization anchor, and 19.83 million circulating supply. The Benchmark is permanent within the framework’s current protocol version.

eCash (XEC). A Layer-1 Coin on the Foundation's tracked set. eCash operates a dual-layer architecture combining Nakamoto-consensus proof-of-work with an Avalanche pre-consensus layer for sub-second finality. Forked from Bitcoin Cash in 2020.

Electric Capital. An independent organization that publishes a quarterly cryptocurrency developer-tracking dataset. The Foundation's Active Developers measurement is cross-validated against the Electric Capital dataset for coins where the dataset includes the coin.

Ethereum (ETH). A Layer-1 cryptocurrency not on the Foundation's tracked set. The framework applies the four-metric methodology to Ethereum in calibration tests (Chapter 9), but Ethereum is not admitted to the tracked set, primarily on Decentralized Governance considerations.

Fair Coin Price. The per-coin expression of the Crypto Fair Value, computed as the Crypto Fair Value divided by the candidate coin's circulating supply at the measurement date. The Fair Coin Price is denominated in United States dollars; it is the per-coin figure the framework's measured fundamentals warrant. The Margin of Safety is computed against the Fair Coin Price.

Falsification Conditions. The five specific empirical patterns that, if observed across the framework's operational record, would establish that the methodology requires substantial revision or is wrong. The conditions are developed in Chapter 15 and operate under the operational responses Chapter 15 specifies.

Foundation. Short form of Digital Gold Foundation.

Free Adoption. The third of the Six Pillars of Perfect Money, and one of the five gating Pillars. Free Adoption is the condition that a coin's holder base has been assembled through voluntary acquisition rather than through legal tender requirements, employer or counterparty obligations, or other forms of coercion.

Freedom to Transact. The fifth of the Six Pillars of Perfect Money, and one of the five gating Pillars. Freedom to Transact is the condition that holders can move the coin to other parties without permission from any intermediary that could deny, censor, or reverse the transaction. The Pillar evaluates the protocol's base-layer permissionlessness, with cryptographic privacy strengthening the satisfaction.

Hidden Gems. A term that is not used in the framework. The phrase appears in some earlier-edition cryptocurrency literature as a category for emerging coins; the framework does not adopt the categorization, does not use the phrase, and the framework's tracked set is not described in such terms.

ICP. Ticker for Internet Computer.

Institutional Allocator. The first of the framework's two readers, addressed in Chapter 10. The institutional allocator manages capital under fiduciary duty and reports to a board or equivalent constituency. Examples include fund managers, family-office principals, endowment officers, pension administrators, sovereign-wealth-fund executives, and state treasurers operating under statutory authority.

Institutional Blue Chip. A term that is not used in the framework. Some earlier-edition cryptocurrency literature has used the phrase to describe Bitcoin and Ethereum as a category; the framework does not adopt the categorization,

does not use the phrase, and the framework's conviction-category structure operates on coin-specific Margin of Safety and trajectory readings rather than on categorical labels.

Internet Computer (ICP). A Layer-1 Coin on the Foundation's tracked set. Internet Computer operates the chain-key cryptography architecture supporting both monetary operations and computational workloads at the protocol level. Launched May 2021.

Intrinsic Value. Graham's term for the value an asset's fundamental properties warrant, distinct from the market price the asset trades at. For equity securities, Graham derived intrinsic value from cash flows, earnings, and balance-sheet position. The framework's bearer-asset translation of intrinsic value for Layer-1 Coins is the Fair Coin Price, derived from the four monetary-function metrics against the Benchmark.

Layer-1 Coin. A cryptocurrency that operates its own blockchain, supports its own native wallet derivation, and is the protocol's native coin. The framework's analytical attention is directed at Layer-1 Coins; tokens are outside the framework's scope.

Margin of Safety (MoS). Graham's central concept, translated by the framework for cryptocurrency allocation. The Margin of Safety is the percentage gap between the Fair Coin Price and the current market price, divided by the Fair Coin Price. A positive Margin of Safety indicates the candidate is undervalued relative to its measured fundamentals; a negative Margin of Safety indicates the candidate is overvalued. The framework operates under two operational thresholds: thirty-five percent for coins with fully documented measurements, fifty percent for coins with any unreliable input.

Market-capitalization anchor. The \$1.983 trillion value at which Bitcoin's market capitalization was measured at the December 31, 2024 calibration moment. The anchor multiplies the Composite Score to produce the Crypto Fair Value. It is one of the six Benchmark values; it does not change.

Monero (XMR). A Layer-1 Coin on the Foundation's tracked set. Monero operates the cryptographic privacy architecture (ring signatures, stealth addresses, confidential transactions) that produces transaction-graph privacy at the protocol level. Launched April 2014.

MoS. See Margin of Safety.

MultiversX (EGLD). A Layer-1 Coin on the Foundation's tracked set. MultiversX operates the adaptive state sharding architecture for high-throughput proof-of-stake consensus. Launched July 2020 (originally as Elrond Network; rebranded November 2022).

Nano (XNO). A Layer-1 Coin on the Foundation's tracked set. Nano operates a feeless transaction architecture using a directed acyclic graph (DAG) ledger structure with delegated proof-of-stake consensus. Launched 2015 (originally as RaiBlocks).

Near Protocol (NEAR). A Layer-1 Coin on the Foundation's tracked set. Near Protocol operates a sharded proof-of-stake consensus with human-readable account-naming. Launched April 2020.

Pillar. Short form of Pillar of Perfect Money. See Six Pillars of Perfect Money.

Protocol Version. The framework's architectural mechanism for admitting methodology evolution while preserving the

temporal reproducibility historical readings require. A protocol version specifies the methodology in effect for a defined period; transitions between versions are documented and operate prospectively. The Benchmark values are protocol-version-invariant.

Ravencoin (RVN). A Layer-1 Coin on the Foundation's tracked set. Ravencoin launched in January 2018 from a Bitcoin codebase fork, without pre-mine, initial coin offering, or founder allocations.

Reproduction Protocol. The step-by-step procedure for independent verification of any Coin Report, developed in Chapter 13 and specified at operational depth in Appendix B. The protocol supports the framework's reproducibility commitment.

Six Pillars of Perfect Money. The framework's qualitative discipline for evaluating a candidate Layer-1 Coin's monetary properties. Five of the Pillars are admission gates: Scarcity, Free Adoption, Decentralized Governance, Freedom to Transact, and Adequate Circulation. One Pillar is developmental: Stable Pricing.

Scarcity. The first of the Six Pillars of Perfect Money, and one of the five gating Pillars. Scarcity is the condition that the supply of a coin is fixed or grows on a known, predictable schedule that cannot be altered by the discretion of any central authority.

Sophisticated Individual. The second of the framework's two readers, addressed in Chapter 11. The sophisticated individual allocates personal capital under personal discipline, has internalized Graham's framework across the equity-

investing decades, and applies the framework to cryptocurrency under the personal-capital constraint Chapter 11 develops.

Stable Pricing. The second of the Six Pillars of Perfect Money, and the developmental Pillar. Stable Pricing is the condition that a coin's purchasing power varies within bounds that allow it to function as a unit of account across the timescale of ordinary commerce. The Pillar is reported developmentally rather than as an admission gate, because no Layer-1 Coin in the current tracked set has yet achieved the price stability the Pillar's full satisfaction requires.

Tracked Set. The Foundation's operationally maintained list of Layer-1 Coins for which the Foundation publishes quarterly Coin Reports. The tracked set currently contains twelve coins, plus Digital Gold (DGD) presented as a separate item. The canonical presentation is "twelve coins plus DGD," never "thirteen."

Trajectory. The sequence of point-in-time Margin of Safety readings across the trailing four quarters at a measurement date. The trajectory distinguishes coins whose Margins of Safety are rising (the framework's reading moving in the position's favor), flat (the reading stable), or falling (the reading moving against the position). The trajectory integrates with the point-in-time reading to produce the conviction category.

Voting Machine. Graham's term for the market's short-term price-discovery mechanism, contrasted with the Weighing Machine. The Voting Machine's outputs reflect aggregate buyer and seller decisions in the current window, influenced by narrative cycles and institutional flows. Prices it produces may differ substantially from the values the asset's fundamentals warrant.

Weighing Machine. Graham's term for the market's long-term price-discovery mechanism, contrasted with the Voting Machine. The Weighing Machine's outputs reflect the fundamentals an asset produces over multi-year horizons. The framework's Fair Coin Price is a Weighing Machine estimate; the Margin of Safety operates on the gap between the Weighing Machine reading and the Voting Machine output.

XCH. Ticker for Chia.

XEC. Ticker for eCash.

XMR. Ticker for Monero.

XNO. Ticker for Nano.

ZClassic (ZCL). A Layer-1 Coin on the Foundation's tracked set. ZClassic launched in November 2016 as a fork of Zcash that removed the founder's reward allocation, combining zero-knowledge cryptographic privacy with launch-distribution discipline.

ABOUT THE AUTHOR

John Wright Gotts is the founder of the Digital Gold Foundation and the architect of the Crypto Fair Value framework. That framework supplies the analytical infrastructure under which the Foundation maintains the Digital Gold Standard Benchmark, produces the quarterly Coin Reports on the twelve tracked Layer-1 Coins, supports the CFV CoinFunds, and operates the verification channels through which independent analysts continue to validate the framework's outputs.

This Crypto Fair Value framework was developed across the years preceding the December 2024 calibration moment. Its first comprehensive presentation, *Cryptocurrency Analysis*, runs to over one thousand pages and supplies the methodology at the depth a framework of this kind requires. *The Intelligent Investor (Crypto Edition)* is the second volume of the four-volume corpus, applying the framework's analytical infrastructure to the allocation discipline Benjamin Graham developed for equity investing across the twentieth century. The third and fourth volumes are forthcoming.

The Digital Gold Foundation is organized as a public-benefit corporation. Its institutional architecture is designed to support regulatory engagement, fiduciary defensibility, and the multi-decade horizons monetary instruments require. Institutional details, including the governance structure, the operational principles, the relationships among the Foundation, the framework, the CoinFunds, and Digital Gold itself, are developed in *Cryptocurrency Analysis* Chapter 14 at the depth that document supplies.

Gotts has positioned the framework as accountable to adversarial review rather than asserted on the author's authority. A reproduction protocol developed in this volume's Chapter 13 and specified at operational depth in Appendix B is the framework's principal mechanism for that accountability. Falsification conditions developed in Chapter 15 are the framework's commitment to declare itself wrong, in writing, in advance, when the conditions are satisfied. These commitments are part of the framework's structural integrity; they are what distinguish the framework from analytical methodologies whose proponents accommodate every adverse outcome through ad-hoc adjustment.

Continued operation of the framework depends on the practitioners who deploy capital under it, the institutional reviewers who engage independent analysts to verify the outputs, the adversarial reviewers who continue to submit reproductions and discrepancies for the Foundation's response, and the state-treasurer applications that operate under statutory authorities incorporating the framework's tracked set and verification protocols. Arizona Senate Bill 1649's statutory architecture is the first sustained state-treasurer application; further state-treasurer applications and additional institutional integrations are expected to develop as the framework's operational record continues.

Inquiries regarding the framework, the Foundation, the Coin Reports, or independent verification submissions are directed to the Digital Gold Foundation through the channels published at digitalgoldfoundation.org.

This framework is not Gotts's alone; it is the work of the Foundation and the broader community of practitioners, analysts, and reviewers who have engaged the framework across its operational history. The author's role is to bear public

responsibility for the framework's integrity, to maintain the commitments the framework's accountability structure requires, and to continue developing the corpus the four-volume project produces. Continued operation of the framework is the work; the work is what the author offers the practitioner who has read this book.

INDEX

Terms, persons, and concepts in this volume

This index covers the principal terms, persons, and concepts treated in this volume. Locations are given by chapter (Ch 1 through Ch 15), front matter (Preface), back matter (Acknowledgments, Bibliography, Glossary, About the Author), and the two appendices (App A: Benchmark Specification; App B: Reproduction Protocol). Where a term has a canonical form different from the one a reader may search under, a cross-reference directs the reader to the canonical entry.

A

Active Developers. Ch 1, Ch 3, Ch 4, Ch 5, Ch 6, Ch 7, Ch 8, Ch 9, Ch 13, Ch 14, App A, App B, Bibliography, Glossary.

Adequate Circulation. Ch 3, Ch 4, Ch 8, Glossary.

admission gates. *See* Six Pillars of Perfect Money.

Adoption (metric). Ch 1, Ch 3, Ch 4, Ch 5, Ch 6, Ch 7, Ch 8, Ch 9, Ch 13, Ch 14, App A, App B, Bibliography, Glossary.

Alden, Lyn. Ch 14, Bibliography.

Annual Transaction Value. Ch 1, Ch 3, Ch 4, Ch 5, Ch 6, Ch 7, Ch 8, Ch 9, Ch 13, Ch 14, App A, App B, Glossary.

Annual Transactions. Ch 1, Ch 3, Ch 4, Ch 5, Ch 6, Ch 7, Ch 8, Ch 9, Ch 13, Ch 14, App A, App B, Glossary.

architectural test. Preface, Ch 2, Ch 3, Ch 4, Ch 7, Bibliography, Glossary.

Arizona Senate Bill 1649. Ch 1, Ch 4, Ch 8, Ch 10, Ch 14, Acknowledgments, About the Author.

Austrian tradition. Preface, Ch 3, Acknowledgments, Bibliography.

Avalanche pre-consensus. Ch 4, Ch 8, Glossary.

B

basket calibration (rejected). Ch 5.

bearer-asset translation. Ch 7, Ch 14, Glossary.

below-threshold coins. Ch 7, Ch 10, Ch 12, Glossary.

Benchmark. *See* Digital Gold Standard Benchmark.

Bitcoin. Preface, Ch 1, Ch 2, Ch 3, Ch 4, Ch 5, Ch 6, Ch 7, Ch 8, Ch 9, Ch 14, App A, Bibliography, Glossary.

Bitcoin, calibration source. *See* Digital Gold Standard Benchmark.

Bitcoin, December 2024 state. Ch 1, Ch 4, Ch 5, Ch 6, Ch 9, Ch 14, Glossary.

Bitcoin maximalism. *See* Alden, Lyn.

Bitcoin, October 2025 peak. Ch 1, Ch 5, Ch 7, Ch 9, Ch 14.

block explorer. Ch 2.

Buffett, Warren. Preface, Ch 1, Bibliography.

C

Cambridge Centre for Alternative Finance. App B, Bibliography.

Cardano (ADA). Ch 1, Ch 2, Ch 4, Ch 6, Ch 8, Glossary.

Cash Flow Value (disowned term). Glossary.

CFV CoinFunds. Ch 1, Ch 4, Ch 7, Ch 9, Glossary, About the Author.

Chainalysis. App B.

ChainLocks. Ch 4, Ch 8, Glossary.

Chia (XCH). Ch 1, Ch 4, Ch 8, Glossary.

Chialisp. Ch 4, Ch 8.

circulating supply. Ch 3, Ch 5, Ch 6, Ch 7, Ch 9, Ch 13, App A, App B, Glossary.

Coin Metrics. App B.

Coin Reports. Preface, Ch 1, Ch 3, Ch 4, Ch 6, Ch 7, Ch 8, Ch 9, Ch 10, Ch 11, Ch 12, Ch 13, Ch 14, Ch 15, Acknowledgments, App A, App B, Bibliography, Glossary, About the Author.

Composite Score. Ch 1, Ch 3, Ch 4, Ch 5, Ch 6, Ch 7, Ch 8, Ch 9, Ch 10, Ch 12, Ch 13, Ch 15, App A, App B, Glossary.

concentration limits. Ch 10, Ch 11, Ch 12.

conviction categories. Ch 7, Ch 8, Ch 10, Ch 11, Ch 12, Glossary.

conviction transitions. Ch 12.

correlation risk. Ch 12.

cross-validation. Ch 1, Ch 5, Ch 6, Ch 7, Ch 10, Ch 13, App A, Bibliography, Glossary.

Crypto Fair Value (CFV). Preface, Ch 1, Ch 3, Ch 4, Ch 5, Ch 6, Ch 7, Ch 8, Ch 9, Ch 10, Ch 12, Ch 13, App A, App B, Bibliography, Glossary, About the Author.

Cryptocurrency Analysis (Volume 1). Preface, Ch 1, Ch 4, Ch 5, Ch 6, Ch 9, Ch 10, Ch 13, Ch 14, Acknowledgments, App A, App B, Bibliography, About the Author.

D

DASH. *See* Digital Cash.

Decentralized Governance. Ch 3, Ch 4, Ch 8, Ch 9, Ch 14, Bibliography, Glossary.

DFINITY Foundation. Ch 4.

DigiByte (DGB). Ch 1, Ch 3, Ch 4, Ch 8, Glossary.

Digital Cash (DASH). Ch 1, Ch 3, Ch 4, Ch 6, Ch 8, Glossary.

Digital Gold (DGD). Preface, Ch 1, Ch 4, Ch 10, Acknowledgments, App A, Bibliography, Glossary, About the Author.

Digital Gold Foundation. Preface, Acknowledgments, Bibliography, Glossary, About the Author.

Digital Gold Standard Benchmark. Preface, Ch 1, App A, Bibliography, Glossary, About the Author.

Digital Gold White Paper. Ch 4, Bibliography.

discrepancies, submission to Foundation. Ch 13, App A, App B, About the Author.

Dodd, David. Ch 7, Bibliography.

drawdown risk. Ch 12.

E

eCash (XEC). Ch 1, Ch 4, Ch 8, Glossary.

Electric Capital developer-tracking dataset. Ch 5, Ch 7, Ch 13, App A, App B, Bibliography, Glossary.

Elrond. *See* MultiversX.

emotional discipline (personal capital). Ch 11, Ch 12.

engineering origin of Margin of Safety. Ch 4, Ch 7, Ch 10, Ch 15, Glossary.

entity adjustment. Ch 1, Ch 5, Ch 8, Ch 9, Ch 13, App A, App B, Glossary.

ERC-20 tokens. Ch 2.

estate planning. Ch 1, Ch 11, Ch 12.

Ethereum. Preface, Ch 1, Ch 2, Ch 3, Ch 4, Ch 5, Ch 9, Ch 14, Glossary.

F

Fair Coin Price. Ch 1, Ch 3, Ch 4, Ch 5, Ch 6, Ch 7, Ch 8, Ch 9, Ch 10, Ch 11, Ch 12, Ch 13, Ch 14, Ch 15, App A, App B, Glossary.

falsification conditions. Preface, Ch 15.

family office. Preface, Ch 1, Acknowledgments, Glossary.

fiduciary duty. Preface, Ch 1, Ch 10, Glossary.

fifty-one-percent attack. Ch 4, Ch 8, Ch 15.

Foundation. *See* Digital Gold Foundation.

Free Adoption. Ch 3, Ch 4, Ch 9, Glossary.

Freedom to Transact. Ch 3, Ch 4, Glossary.

G

Gensler, Gary. Ch 14, Bibliography.

Glassnode. App B.

Gotts, John Wright. Acknowledgments, Bibliography, Glossary, About the Author.

Graham, Benjamin. Preface, Ch 1, Ch 7, Ch 10, Ch 11, Ch 14, Ch 15, Acknowledgments, Bibliography, Glossary, About the Author.

H

Hayek, Friedrich. Ch 3, Acknowledgments, Bibliography.

Hidden Gems (disowned term). Glossary.

high-conviction coins. Ch 7, Ch 8, Ch 10, Ch 11, Ch 12, Glossary.

holder distribution. Ch 3, Ch 14, Glossary.

holding through divergence. Ch 7, Ch 11, Ch 12.

Hoskinson, Charles. Ch 4.

Howey test. Ch 4, Ch 14, Bibliography.

Hydra (Cardano layer-2). Ch 4, Ch 8.

I

ICP. *See* Internet Computer.

independent verification. Preface, Ch 13, App A, App B, Glossary, About the Author.

Input Output (IOG). Ch 4.

InstantSend. Ch 4, Ch 6, Ch 8, Glossary.

institutional allocator. Preface, Ch 1, Ch 2, Ch 10, Ch 11, Ch 12, Ch 13, Ch 15, Acknowledgments, Glossary.

Institutional Blue Chip (disowned term). Glossary.

Internet Computer (ICP). Ch 1, Ch 4, Glossary.

intrinsic value. Preface, Ch 7, Ch 14, Bibliography, Glossary.

K

kilogram in Sèvres (analogy). Ch 1, Ch 5, Ch 7.

Krugman, Paul. Ch 7.

L

Layer-1 Coin. Preface, Ch 1, Ch 2, Ch 3, Ch 4, Ch 5, Ch 7, Ch 8, Ch 9, Ch 14, Bibliography, Glossary, About the Author.

layer-2 infrastructure. Ch 4, Ch 5, Ch 8, Ch 13, Ch 14, App A, App B, Glossary.

Lightning Network. Ch 5, Ch 9, App A.

liquidity risk at scale. Ch 12.

low-conviction coins. Ch 7, Ch 10, Ch 12, Glossary.

M

Margin of Safety (MoS). Preface, Ch 1, Ch 2, Ch 4, Ch 5, Ch 6, Ch 7, Ch 8, Ch 9, Ch 10, Ch 11, Ch 12, Ch 13, Ch 14, App A, App B, Bibliography, Glossary.

market-capitalization anchor. Ch 5, Ch 6, Ch 7, Ch 13, App A, App B, Glossary.

masternode (DASH). Ch 3, Ch 4, Ch 8, Glossary.

measurement date. Ch 4, Ch 5, Ch 6, Ch 7, Ch 8, Ch 9, Ch 10, Ch 11, Ch 12, Ch 13, Ch 14, Ch 15, App A, App B, Glossary.

measurement-grade tolerance. Ch 6, Ch 13, App A.

Menger, Carl. Ch 3, Acknowledgments, Bibliography.

Mises, Ludwig von. Ch 3, Acknowledgments, Bibliography.

Model T (Bitcoin analogy). Ch 1, Ch 14.

Monero (XMR). Ch 1, Ch 2, Ch 3, Ch 4, Glossary.

MultiversX (EGLD). Ch 1, Ch 4, Glossary.

N

Nakamoto, Satoshi. Ch 4, Ch 8, Bibliography, Glossary.

Nano (XNO). Ch 1, Ch 4, Glossary.

Near Protocol (NEAR). Ch 1, Ch 4, Glossary.

negative Margin of Safety. Ch 7, Glossary.

O

Odocrypt mining algorithm. Ch 4, Ch 8, Glossary.

on-chain analytics. Ch 1, Ch 3, Ch 4, Ch 5, Ch 8, Ch 13, App A, App B.

one-third discount (Graham). Ch 7.

Ouroboros consensus. Ch 4, Ch 8, Glossary.

P

peer-to-peer transactions. Ch 3.

Pillar of Perfect Money. *See* Six Pillars of Perfect Money.

Plomin hard fork (Cardano). Ch 4, Ch 8.

Popper, Karl. Ch 15, Bibliography.

PrivateSend (DASH). Ch 4, Ch 8, Glossary.

proof-of-space-and-time consensus. Ch 2, Ch 4, Ch 8, Glossary.

proof-of-stake consensus. Ch 2, Ch 4, Ch 8, Ch 9, Glossary.

proof-of-work consensus. Ch 1, Ch 2, Ch 4, Ch 8, Ch 14, Glossary.

protocol-version architecture. Ch 5, Ch 15, Acknowledgments, App A, App B, Glossary.

publication-grade tolerance. Ch 1, Ch 6, Ch 13, Ch 15, App A, App B.

Q

quarterly Coin Reports. *See* Coin Reports.

quarterly rebalancing cadence. Ch 10, Ch 11, Ch 12.

R

RaiBlocks. *See* Nano.

RandomX. Ch 4.

Ravencoin (RVN). Ch 1, Ch 4, Glossary.

rebalancing. Preface, Ch 10, Ch 11, App A.

reporting constituency. Ch 1, Ch 10, Ch 11, Ch 12, Ch 13, Ch 15.

reproducibility. Preface, Ch 1, Ch 3, Ch 10, Ch 11, Ch 13, Ch 15, Acknowledgments, App A, App B, Glossary.

reproduction protocol. Preface, Ch 1, Ch 6, Ch 8, Ch 10, Ch 13, Ch 15, Acknowledgments, App A, About the Author.

Rothbard, Murray. Ch 3, Acknowledgments, Bibliography.

Roubini, Nouriel. Ch 14, Bibliography.

S

Scarcity (Pillar). Ch 3, Ch 4, Glossary.

SEC (Securities and Exchange Commission). Ch 5, Ch 14, Bibliography.

Six Pillars of Perfect Money. Preface, Ch 1, Ch 2, Ch 3, Ch 4, Ch 5, Ch 7, Ch 14, Acknowledgments, Bibliography, Glossary.

sophisticated individual. Preface, Ch 1, Ch 2, Ch 11, Ch 12, Glossary.

spot exchange-traded products. Ch 1, Ch 5, Ch 14, App B.

Stable Pricing (developmental Pillar). Ch 3, Glossary.

state treasurer. Preface, Ch 1, Ch 10, Acknowledgments.

T

tokens (vs. coins). Preface, Ch 2, Ch 3, Ch 4, Ch 8, Glossary.

tolerance bands. Ch 13, Ch 15, App A, App B.

tracked set. Preface, Ch 1, Ch 2, Ch 3, Ch 4, Ch 6, Ch 7, Ch 8, Ch 9, Ch 10, Ch 11, Ch 12, Ch 14, Ch 15, Bibliography, Glossary, About the Author.

trajectory analysis. Ch 1, Ch 7, Ch 8, Ch 10, Ch 12, Glossary.

V

voluntary holder base. Ch 1, Ch 3, Ch 6, Ch 9, Ch 14, Bibliography.

voting machine and weighing machine (Graham). Ch 7, Ch 12.

W

weighing machine. *See* voting machine and weighing machine.

X

XRP. Preface, Ch 9.

Z

ZClassic (ZCL). Ch 1, Ch 4, Glossary.

zk-SNARK cryptography. Ch 4.